

Representation Frontiers for the Fabius–Rvachev System

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Contents

I	Resolvent, Continued-Fraction, and Transform Representations of the Fabius–Rvachev System	9
1	Executive summary and novelty boundary	10
1.1	What was audited	10
1.2	Non-duplication protocol	10
1.3	Main new results	10
2	Normalization and structural dictionary	11
2.1	The functions	11
2.2	Fourier convention and basic transforms	11
2.3	The probability model	12
3	Audited representation atlas	12
3.1	Real-space and probability representations of ρ	12
3.2	Product, zero, gamma, and exponential representations of Φ	13
3.3	Moment, cumulant, and Bell-polynomial representations	14
3.4	Transform representations of F and F^{-1}	14
3.5	A compact atlas table	14
4	The Cauchy–Stieltjes transform	15
5	A dyadic functional equation for the resolvent	16
5.1	Logarithmic fixed-point identity	16
5.2	Functional-differential equation	17
5.3	All-order Thue–Morse derivative tree	17
5.4	Uniqueness in the Stieltjes class	18
6	Finite logarithmic resolvents and the spline boundary jump	19
7	Exact transforms of F, one-sided up, and F^{-1}	20

7.1	The Laplace and Fourier transforms of F	20
7.2	One-sided transforms of the up-function	21
7.3	The inverse Fabius function as a transport map	21
8	Logarithmic derivatives of the Fourier image	22
8.1	Three meromorphic expansions	22
8.2	Digamma and Hurwitz-zeta hierarchy	22
9	A Fredholm determinant for the Fourier product	23
10	Jacobi continued fractions and exact rational coefficients	24
10.1	Exact moment recurrence	24
10.2	Monic orthogonal polynomials	24
10.3	The Jacobi and Stieltjes fractions	24
10.4	Hankel determinants	25
10.5	Exact coefficients and numerical evidence	26
10.6	The limiting coefficient	27
10.7	Padé and Gaussian-quadrature representations	28
11	Further combined representations	29
11.1	The Cauchy transform of the Fabius measure	29
11.2	A determinant form of the Fabius Laplace transform	29
11.3	A resolvent–product–continued-fraction triangle	29
12	Numerical experiments and reproducibility	30
12.1	Exact arithmetic	30
12.2	Numerical verification of the resolvent equation	30
12.3	Numerical comparison of logarithmic derivatives	30
13	Conjectures and frontier problems	30
14	Promising research program	32
14.1	Analytic program	32
14.2	Orthogonal-polynomial program	32
14.3	Operator and special-function program	32
14.4	Inverse-function program	33
15	Conclusion	33
A	Reproducibility instructions	33
B	Corpus audit ledger	34

C Selected exact data 36

II Representation Atlas and New Analytic Bridges for the Fabius and Rvachev Functions 38

1 Scope, source audit, and status discipline 39

- 1.1 Repository boundary 39
- 1.2 Collision quarantine 39
- 1.3 Status legend and limits of the novelty statement 39

2 Normalization and the master probability diagram 40

- 2.1 The compact law and its CDF 40
- 2.2 Fixed-point integral equations 40
- 2.3 The signed global extension and derivative tower 41

3 Infinite convolutions, finite splines, and head–tail representations 41

- 3.1 Box convolution 41
- 3.2 General truncated-power spline 42
- 3.3 Hypercube and section representations 42

4 Fourier-product atlas and spectral representations 42

- 4.1 Conventions and the three canonical products 42
- 4.2 Logarithmic Taylor series and partial fractions 43
- 4.3 Linear and quadratic heat traces 43
- 4.4 Frullani–Lambert integral for the Fourier product 44
- 4.5 Fredholm determinant and spectral zeta 44

5 Fourier inversion, sampling, and interval series 44

- 5.1 Whole-line inversion 44
- 5.2 Poisson partition and half-integer spectrum 45
- 5.3 Paley–Wiener sampling of the Fourier image 45
- 5.4 Fourier–Legendre representation 45

6 Laplace, Mellin, moment, and q -series representations 45

- 6.1 Moment-generating and Laplace products 45
- 6.2 Mellin moments, including negative orders 46
- 6.3 Cumulants, Bernoulli numbers, and Bell polynomials 46
- 6.4 Wick rotation, Newton recurrence, and q -MZV formula 46
- 6.5 Appell and logistic representations already in the corpus 47

7 Thue–Morse, dyadic arithmetic, and finite product-to-sum bridges 47

8 Endpoint Laplace phases, Lambert W, and inverse asymptotics	48
New theorem packages	49
9 The dyadic gamma factorization	49
9.1 Definition, convergence, and poles	49
9.2 Reflection factorization of the Rvachev transform	49
9.3 Canonical reciprocal product and power series	50
9.4 Dyadic digamma and Stieltjes resolvent	50
9.5 Malmsten and heat-trace integrals	50
9.6 Geometric- q extension	51
10 Gaussian variance mixtures and the logistic dual	51
10.1 The existing Thorin variable	51
10.2 Equality in law	51
10.3 Convolution semigroup, density, and moments	52
10.4 Closed Levy density and self-decomposability	53
10.5 Complete monotonicity in the squared coordinate	53
11 Jensen means, factorial products, and binary digit sums	54
11.1 Exact circular mean	54
11.2 Digit-sum derivative	54
11.3 Asymptotic growth	54
11.4 Subunit circular energy	55
12 Non-holonomicity from the arithmetic zero divisor	55
13 Fourier representations of the bounded Fabius function	56
13.1 The density transform	56
13.2 The global CDF as a tempered distribution	56
13.3 Sine integral and odd-harmonic series	57
13.4 Gamma versions	57
14 Quantile and inverse-Fabius transform representations	57
14.1 Quantile pushforward and product transforms	57
14.2 A nonlinear Fourier transform of the inverse	58
14.3 Sine series of the inverse in terms of F	58
15 Compact representation atlas	59
16 Numerical experiments and reproducibility	60
16.1 Dyadic gamma factorization	61

16.2	Jensen mean	61
16.3	Logistic series versus Gaussian mixture	62
16.4	Fabius sine series	62
16.5	Reproduction command	63
17	Conjectures and testable frontier statements	63
18	Promising research directions	64
18.1	Formalization roadmap	64
18.2	Spectral geometry and complex analysis	65
18.3	Probability and semigroups	65
18.4	Base- b and geometric- q families	65
18.5	Inverse spectral analysis	65
18.6	Computer-assisted certification	65
19	Conclusion	65
A	Audited living document families	66
B	Numerical package contents	66
C	Selected derivations in one line	67
III	Dyadic Multiresolution and Product-Series Representations in the Fabius–Rvachev System	69
1	Audit protocol, scope, and novelty boundary	70
1.1	Pinned corpus and cumulative audit	70
1.2	Explicit exclusion map	70
1.3	Status ledger	71
2	Normalization and the function triad	71
2.1	Bounded Fabius function, density, and Rvachev function	71
2.2	Fourier convention	72
2.3	Probabilistic model	72
3	A representation atlas from the existing theory	72
3.1	Infinite convolution, cube integrals, and finite heads	73
3.2	Sinc, hyperbolic-sinc, and renormalization products	73
3.3	Moments, cumulants, Bell polynomials, and Bernoulli numbers	74
3.4	Fourier inversion, twisted Poisson summation, and cosine series	75
3.5	Fourier images of p , F , and $1 - F$	75

3.6	Quantile, Stieltjes, and correlation representations	76
3.7	Legendre and Chebyshev coefficient integrals	76
4	The projective dyadic mass array	77
4.1	Definition and exact algebra	77
4.2	Fixed-cell uniform mixtures	77
4.3	Rational exponential-polynomial limits for the transforms	78
4.4	A second-order quadrature principle	79
4.5	Moments, L^q norms, and discrete entropies	80
5	Haar and Faber–Schauder representations	80
5.1	Rational unnormalized Haar increments	80
5.2	Fourier image of the Haar expansion	81
5.3	Faber–Schauder series of the Fabius function	81
5.4	A sinc-squared series for the Fourier image of F	82
6	Walsh–Hadamard and Thue–Morse product representations	83
6.1	Paley Walsh functions and the finite Hadamard transform	83
6.2	Reflection selection rules	84
6.3	Finite-product Fourier transform of a Walsh mode	84
6.4	A series of finite products for the infinite sinc product	85
6.5	Exact Hamming-weight cancellation	85
6.6	The Gaussian-binomial cancellation spectrum	86
6.7	All higher Walsh moments: a Bell–Bernoulli formula	86
7	Bernstein polynomials and positive beta-mixture limits	88
7.1	Rational Bernstein approximants	88
7.2	Derivative as a beta-density mixture	88
7.3	Hypergeometric transform representations	89
8	Dual quantile cells and equal-weight sinc mixtures	89
8.1	The inverse dyadic width array	89
8.2	Exact chord integration and the Fourier image	90
8.3	Moment and Laplace limits from inverse dyadic values	91
8.4	Dual L^q spectra and entropy constants	91
9	Haar scale energies and the sinc^4 continuum expansion	92
9.1	Three exact forms of the energy	92
9.2	Local Taylor expansion of a Haar coefficient	92
9.3	A frequency multiplier and a superalgebraic alias remainder	93
9.4	All orders and Bell–Bernoulli coefficients	93

9.5	Derivative norms from half-integer samples of the sinc product	94
9.6	A meromorphic generating function for the rational energies	94
9.7	Beyond all orders: a phase-locked alias frontier	95
10	Multiresolution of the inverse Fabius function	95
10.1	Faber–Schauder series and reflection	95
10.2	Fourier and Walsh images of the inverse	96
10.3	Boundary coefficients and slow variation	96
10.4	Boundary domination of inverse interpolation	97
11	Comparison of the representation families	97
11.1	A commuting diagram of finite transforms	98
11.2	Computational observations	98
12	Further deductions and frontier directions	99
12.1	Exact arithmetic of multiresolution coefficients	99
12.2	Mass-polynomial renormalization	99
12.3	Hamming layers and Walsh pressure	99
12.4	Higher-order rational multiwavelets	100
12.5	Beta mixtures and confluent dyadic extrapolation	100
12.6	Quantile-width extremes and Lambert phases	100
12.7	Base- b and nonuniform scale systems	101
12.8	Formalization route	101
13	Numerical and exact-symbolic verification	101
13.1	Exact checks	102
13.2	Transform convergence	102
13.3	Haar energy asymptotics	103
13.4	Reproducibility files	103
14	Conclusions	104
A	Compact formula index	105
B	Proof details for two auxiliary identities	106
B.1	Derivation of the local Haar expansion	106
B.2	Absolute convergence of the first-moment Hamming layers	106
C	Reproducibility ledger	106

Provenance

This volume preserves the mathematical bodies of the following drafts (labels, citation keys, and asset paths mechanically prefixed per part; top-level counters, reproduction paths, and raster figure selection adapted to the consolidated layout). No mathematical absorbed mathematical content is altered; later formalization notes are explicitly marked. The absorbed drafts are deleted from the working tree; git history is the archive.

- Fabius_Rvachev_Representation_Frontiers/rvachev_fabius_representation_frontiers.tex
SHA-256 8fdb498ec33cde2db9e7dc82c0c928946f632beb8a0ffd216afe310eaea0ea2
- fabius_rvachev_representation_frontier/fabius_rvachev_representation_frontier_report.tex
SHA-256 dfae837099df1578665a7bb4dabd9417cd160a2165987785062097e5046177ec
- Fabius_Rvachev_Multiresolution_Report/fabius_rvachev_multiresolution_report.tex
SHA-256 54b4817fae50e2594a42d6f5ee06152b2c3c17a1e653157baf6ddd15002333ac

Part I

Resolvent, Continued-Fraction, and Transform Representations of the Fabius–Rvachev System

Abstract

This report audits the LaTeX corpus in the repository subtree `Analysis/FabiusFunction/docs`, including its living expositions, consolidated frontier volume, active draft families, archive lineages, and paper transcriptions [5]. The repository already contains an unusually broad theory of the Fabius function, Rvachev’s up-function, their dyadic values, Thue–Morse expansions, Fourier products, moment and cumulant algebra, inverse asymptotics, and spectral decay. The objective here is therefore not to reproduce that material, but to organize it as a representation atlas and then add constructions that are absent from the audited corpus.

The principal new object is the Cauchy–Stieltjes transform

$$R(z) = \int_{-1}^1 \frac{\text{up}(x)}{z - x} dx.$$

We derive its hyperbolic-product Laplace representation, its one-sided Fourier–Laplace representation, an exact logarithmic fixed-point identity, the functional-differential law

$$R'(z) = 2(R(2z + 1) - R(2z - 1)),$$

and an all-orders Thue–Morse derivative tree. Finite dyadic convolution laws yield an explicit 2^N -term logarithmic formula for R_N , whose boundary jump is exactly the finite Thue–Morse spline for the density. This gives a new “logarithm-to-spline” bridge.

We also construct the Jacobi continued fraction of R , equivalently the Stieltjes continued fraction of the ordinary even-moment generating function. Its coefficients are positive rational numbers; the first forty are computed exactly. The first coefficients are

$$\beta_1 = \frac{1}{9}, \quad \beta_2 = \frac{32}{225}, \quad \beta_3 = \frac{619}{3675}, \quad \beta_4 = \frac{835232}{4640643}.$$

The Denisov–Rakhmanov theorem proves $\beta_n \rightarrow 1/4$, while exact computation through $n = 40$ supports strict monotonicity and a logarithmically corrected convergence law. Further new representations include a trace-class Fredholm determinant for the Fourier image, three equivalent meromorphic logarithmic-derivative expansions, a digamma/Hurwitz- zeta hierarchy, exact one-sided Laplace and Fourier transforms of up and F , and pushforward transform identities for the inverse Fabius function. A commented Python program, exact CSV data, numerical checks, and figures accompany the report.

1 Executive summary and novelty boundary

1.1 What was audited

The repository contains several kinds of LaTeX source that must not be conflated:

1. a living primary synthesis of the formalized and analytic theory;
2. a large consolidated non-formalized frontier volume;
3. current draft reports on arithmetic rays, half-integer spectra, self-sampling, inverse asymptotics, Newton/exponent-sequence hierarchies, Fourier decay, and related topics;
4. archived earlier lineages, retained because some contain historically distinctive formulas even when superseded;
5. seven paper-source transcriptions or corrected transcriptions.

The repository’s own pinned audit ledger records 67 LaTeX paths at the following commit:

32d6d36c51d803289e6d6a0dc0c37753766eba47.

The live tree inspected on 27 August 2026 contains additional current drafts, notably the self-sampling, inverse, and Newton/Rvachev reports. Appendix B records the pinned ledger and the live additions used in this study.

1.2 Non-duplication protocol

Every formula in the atlas is tagged conceptually as follows.

Tag	meaning
K	already explicit in the audited repository or classical source; included for orientation
D	an immediate transform or recombination of known identities; useful but not claimed as new
N	apparently absent from the audited corpus; proved or conjectured here

The word “new” throughout this report means *new relative to the audited repository*. No claim of global bibliographic priority is made without a separate literature search. In particular, the current frontier report explicitly proposes a Jacobi continued fraction for the moment generating function or Cauchy transform as an open direction. The present report constructs that fraction and develops its consequences.

1.3 Main new results

The most substantial additions are:

1. the Cauchy transform R and its moment, Laplace-product, Fourier–Laplace, boundary-value, and logarithmic fixed-point representations;
2. the exact resolvent equation and its all-orders Thue–Morse derivative tree;
3. a finite-level 2^N -term logarithmic formula converging to R , paired by boundary jumps with the finite Thue–Morse spline density;

4. the Jacobi J -fraction of R , the contracted Stieltjes S -fraction of the even moments, exact rational coefficients through order 40, Gaussian quadrature convergents, and the theorem $\beta_n \rightarrow 1/4$;
5. a Fredholm determinant realization of the Fourier image and a trace-resolvent interpretation of its logarithmic derivative;
6. cotangent, tangent, valuation-weighted partial-fraction, digamma, and Hurwitz-zeta representations of the logarithmic derivatives of the Fourier image;
7. exact one-sided Laplace and Fourier transforms of up , exact transforms of F on $[0, 1]$, and a master pushforward identity for F^{-1} .

2 Normalization and structural dictionary

2.1 The functions

Let $F : [0, 1] \rightarrow [0, 1]$ denote the bounded Fabius function [4, 2], extended by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$. On the unit interval it satisfies

$$F(0) = 0, \quad F(1) = 1, \quad F(1 - x) = 1 - F(x), \quad (1)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}). \quad (2)$$

The Rvachev function is written

$$\rho(x) := \text{up}(x) = \begin{cases} F(x+1), & -1 \leq x \leq 0, \\ F(1-x), & 0 \leq x \leq 1, \\ 0, & |x| \geq 1. \end{cases} \quad (3)$$

Then ρ is even, strictly positive on $(-1, 1)$, normalized by $\int \rho = 1$, flat at ± 1 , and satisfies

$$\rho'(x) = 2(\rho(2x+1) - \rho(2x-1)). \quad (4)$$

2.2 Fourier convention and basic transforms

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} \, dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}(\xi) e^{2\pi i \xi x} \, d\xi. \quad (5)$$

Set

$$\Phi(\xi) := \widehat{\rho}(\xi), \quad \text{sinc}_{\pi}(z) := \frac{\sin(\pi z)}{\pi z}, \quad \text{sinc}_{\pi}(0) = 1. \quad (6)$$

The classical products are

$$\Phi(z) = \prod_{n=0}^{\infty} \text{sinc}_{\pi}\left(\frac{z}{2^n}\right) = \prod_{m=1}^{\infty} \cos\left(\frac{\pi z}{2^m}\right)^m. \quad (7)$$

The scaling law is

$$\Phi(z) = \text{sinc}_{\pi}(z) \Phi(z/2). \quad (8)$$

2.3 The probability model

Let $U_1, U_2, \dots$ be independent random variables, uniform on $[-1, 1]$, and set

$$X = \sum_{k=1}^{\infty} 2^{-k} U_k. \quad (9)$$

Then X has density ρ . Equivalently,

$$X \stackrel{d}{=} \frac{U + X'}{2}, \quad (10)$$

where $U \sim \text{Unif}[-1, 1]$ and X' is an independent copy of X . The moment generating function is

$$M(t) := \mathbb{E} e^{tX} = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} = \Phi\left(\frac{it}{2\pi}\right). \quad (11)$$

The even moments are

$$\mu_{2n} := \mathbb{E} X^{2n} = \int_{-1}^1 x^{2n} \rho(x) \, dx, \quad \mu_{2n+1} = 0. \quad (12)$$

Finally, if

$$Y = \frac{X + 1}{2}, \quad (13)$$

then F is the distribution function of Y , and

$$F'(x) = 2\rho(2x - 1), \quad 0 < x < 1. \quad (14)$$

3 Audited representation atlas

This section compresses the principal representation families already present in the repository and places the new formulas beside them. Proofs of the new entries begin in Section 4.

3.1 Real-space and probability representations of ρ

Infinite convolution (K). Since $2^{-k}U_k$ is uniform on $[-2^{-k}, 2^{-k}]$,

$$\rho = \bigstar_{k=1}^{\infty} \left(2^{k-1} \mathbf{1}_{[-2^{-k}, 2^{-k}]} \right). \quad (15)$$

This is the real-space counterpart of the sinc product.

Infinite cube integral (D). For every bounded continuous test function g ,

$$\int_{-1}^1 g(x) \rho(x) \, dx = \int_{[-1, 1]^{\mathbb{N}}} g\left(\sum_{k \geq 1} 2^{-k} u_k\right) \prod_{k \geq 1} \frac{du_k}{2}. \quad (16)$$

Formally, this may be written as a Dirac-delta integral for $\rho(x)$.

Finite Thue–Morse splines (K). Put

$$X_N = \sum_{k=1}^N 2^{-k} U_k, \quad a_{N,j} = \frac{2^N - 1 - 2j}{2^N}, \quad \varepsilon_j = (-1)^{\text{wt}_2(j)}. \quad (17)$$

The density ρ_N of X_N is

$$\rho_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j (x + a_{N,j})_+^{N-1}. \quad (18)$$

For each fixed derivative order, $\rho_N \rightarrow \rho$ uniformly after N is sufficiently large relative to that order. The repository develops several centered and one-sided variants with explicit error control.

Fourier inversion and cosine form (K). Because Φ is real and even,

$$\rho(x) = \int_{\mathbb{R}} \Phi(\xi) e^{2\pi i x \xi} d\xi, \quad (19)$$

$$\rho(x) = 2 \int_0^\infty \Phi(\xi) \cos(2\pi x \xi) d\xi. \quad (20)$$

The audited corpus also contains Poisson-summation formulas, Legendre expansions, periodized cosine series, dyadic spline limits, and endpoint transseries. They are not rederived here.

3.2 Product, zero, gamma, and exponential representations of Φ

Canonical zero product (K). Let $\nu_2(n)$ be the exponent of 2 in n . The nonzero integer n is a zero of multiplicity $1 + \nu_2(|n|)$, and

$$\Phi(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)}. \quad (21)$$

Gamma product (D). Euler’s reflection identity gives

$$\Phi(z) = \prod_{k=0}^{\infty} \frac{1}{\Gamma(1 + z/2^k) \Gamma(1 - z/2^k)}. \quad (22)$$

This is the natural entry point for the Barnes and multiple-gamma compressions studied in current repository drafts.

Spectral-zeta exponential (K). For $|z| < 1$,

$$\log \Phi(z) = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m(1 - 2^{-2m})} z^{2m}. \quad (23)$$

The coefficient sequence is the spectral zeta function

$$Z_\Phi(s) = \sum_{n=1}^{\infty} \frac{1 + \nu_2(n)}{n^s} = \frac{\zeta(s)}{1 - 2^{-s}}. \quad (24)$$

Fredholm determinant (N). Section 9 upgrades (21) to

$$\Phi(z) = \det(I - z^2 \mathcal{K}), \quad (25)$$

for an explicit positive trace-class diagonal operator $\mathcal{K}$. This makes (24) literally the trace zeta function of $\mathcal{K}$.

3.3 Moment, cumulant, and Bell-polynomial representations

The moment generating function has the convergent Taylor series

$$M(t) = \sum_{n=0}^{\infty} \mu_{2n} \frac{t^{2n}}{(2n)!}. \quad (26)$$

Its logarithm is

$$\log M(t) = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!(4^m - 1)} t^{2m}, \quad (27)$$

so all moments are complete Bell polynomials in explicit Bernoulli/Mersenne cumulants. These formulas, together with reciprocal Bell expansions for $1/\Phi$, are developed extensively in the repository.

3.4 Transform representations of F and F^{-1}

The CDF inversion formula gives

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \Phi(\xi) \frac{\sin(2\pi(2x-1)\xi)}{\xi} d\xi, \quad 0 < x < 1. \quad (28)$$

Section 7 adds closed Laplace and Fourier transforms of F as a function, as well as transforms of its derivative measure and its inverse.

3.5 A compact atlas table

Table 1: Representation atlas. Tags refer to the audited-corpus novelty boundary.

Object	Representation family	Tag	Location
ρ	fold of F , compact-support DDE, infinite convolution, random dyadic sum	K	§2–3
ρ	finite Thue–Morse truncated-power splines	K	(18)
ρ	Fourier inversion, cosine integral, Poisson and Legendre series	K	(19)
ρ	boundary jump of a finite logarithmic resolvent sum	N	Thm. 6.1
Φ	sinc product and weighted cosine product	K	(7)
Φ	canonical integer-zero product	K	(21)
Φ	gamma/Barnes products	K/D	(22)
Φ	zeta-exponential series	K	(23)
Φ	trace-class Fredholm determinant	N	Thm. 9.2
Φ'/Φ	cotangent dyadic sum	N	(63)
Φ'/Φ	weighted tangent dyadic sum	N	(64)
Φ'/Φ	valuation-weighted partial fractions	N	(65)
$(\log \Phi)^{(r)}$	digamma and Hurwitz-zeta hierarchy	N	Thm. 8.2
M	hyperbolic-sinc product, moment series, Bell cumulants	K	(11)–(27)
M	positive Fredholm determinant	N	(74)

Object	Representation family	Tag	Location
R	moment Laurent series	N	(31)
R	Laplace transform of the hyperbolic product	N	(32)
R	one-sided Fourier–Laplace integral	N	(33)
R	logarithmic fixed-point expectation	N	(36)
R	dyadic functional-differential equation	N	(37)
$R^{(n)}$	finite Thue–Morse affine orbit of R	N	(38)
R	finite 2^N -logarithm limit	N	(44)
R	Jacobi continued fraction	N	(85)
R	Gaussian-quadrature partial fractions	N	(94)
$\sum \mu_{2n} t^n$	Stieltjes continued fraction and product integral	N	(87)
F	CDF sine-product integral	D	(28)
F	exact Laplace transform on $[0, 1]$	N	(48)
F	exact Fourier transform on $[0, 1]$	N	(49)
$\rho _{[0,1]}$	exact one-sided Laplace/Fourier/sine transforms	N	Thm. 7.2
F^{-1}	pushforward expectation, Laplace, Fourier, and Cauchy transforms	N	Thm. 7.3

4 The Cauchy–Stieltjes transform

Definition 4.1. For $z \in \mathbb{C} \setminus [-1, 1]$, define

$$R(z) = \int_{-1}^1 \frac{\rho(x)}{z - x} dx = \mathbb{E} \frac{1}{z - X}. \quad (29)$$

Remark 4.2 (Formal Cauchy–Stieltjes core). `CauchyTransform.lean` defines the report-oriented transform as `rvachevCauchyTransform`; `Mathlib`’s `resolventTransform` uses $(x - z)^{-1}$, so the formal definition is its negative. The declarations `rvachevCauchyTransform_apply` and `rvachevCauchyTransform_eq_integral_rvachevUp` give respectively the measure and density forms of (29), while `analyticOn_rvachevCauchyTransform` and `hasDerivAt_rvachevCauchyTransform` prove holomorphy off the complexified support interval and the measure-form first derivative

$$R'(z) = - \int (z - x)^{-2} d\mu_{\text{up}}(x).$$

Expanding $\mu_{\text{up}} = \rho(x) dx$ recovers the corresponding density formula. The total Lean definition follows the Bochner-integral convention at points where the kernel is not integrable; the analytic Cauchy-transform interpretation is asserted only off the cut.

The transform R is odd because ρ is even. It is an anti-Herglotz function: when $\text{Im } z > 0$,

$$\text{Im } R(z) = - \text{Im } z \int_{-1}^1 \frac{\rho(x)}{|z - x|^2} dx < 0. \quad (30)$$

In particular, R has no zeros off the cut. For real $z > 1$, $R(z) > 0$; for $z < -1$, $R(z) < 0$.

Theorem 4.3 (Four complementary representations of R). *The following formulas hold.*

(i) For $|z| > 1$,

$$R(z) = \sum_{n=0}^{\infty} \frac{\mu_{2n}}{z^{2n+1}}. \quad (31)$$

(ii) For $\text{Re } z > 1$,

$$R(z) = \int_0^{\infty} e^{-zt} \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} dt. \quad (32)$$

(iii) For $\operatorname{Im} z > 0$,

$$R(z) = -2\pi i \int_0^\infty e^{2\pi i z \xi} \Phi(\xi) \, d\xi. \quad (33)$$

For $\operatorname{Im} z < 0$, take the conjugate counterpart.

(iv) For $-1 < x < 1$, the non-tangential boundary values satisfy

$$\rho(x) = -\frac{1}{\pi} \lim_{y \downarrow 0} \operatorname{Im} R(x + iy), \quad (34)$$

$$\operatorname{PV} \int_{-1}^1 \frac{\rho(t)}{x - t} \, dt = \lim_{y \downarrow 0} \operatorname{Re} R(x + iy) = 2\pi \int_0^\infty \Phi(\xi) \sin(2\pi x \xi) \, d\xi. \quad (35)$$

Proof. For $|z| > 1$, expand $(z - X)^{-1} = z^{-1} \sum_{m \geq 0} (X/z)^m$, use $|X| \leq 1$, and then symmetry. This gives (31).

For $\operatorname{Re} z > 1$,

$$\frac{1}{z - X} = \int_0^\infty e^{-(z-X)t} \, dt.$$

Fubini's theorem is justified because $X \leq 1$. Taking expectation and using (11) yields (32).

For $\operatorname{Im} z > 0$,

$$\frac{1}{z - x} = -2\pi i \int_0^\infty e^{2\pi i (z-x)\xi} \, d\xi.$$

The exponential damping from $\operatorname{Im} z > 0$, together with the rapid decay of Φ , permits Fubini and gives (33). Finally, Plemelj's boundary formula gives (34); taking real parts of (33) gives (35). $\square$

Remark 4.4. The four formulas emphasize different structures. The Laurent series exposes exact rational moments; the Laplace integral exposes the hyperbolic product; the Fourier–Laplace integral exposes spectral decay; and the boundary values recover both ρ and its Hilbert transform. Thus R is a single analytic object joining the moment, product, and inversion theories.

5 A dyadic functional equation for the resolvent

5.1 Logarithmic fixed-point identity

Theorem 5.1 (Logarithmic fixed point). *For real $z > 1$,*

$$R(z) = \int_{-1}^1 \rho(y) \log \left(\frac{2z + 1 - y}{2z - 1 - y} \right) \, dy. \quad (36)$$

The identity extends by analytic continuation to any simply connected subdomain of $\mathbb{C} \setminus [-1, 1]$, with the logarithm chosen by continuation from large positive z .

Proof. Use the fixed point (10) and condition on X' :

$$\begin{aligned} R(z) &= \mathbb{E} \left[\int_{-1/2}^{1/2} \frac{du}{z - u - X'/2} \right] \\ &= \mathbb{E} \log \left(\frac{z + 1/2 - X'/2}{z - 1/2 - X'/2} \right), \end{aligned}$$

which is (36) after multiplying numerator and denominator by 2. $\square$

5.2 Functional-differential equation

Theorem 5.2 (Resolvent equation). *For every $z \in \mathbb{C} \setminus [-1, 1]$,*

$$\boxed{R'(z) = 2(R(2z+1) - R(2z-1))}. \quad (37)$$

Proof. Differentiate (36) first for $z > 1$:

$$\begin{aligned} R'(z) &= \mathbb{E} \left(\frac{2}{2z+1-X'} - \frac{2}{2z-1-X'} \right) \\ &= 2(R(2z+1) - R(2z-1)). \end{aligned}$$

Both sides are holomorphic on $\mathbb{C} \setminus [-1, 1]$, so the identity theorem gives the stated continuation. $\square$

This is the transform-side analogue of the Rvachev equation (4). It is not merely a Fourier rephrasing: it acts on the spectral parameter of a Cauchy transform and therefore interacts directly with moments, Padé approximation, and orthogonal polynomials.

5.3 All-order Thue–Morse derivative tree

Theorem 5.3 (Thue–Morse derivative orbit). *For every integer $n \geq 0$,*

$$\boxed{R^{(n)}(z) = 2^{n(n+1)/2} \sum_{j=0}^{2^n-1} \varepsilon_j R(2^n z + 2^n - 1 - 2j)}. \quad (38)$$

Consequently,

$$\int_{-1}^1 \frac{\rho(x)}{(z-x)^{n+1}} dx = \frac{(-1)^n 2^{n(n+1)/2}}{n!} \sum_{j=0}^{2^n-1} \varepsilon_j R(2^n z + 2^n - 1 - 2j). \quad (39)$$

Proof. The case $n = 0$ is trivial and $n = 1$ is (37). Assume the formula at order n . Differentiating contributes a factor 2^n , and then applying (37) contributes another factor 2. The exponent therefore changes by $n + 1$, from $n(n+1)/2$ to $(n+1)(n+2)/2$.

For the shifts, write $a_{n,j} = 2^n - 1 - 2j$. The two children are

$$2a_{n,j} + 1 = 2^{n+1} - 1 - 2(2j), \quad 2a_{n,j} - 1 = 2^{n+1} - 1 - 2(2j+1).$$

The signs obey $\varepsilon_{2j} = \varepsilon_j$ and $\varepsilon_{2j+1} = -\varepsilon_j$, exactly matching the subtraction in (37). This proves (38). Formula (39) follows from

$$R^{(n)}(z) = (-1)^n n! \int_{-1}^1 \frac{\rho(x)}{(z-x)^{n+1}} dx.$$

$\square$

Proposition 5.4 (Generic affine-difference orbit; formalized). *Over a commutative ring, let*

$$(\Delta_{b,c} f)(z) = f(bz+c) - f(bz-c), \quad \xi_{n,T}(z) = b^n z + c \sum_{k=0}^{n-1} b^k - 2c \sum_{k \in T} b^k.$$

For every function with values in an additive commutative group,

$$(\Delta_{b,c}^n f)(z) = \sum_{T \subseteq \{0, \dots, n-1\}} (-1)^{|T|} f(\xi_{n,T}(z)). \quad (40)$$

Moreover, suppose the ring is a nontrivially normed field, let S be open and invariant under $z \mapsto bz \pm c$, and let f take values in a normed vector space. If $f'(z) = a \Delta_{b,c} f(z)$ for $z \in S$, then no separate smoothness assumption is needed: for every $n \geq 0$ and $z \in S$,

$$f^{(n)}(z) = a^n b^{\binom{n}{2}} (\Delta_{b,c}^n f)(z). \quad (41)$$

Constant scalar multiplication commutes with every iterate. Taking S to be the whole field gives the global form, while $b = 2, c = 1$ gives

$$(\Delta_{2,1}^n f)(z) = \sum_{j=0}^{2^n-1} \varepsilon_j f(2^n z + 2^n - 1 - 2j). \quad (42)$$

Lean proof and boundary. The following declarations in `AffineDifferenceOrbit.lean` prove the statements above:

```
affineDifference_iterate_const_smul
affineDifference_iterate_apply
iteratedDeriv_eq_affineDifference_iterate_on
iteratedDeriv_eq_affineDifference_iterate
affineDifference_iterate_two_one_apply.
```

The first formula is a Boolean-cube induction. The second differentiates that finite operator orbit on the invariant open set; its one-step derivative hypothesis bootstraps all regularity needed. The last formula reindexes subsets by binary support, so its signs are precisely the Thue–Morse signs.

Thus the conditional passage from (37) to (38) is formalized in greater generality. On the density side, `iteratedDeriv_rvachevUp_eq_thueMorse_sum` applies the same engine to the already formalized differential equation for ρ , globally on the real line. The transform definition, density formula, natural-domain holomorphy, first Cauchy-kernel derivative, and affine identity $S(z) = 2R(2z - 1)$ are now formalized in `CauchyTransform.lean`. That module does not prove (36) or (37); the application of the generic orbit theorem to the complex resolvent R , and the kernel identity (39), remain formalization obligations. $\square$

Taking boundary imaginary parts in (38) recovers the familiar iterated derivative formula for ρ . The new point is that the same finite Thue–Morse orbit controls every higher Cauchy kernel, not only the density derivatives.

5.4 Uniqueness in the Stieltjes class

Proposition 5.5. *Among Cauchy transforms of probability measures supported on $[-1, 1]$, R is the unique solution of (37) satisfying*

$$R(z) = \frac{1}{z} + \mathcal{O}(z^{-3}) \quad (z \rightarrow \infty). \quad (43)$$

Proof. Let $G(z) = \sum_{r \geq 0} m_r z^{-r-1}$ be the Laurent expansion at infinity of any such Cauchy transform. The normalization gives $m_0 = 1$ and $m_1 = 0$. Substitute this expansion into (37). In the coefficient of z^{-r-2} , the contribution containing m_r is

$$-(r+1)m_r + \frac{r+1}{2^r} m_r = -(r+1)(1-2^{-r})m_r,$$

while every other contribution contains only $m_0, \dots, m_{r-2}$. Since the displayed factor is nonzero for $r \geq 1$, the differential equation determines all moments recursively. The transform R is one solution, so G and R have identical moments. Probability measures on the compact interval $[-1, 1]$ are determinate by their moments (equivalently, polynomials are dense in $C[-1, 1]$); hence the two measures and their Cauchy transforms coincide. $\square$

6 Finite logarithmic resolvents and the spline boundary jump

The finite random sum X_N has two exact representations: its density is a Thue–Morse spline, while its Cauchy transform is a Thue–Morse sum of logarithms. The second formula appears to be absent from the audited corpus.

Theorem 6.1 (Finite logarithmic formula). *Let*

$$R_N(z) = \mathbb{E} \frac{1}{z - X_N}, \quad X_N = \sum_{k=1}^N 2^{-k} U_k.$$

For $z \notin [-1 + 2^{-N}, 1 - 2^{-N}]$, with logarithms continued consistently from $z = +\infty$,

$$R_N(z) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j (z + a_{N,j})^{N-1} \text{Log}(z + a_{N,j}). \quad (44)$$

Moreover, $R_N \rightarrow R$ locally uniformly on $\mathbb{C} \setminus [-1, 1]$, so

$$R(z) = \lim_{N \rightarrow \infty} \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j (z + a_{N,j})^{N-1} \text{Log}(z + a_{N,j}). \quad (45)$$

Finally, the upper boundary jump of (44) is exactly the spline (18):

$$\rho_N(x) = -\frac{1}{\pi} \text{Im } R_N(x + i0). \quad (46)$$

Proof. For $a > 0$, convolution with the uniform density on $[-a, a]$ sends a Cauchy transform $G'(z)$ to

$$\frac{1}{2a} \int_{-a}^a G'(z - u) \, du = \frac{G(z + a) - G(z - a)}{2a}. \quad (47)$$

Start from the point mass at zero, whose Cauchy transform is $1/z$. An N -fold antiderivative is

$$H_N(z) = \frac{z^{N-1} \text{Log } z}{(N-1)!}; \quad H_N^{(N)}(z) = \frac{1}{z}.$$

Applying (47) successively with $a_k = 2^{-k}$ gives

$$R_N(z) = \frac{1}{\prod_{k=1}^N 2a_k} \sum_{\sigma \in \{\pm 1\}^N} \left(\prod_{k=1}^N \sigma_k \right) H_N \left(z + \sum_{k=1}^N \sigma_k a_k \right).$$

Since

$$\frac{1}{\prod_{k=1}^N 2a_k} = 2^{N(N-1)/2},$$

and binary subsets identify the shifts and signs with $a_{N,j}$ and ε_j , this is (44).

The residual tail $X - X_N$ has support in $[-2^{-N}, 2^{-N}]$, so $X_N \rightarrow X$ almost surely and uniformly in support. The kernels $(z - x)^{-1}$ are uniformly continuous on compact subsets away from $[-1, 1]$; hence $R_N \rightarrow R$ locally uniformly.

For the boundary value, $\text{Im Log}(x + a + i0) = \pi$ when $x + a < 0$ and zero when $x + a > 0$. The alternating finite difference annihilates every polynomial of degree less than N , so the sum of the full powers $(x + a_{N,j})^{N-1}$ is zero. Replacing the negative-part sum by minus the positive-part sum yields exactly (18). This proves (46). $\square$

Remark 6.2 (Two complementary approximation geometries). The finite-convolution approximant R_N is a sum of logarithms and preserves the dyadic construction. The Jacobi approximants in Section 10 are rational functions and preserve moments. Both converge locally uniformly off the support, but their poles, branch cuts, and error mechanisms are completely different. This duality suggests hybrid algorithms: use finite convolution to resolve endpoint geometry and a short Jacobi fraction to compress the remaining moment error.

7 Exact transforms of F , one-sided up, and F^{-1}

7.1 The Laplace and Fourier transforms of F

Theorem 7.1 (Transform of the Fabius function). *For complex s , with the value at $s = 0$ understood by continuity,*

$$\int_0^1 e^{-sx} F(x) \, dx = \frac{e^{-s/2} M(s/2) - e^{-s}}{s}. \quad (48)$$

Equivalently, for real or complex ξ ,

$$\int_0^1 F(x) e^{-2\pi i \xi x} \, dx = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}. \quad (49)$$

At $\xi = 0$, the value is $1/2$.

Proof. The distributional Laplace transform of Y is

$$\int_0^1 e^{-sx} \, dF(x) = \mathbb{E} e^{-sY} = e^{-s/2} M(s/2). \quad (50)$$

Integration by parts gives

$$\int_0^1 e^{-sx} \, dF(x) = e^{-s} + s \int_0^1 e^{-sx} F(x) \, dx,$$

which is (48). Substitute $s = 2\pi i \xi$ and use $M(\pi i \xi) = \Phi(\xi/2)$. $\square$

Separating real and imaginary parts gives the explicit sine and cosine transforms

$$\int_0^1 F(x) \cos(2\pi \xi x) \, dx = \frac{\sin(2\pi \xi) - \Phi(\xi/2) \sin(\pi \xi)}{2\pi \xi}, \quad (51)$$

$$\int_0^1 F(x) \sin(2\pi \xi x) \, dx = \frac{\Phi(\xi/2) \cos(\pi \xi) - \cos(2\pi \xi)}{2\pi \xi}. \quad (52)$$

7.2 One-sided transforms of the up-function

On $[0, 1]$, $\rho(x) = F(1 - x) = 1 - F(x)$. Therefore the previous theorem immediately yields the following closed forms.

Theorem 7.2 (One-sided transform package). *For complex s , with the removable value at zero,*

$$\int_0^1 e^{-sx} \rho(x) \, dx = \frac{1 - e^{-s/2} M(s/2)}{s}. \quad (53)$$

Consequently,

$$\int_0^1 e^{-2\pi i \xi x} \rho(x) \, dx = \frac{1 - e^{-\pi i \xi} \Phi(\xi/2)}{2\pi i \xi}. \quad (54)$$

The half-line cosine and sine transforms are

$$\int_0^1 \rho(x) \cos(2\pi \xi x) \, dx = \frac{1}{2} \Phi(\xi), \quad (55)$$

$$\int_0^1 \rho(x) \sin(2\pi \xi x) \, dx = \frac{1 - \Phi(\xi/2) \cos(\pi \xi)}{2\pi \xi}. \quad (56)$$

The sum of (53) at s and $-s$ recovers $M(s)$ precisely because

$$M(s) = \frac{2 \sinh(s/2)}{s} M(s/2), \quad (57)$$

which is the hyperbolic continuation of (8).

7.3 The inverse Fabius function as a transport map

Let $I = F^{-1}$ on $[0, 1]$. Since I pushes Lebesgue measure forward to the law of Y , every integral of a composition with I reduces to the up-law.

Theorem 7.3 (Inverse pushforward master identity). *For every Borel function g for which one side is integrable,*

$$\int_0^1 g(I(u)) \, du = \int_0^1 g(x) \, dF(x) = \int_{-1}^1 g\left(\frac{x+1}{2}\right) \rho(x) \, dx. \quad (58)$$

In particular,

$$\int_0^1 e^{-sI(u)} \, du = e^{-s/2} M(s/2), \quad (59)$$

$$\int_0^1 e^{-2\pi i \xi I(u)} \, du = e^{-\pi i \xi} \Phi(\xi/2), \quad (60)$$

$$\int_0^1 \frac{du}{z - I(u)} = 2R(2z - 1), \quad z \in \mathbb{C} \setminus [0, 1], \quad (61)$$

$$\int_0^1 I(u)^n \, du = 2^{-n} \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} \mu_{2r}. \quad (62)$$

Proof. If U is uniform on $[0, 1]$, then $I(U)$ has CDF F , hence the same law as Y . This proves (58). The remaining formulas follow by taking $g(x) = e^{-sx}$, $e^{-2\pi i \xi x}$, $(z - x)^{-1}$, and x^n , respectively, and then using (13) and symmetry of X . $\square$

Remark 7.4. Formula (61) transports the entire Jacobi-fraction theory of R to the inverse Fabius function. Thus a function whose pointwise inverse asymptotics involve Lambert W nevertheless has a globally simple Cauchy transform when sampled with uniform measure in its range variable.

8 Logarithmic derivatives of the Fourier image

8.1 Three meromorphic expansions

Theorem 8.1 (Cotangent, tangent, and zero-set formulas). *Away from the integer zero set of Φ ,*

$$\frac{\Phi'(z)}{\Phi(z)} = \sum_{k=0}^{\infty} \left[\frac{\pi}{2^k} \cot\left(\frac{\pi z}{2^k}\right) - \frac{1}{z} \right], \quad (63)$$

$$= -\pi \sum_{m=1}^{\infty} \frac{m}{2^m} \tan\left(\frac{\pi z}{2^m}\right), \quad (64)$$

$$= -2z \sum_{n=1}^{\infty} \frac{1 + \nu_2(n)}{n^2 - z^2}. \quad (65)$$

All three series converge locally normally on compact subsets avoiding the poles.

Proof. Differentiate the logarithm of each of the three products (7) and (21). For large k , the bracket in (63) and the term in (64) are $\mathcal{O}(4^{-k})$ uniformly on compact sets. The terms in (65) are bounded by a constant times $(1 + \nu_2(n))/n^2$, which is summable by (24) at $s = 2$. $\square$

The three forms have different numerical regimes. The tangent sum is usually the fastest away from its poles; the cotangent sum tracks the original sinc factors; and the zero-set sum exposes multiplicities and supports rigorous rational tail bounds.

8.2 Digamma and Hurwitz-zeta hierarchy

The valuation weight has the divisor decomposition

$$1 + \nu_2(n) = \sum_{k=0}^{\infty} \mathbf{1}_{2^k | n}. \quad (66)$$

This turns the zero sum into a dyadic superposition of classical special functions.

Theorem 8.2 (Digamma and Hurwitz-zeta representations). *Let $\psi = \Gamma'/\Gamma$. Then*

$$\frac{\Phi'(z)}{\Phi(z)} = \sum_{k=0}^{\infty} 2^{-k} \left[\psi\left(1 - \frac{z}{2^k}\right) - \psi\left(1 + \frac{z}{2^k}\right) \right]. \quad (67)$$

For every integer $r \geq 2$,

$$\frac{d^r}{dz^r} \log \Phi(z) = (r-1)! \sum_{k=0}^{\infty} 2^{-kr} \left[-\zeta\left(r, 1 - \frac{z}{2^k}\right) + (-1)^{r-1} \zeta\left(r, 1 + \frac{z}{2^k}\right) \right]. \quad (68)$$

Proof. From (21), for $r \geq 1$, paired symmetrically when $r = 1$,

$$(\log \Phi)^{(r)}(z) = (r-1)! \sum_{n \geq 1} (1 + \nu_2(n)) \left[-\frac{1}{(n-z)^r} + \frac{(-1)^{r-1}}{(n+z)^r} \right]. \quad (69)$$

Insert (66), write $n = 2^k m$, and sum over $m \geq 1$. For $r \geq 2$, the inner sums are Hurwitz zeta functions. For $r = 1$, the convergent paired sum is the standard digamma difference. $\square$

Corollary 8.3 (Local jet data at nonzero zeros). *Let $N \neq 0$ be an integer and $d_N = 1 + \nu_2(|N|)$. Then*

$$\frac{\Phi'(z)}{\Phi(z)} = \frac{d_N}{z - N} + H_N(z), \quad (70)$$

where H_N is holomorphic near N , and every derivative of H_N can be written as an absolutely convergent valuation-weighted Hurwitz-zeta sum. Thus the complete local Taylor jet of $\Phi(z)/(z - N)^{d_N}$ is algorithmically explicit.

9 A Fredholm determinant for the Fourier product

The canonical product can be realized as an ordinary Fredholm determinant, not merely as a formal spectral product.

Definition 9.1. Let

$$\mathcal{I} = \{(n, r) : n \geq 1, 1 \leq r \leq 1 + \nu_2(n)\}, \quad (71)$$

and define the positive diagonal operator $\mathcal{K}$ on $\ell^2(\mathcal{I})$ by

$$\mathcal{K}e_{n,r} = \frac{1}{n^2}e_{n,r}. \quad (72)$$

Theorem 9.2 (Fredholm determinant representation). *The operator $\mathcal{K}$ is trace class and*

$$\boxed{\Phi(z) = \det(I - z^2 \mathcal{K})}, \quad (73)$$

$$\boxed{M(t) = \det\left(I + \frac{t^2}{4\pi^2} \mathcal{K}\right)}. \quad (74)$$

Furthermore,

$$\mathrm{tr}(\mathcal{K}^m) = \frac{\zeta(2m)}{1 - 2^{-2m}}, \quad (75)$$

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \, \mathrm{tr}(\mathcal{K}(I - z^2 \mathcal{K})^{-1}). \quad (76)$$

Proof. By (24),

$$\mathrm{tr} \mathcal{K} = \sum_{n \geq 1} \frac{1 + \nu_2(n)}{n^2} = \frac{\zeta(2)}{1 - 2^{-2}} < \infty,$$

so $\mathcal{K}$ is trace class. Its Fredholm determinant is the product over its eigenvalues, with multiplicity:

$$\det(I - z^2 \mathcal{K}) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{1 + \nu_2(n)} = \Phi(z).$$

Substituting $z = it/(2\pi)$ gives (74). Formula (75) is again (24), and differentiating the Fredholm logarithm gives (76). $\square$

Remark 9.3. This representation converts several scalar identities into operator identities. For example, the zeta expansion (23) is the standard trace expansion

$$\log \det(I - z^2 \mathcal{K}) = - \sum_{m \geq 1} \frac{z^{2m}}{m} \mathrm{tr}(\mathcal{K}^m).$$

It also suggests perturbation theory for generalized exponent sequences: changing the multiplicity sequence changes only the diagonal spectral multiplicities.

10 Jacobi continued fractions and exact rational coefficients

10.1 Exact moment recurrence

Proposition 10.1 (Moment recurrence). *The even moments satisfy $\mu_0 = 1$ and, for $n \geq 1$,*

$$(4^n - 1)\mu_{2n} = \sum_{j=1}^n \binom{2n}{2j} \frac{\mu_{2n-2j}}{2j+1}. \quad (77)$$

In particular, every μ_{2n} is rational.

Proof. Raise (10) to the $2n$ -th power and take expectations:

$$4^n \mu_{2n} = \sum_{j=0}^n \binom{2n}{2j} \mathbb{E} U^{2j} \mu_{2n-2j}.$$

Since $\mathbb{E} U^{2j} = 1/(2j+1)$, the $j=0$ term is μ_{2n} . Move it to the left. $\square$

10.2 Monic orthogonal polynomials

Let P_n be the monic orthogonal polynomials for the probability measure $\rho(x) dx$, and let

$$h_n = \int_{-1}^1 P_n(x)^2 \rho(x) dx. \quad (78)$$

Evenness forces the diagonal Jacobi parameters to vanish, so

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_{n+1}(x) = xP_n(x) - \beta_n P_{n-1}(x), \quad \beta_n = \frac{h_n}{h_{n-1}} > 0. \quad (79)$$

The first polynomials are

$$P_0(x) = 1, \quad (80)$$

$$P_1(x) = x, \quad (81)$$

$$P_2(x) = x^2 - \frac{1}{9}, \quad (82)$$

$$P_3(x) = x^3 - \frac{19}{75}x, \quad (83)$$

$$P_4(x) = x^4 - \frac{62}{147}x^2 + \frac{619}{33075}. \quad (84)$$

10.3 The Jacobi and Stieltjes fractions

Theorem 10.2 (Jacobi continued fraction). *The Cauchy transform has the locally uniform continued fraction*

$$R(z) = \frac{1}{z - \frac{\beta_1}{z - \frac{\beta_2}{z - \frac{\beta_3}{z - \ddots}}}}. \quad (85)$$

The coefficients are positive rational numbers.

Proof. Positivity follows from $\beta_n = h_n/h_{n-1}$, and rationality follows because Gram–Schmidt applied to the rational moment functional (77) uses only rational operations. Markov’s theorem for compactly supported positive measures identifies the continued fraction generated by (79) with the Cauchy transform and gives local uniform convergence off the support [7, 6]. $\square$

Define the ordinary even-moment generating function

$$\mathcal{O}(t) = \sum_{n=0}^{\infty} \mu_{2n} t^n = \int_{-1}^1 \frac{\rho(x)}{1 - tx^2} dx, \quad |t| < 1. \quad (86)$$

Substituting $t = z^{-2}$ into (85) gives the contracted fraction.

Corollary 10.3 (Stieltjes fraction and product integral). *For $|t| < 1$,*

$$\mathcal{O}(t) = \cfrac{1}{1 - \cfrac{\beta_1 t}{1 - \cfrac{\beta_2 t}{1 - \cfrac{\beta_3 t}{1 - \ddots}}}} \quad (87)$$

$$\mathcal{O}(t) = \int_0^{\infty} e^{-u} \prod_{k=1}^{\infty} \frac{\sinh(\sqrt{t} u/2^k)}{\sqrt{t} u/2^k} du. \quad (88)$$

Proof. The continued fraction follows from $\mathcal{O}(z^{-2}) = zR(z)$. For the integral, use

$$\frac{1}{1 - tx^2} = \int_0^{\infty} e^{-u} \cosh(\sqrt{t} xu) du$$

and then average over X , obtaining $M(\sqrt{t} u)$. $\square$

For $s \geq 0$, $\mathcal{O}(-s)$ is completely monotone:

$$(-1)^m \frac{d^m}{ds^m} \mathcal{O}(-s) = m! \int_{-1}^1 \frac{x^{2m} \rho(x)}{(1 + sx^2)^{m+1}} dx > 0. \quad (89)$$

Thus the even moments form a Stieltjes moment sequence, and the fraction (87) belongs to the classical positive Stieltjes class.

10.4 Hankel determinants

Let

$$\Delta_n = \det[\mu_{i+j}]_{i,j=0}^{n-1}, \quad \Delta_0 = 1. \quad (90)$$

Then

$$h_n = \frac{\Delta_{n+1}}{\Delta_n}, \quad \boxed{\beta_n = \frac{\Delta_{n+1} \Delta_{n-1}}{\Delta_n^2}}. \quad (91)$$

Because odd moments vanish, each Δ_n factors after a parity permutation into an even-even and odd-odd moment determinant. This parity factorization is useful for exact arithmetic and should be exploited in future Lean formalization.

Remark 10.4 (Post-snapshot finite Gram–Stieltjes checkpoint). This status note postdates the absorbed-draft hashes and the pinned audit commits above; it does not alter their provenance or novelty boundary. At Lean source checkpoint b3720d4b5, the following generic finite-algebra API is proved.

Layer	Exact Lean declarations and content
Moment functional	<code>FabijsFunction.FiniteMomentGram</code> : <code>momentFunctional</code> , with evaluation on monomials, constants, and powers, support and bounded-range sums, reconstruction from an arbitrary polynomial linear functional, injectivity, and scalar base change.
Moment pairing and Hankel form	<code>momentPairing</code> , <code>momentHankelMatrix</code> , <code>momentHankelDet</code> , and <code>finiteMomentPairing</code> ; <code>finiteMomentPairing_toMatrix</code> identifies the restricted pairing matrix with the Hankel matrix, while <code>finiteMomentPairing_nondegenerate_iff</code> gives nondegeneracy if and only if the determinant is nonzero over an integral domain.
Fraction-free construction	<code>FabijsFunction.GramStieltjes</code> : <code>gramStieltjesNumerator</code> is the final adjugate column encoded as a polynomial. The proved identities give its degree bound, vanishing pairings with all lower powers, pairing with X^n equal to Δ_{n+1} , top coefficient Δ_n , coefficient extraction against every polynomial of degree at most n , and self-pairing $\Delta_n \Delta_{n+1}$.
Monic field normalization	<code>gramStieltjesPolynomial</code> ; assuming $\Delta_n \neq 0$, <code>gramStieltjesPolynomial_isMonicOfDegree</code> gives monicity of exact degree n , <code>eq_gramStieltjesPolynomial_of_isMonicOfDegree_of_orthogonal</code> gives uniqueness among monic degree- n polynomials orthogonal to every lower-degree polynomial, and <code>momentPairing_gramStieltjesPolynomial_self</code> gives the finite self-pairing identity $\langle P_n, P_n \rangle = \Delta_{n+1}/\Delta_n$.

The two modules take an arbitrary scalar moment sequence; they do not identify it with the analytic Fabius moments. They also do not prove the moment recurrence (77), determinant positivity or nonvanishing, a three-term recurrence, the second identity for β_n in (91), root location or simplicity, Gaussian quadrature, either continued fraction, or convergence. Thus the finite self-pairing quotient is now formalized under its explicit field and determinant hypotheses, while the analytic and infinite-fraction assertions of this section remain frontier claims.

10.5 Exact coefficients and numerical evidence

The accompanying Python program computes the moments by (77) and then uses the exact symmetric Stieltjes procedure (79). Selected results are:

n	μ_{2n}	decimal
0	1	1
1	$\frac{1}{9}$	0.111111111111
2	$\frac{19}{675}$	0.0281481481481
3	$\frac{583}{59535}$	0.00979255899891
4	$\frac{132809}{32531625}$	0.00408245822334
5	$\frac{46840699}{24405225075}$	0.00191928977733
6	$\frac{4068990560161}{4133856862760625}$	0.000984308527181
7	$\frac{1204567303451311}{2232691548877164375}$	0.000539513532022
8	$\frac{4146897304424408411}{13301767332333178846875}$	0.00031175536309
9	$\frac{18814360006695807527868793}{100028040755473167511640090625}$	0.000188090857969
10	$\frac{21431473463327429953796293981397}{182171989134769427819794434994453125}$	0.000117644175513

n	β_n (exact when compact)	decimal
1	$\frac{1}{9}$	0.111111111111
2	$\frac{32}{225}$	0.142222222222
3	$\frac{619}{3675}$	0.168435374150
4	$\frac{835232}{4640643}$	0.179981955087
5	$\frac{1789043898625}{9272274723657}$	0.192945523288
6	see data file	0.199759514028
7	see data file	0.205964295192
8	see data file	0.211147954951
9	see data file	0.215261988744
10	see data file	0.218406233337
11	see data file	0.221259971300
12	see data file	0.223692820062

The full numerators and denominators through $n = 40$ are in `up_jacobi_coefficients.csv`. Their size grows rapidly: by moderate order the reduced integers contain thousands of decimal digits. This arithmetic complexity is not a floating-point artifact.

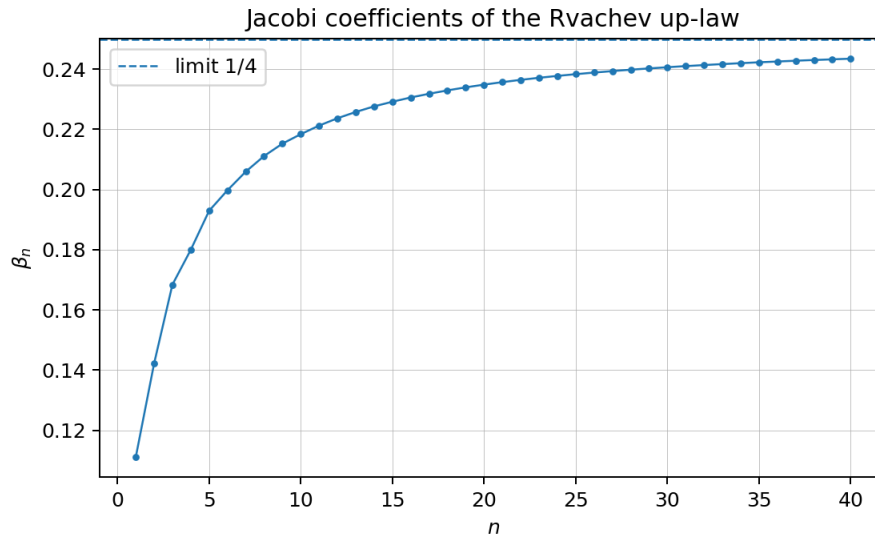


Figure 1: Exact Jacobi coefficients converted to decimals. The dashed line is the proved limit $1/4$.

10.6 The limiting coefficient

Theorem 10.5 (Nevai-class limit). *The Jacobi coefficients satisfy*

$$\lim_{n \rightarrow \infty} \beta_n = \frac{1}{4}. \quad (92)$$

Proof. The measure $\rho(x) dx$ has essential support $[-1, 1]$, and its density is strictly positive for every $x \in (-1, 1)$. The Denisov–Rakhmanov theorem therefore places its orthonormal Jacobi matrix in the Nevai class of $[-1, 1]$ [3, 6]: the off-diagonal orthonormal coefficients tend to $1/2$, and the diagonal coefficients tend to zero. In the monic recurrence (79), β_n is the square of the off-diagonal orthonormal coefficient, hence tends to $1/4$. $\square$

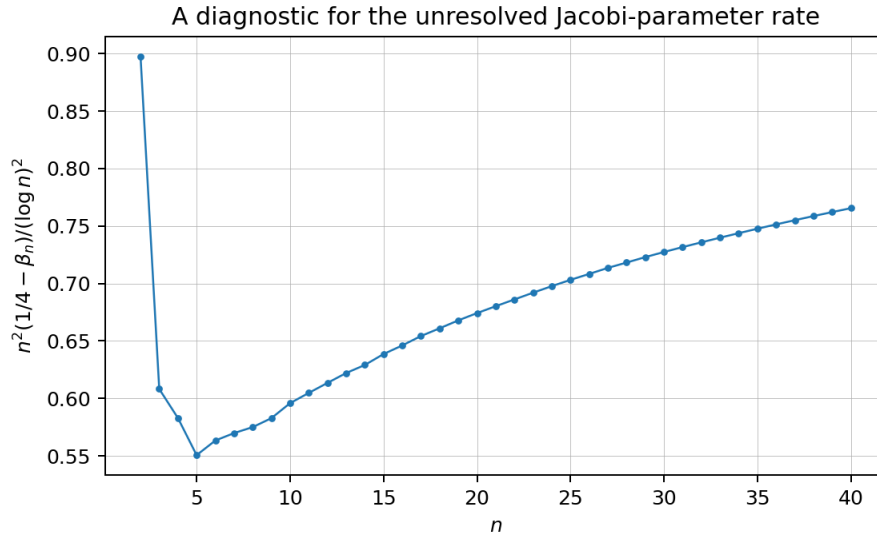


Figure 2: The diagnostic quantity $n^2(1/4 - \beta_n)/(\log n)^2$. The range $n \leq 40$ is evidence, not an asymptotic proof.

10.7 Padé and Gaussian-quadrature representations

Let Q_{N-1} be the polynomial of the second kind

$$Q_{N-1}(z) = \int_{-1}^1 \frac{P_N(z) - P_N(x)}{z - x} \rho(x) \, dx. \quad (93)$$

The N -th Jacobi convergent is Q_{N-1}/P_N . If $x_{1,N}, \dots, x_{N,N}$ are the zeros of P_N , then

$$\frac{Q_{N-1}(z)}{P_N(z)} = \sum_{j=1}^N \frac{w_{j,N}}{z - x_{j,N}}, \quad w_{j,N} > 0, \quad \sum_j w_{j,N} = 1. \quad (94)$$

This is the Cauchy transform of the N -node Gaussian quadrature measure for ρ , and it matches moments through degree $2N - 1$. The exact error identity is

$$R(z) - \frac{Q_{N-1}(z)}{P_N(z)} = \frac{1}{P_N(z)^2} \int_{-1}^1 \frac{P_N(x)^2}{z - x} \rho(x) \, dx. \quad (95)$$

Hence every Gaussian resolvent approximant is a strict lower bound for $R(z)$ when $z > 1$, and a strict upper bound when $z < -1$.

The first nontrivial approximants are completely explicit:

$$R_1^G(z) = \frac{1}{z}, \quad (96)$$

$$R_2^G(z) = \frac{z}{z^2 - 1/9} = \frac{1}{2} \left(\frac{1}{z - 1/3} + \frac{1}{z + 1/3} \right), \quad (97)$$

$$R_3^G(z) = \frac{32/57}{z} + \frac{25/114}{z - \sqrt{19/75}} + \frac{25/114}{z + \sqrt{19/75}}. \quad (98)$$

Thus the continued fraction gives a hierarchy of positive finite atomic models for the up-law, complementary to the continuous finite-convolution models of Section 5.

11 Further combined representations

11.1 The Cauchy transform of the Fabius measure

Let

$$S(z) = \int_0^1 \frac{dF(x)}{z-x} = \mathbb{E} \frac{1}{z-Y}. \quad (99)$$

The affine relation (13) gives

$$\boxed{S(z) = 2R(2z-1)}. \quad (100)$$

Consequently, every representation of R immediately yields one for S . For example, for $|2z-1| > 1$,

$$S(z) = 2 \sum_{n=0}^{\infty} \frac{\mu_{2n}}{(2z-1)^{2n+1}}, \quad (101)$$

and S has a Jacobi fraction obtained by the affine change of variable in (85).

11.2 A determinant form of the Fabius Laplace transform

Combining (48) with (74) gives

$$\int_0^1 e^{-sx} F(x) \, dx = \frac{e^{-s/2} \det \left(I + \frac{s^2}{16\pi^2} \mathcal{K} \right) - e^{-s}}{s}. \quad (102)$$

Similarly,

$$\int_0^1 e^{-sx} \rho(x) \, dx = \frac{1 - e^{-s/2} \det \left(I + \frac{s^2}{16\pi^2} \mathcal{K} \right)}{s}. \quad (103)$$

These are literal combinations of a compact-support transform with a trace-class spectral determinant.

11.3 A resolvent–product–continued-fraction triangle

For $|t| < 1$, the same analytic function $\mathcal{O}(t)$ has three structurally different representations:

$$\boxed{\begin{aligned} \mathcal{O}(t) &= \sum_{n \geq 0} \mu_{2n} t^n \\ &= \int_0^\infty e^{-u} \prod_{k \geq 1} \frac{\sinh(\sqrt{t} u / 2^k)}{\sqrt{t} u / 2^k} \, du \\ &= \frac{1}{1 - \frac{\beta_1 t}{1 - \frac{\beta_2 t}{1 - \ddots}}}. \end{aligned}} \quad (104)$$

The first line is arithmetic, the second is multiplicative and analytic, and the third is orthogonal-polynomial/geometric. Equality of the three creates a direct bridge among the Bernoulli–Bell moment formulas, the infinite hyperbolic product, and the exact rational Jacobi parameters.

12 Numerical experiments and reproducibility

12.1 Exact arithmetic

The script `rvachev_frontier_experiments.py` performs the following operations.

1. It computes $\mu_0, \mu_2, \dots$ exactly using (77).
2. It constructs the monic orthogonal polynomials using the symmetric three-term recurrence and exact rational inner products.
3. It writes all exact β_n through a user-selected order to CSV.
4. It checks (37) independently using a long moment expansion.
5. It compares the cotangent, tangent, and zero-set formulas for Φ'/Φ .
6. It generates Figures 1 and 2.

The default order is 40. On the reference environment the full run takes tens of seconds; the exact integers, rather than floating-point work, dominate at higher order.

12.2 Numerical verification of the resolvent equation

At 45-decimal mpmath precision, using moment expansions through orders 160 and 220, the computed values were:

z	$R'(z)$	$2(R(2z+1) - R(2z-1))$	relative discrepancy
1.25	-0.8312850091714830623446214	same	3.6×10^{-46}
1.75	-0.3679189599146538753803797	same	below printed precision
2.50	-0.1691589760955563001740583	same	4.4×10^{-47}

At $z = 1.25$, the change between the two truncation lengths was about 3.8×10^{-41} , so the identity error is below the visible truncation scale.

12.3 Numerical comparison of logarithmic derivatives

The cotangent and tangent formulas agreed to roughly 44 decimal places at the tested nonzero points. The zero-set formula, truncated at 10^5 , differed by 5.2×10^{-6} , 1.5×10^{-5} , and 2.9×10^{-5} at $z = 0.13, 0.37, 0.73$, respectively, consistent with its algebraic tail. The data file `numerical_checks.txt` records all values.

13 Conjectures and frontier problems

The following conjectures are deliberately separated from the proved results.

Conjecture 13.1 (Strict Jacobi monotonicity). The exact Jacobi coefficients are strictly increasing:

$$0 < \beta_1 < \beta_2 < \dots < \frac{1}{4}. \quad (105)$$

Exact arithmetic verifies the inequalities through $n = 40$. The upper limit is proved by Theorem 10.5, but monotonicity does not follow from the Denisov–Rakhmanov theorem and appears to require a Fabius-specific argument.

Conjecture 13.2 (Logarithmically corrected recurrence rate). There exists a constant $C_{\text{up}} \in (0, \infty)$ such that

$$\frac{1}{4} - \beta_n \sim C_{\text{up}} \frac{(\log n)^2}{n^2}. \quad (106)$$

The scaled quantity in Figure 2 rises from approximately 0.674 at $n = 20$ to 0.766 at $n = 40$. This is too short a range to determine a constant or even to exclude a different slowly varying factor. The conjecture is motivated jointly by the data and by the squared-logarithmic endpoint scale of ρ . A Riemann–Hilbert or strong Szegő analysis adapted to the endpoint flatness is the natural next step.

Conjecture 13.3 (Dyadically regularized Mittag–Leffler expansion). Let $d_n = 1 + \nu_2(|n|)$, and let $\text{PP}_n(z)$ be the principal part of $1/\Phi(z)$ at the nonzero integer n . There exists a canonical dyadic grouping and a finite genus of subtraction polynomials such that

$$\frac{1}{\Phi(z)} = E(z) + \sum_{n \in \mathbb{Z} \setminus \{0\}}^{\text{dyadic}} (\text{PP}_n(z) - \text{canonical subtraction at } n), \quad (107)$$

where E is an explicitly controlled entire function, ideally constant or of very low order.

The local principal parts are computable from Theorem 8.2; the unresolved issue is a global summation scheme compatible with the unbounded dyadic multiplicities.

Problem 13.4 (Arithmetic of β_n). Determine sharp denominator-clearing factors for β_n , h_n , and the parity Hankel determinants. The first coefficients display repeated migration of prime factors between consecutive norms, suggesting that factored norm variables may be arithmetically simpler than reduced β_n themselves.

Problem 13.5 (Endpoint-sensitive Gaussian nodes). Determine the asymptotic distance of the extreme zero of P_n from 1, and compare it with the Chebyshev scale n^{-2} . The density is positive almost everywhere but flatter than every power at the endpoint, so the leading equilibrium distribution is classical while the first correction should retain the Lambert– W /log-square endpoint signature.

Problem 13.6 (A coupled resolvent hierarchy for exponent sequences). The repository’s Newton/Rvachev theory replaces the unit exponent sequence in the sinc product by general integer sequences. Construct the corresponding Cauchy transforms and derive coupled affine functional equations when the exponent sequence is closed under a Pascal or shift operator. The Fredholm model of Section 9 suggests that the Fourier side is easy; the real-space fixed-point law is the main obstacle.

Problem 13.7 (Fractional resolvent dynamics). Equation (37) gives every integer derivative as a finite Thue–Morse orbit. Seek a fractional-order interpolation

$$D_z^\alpha R(z) = \int_{\Gamma_\alpha} K_\alpha(j) R(2^\alpha z + \text{affine shift}(j)) \, dj$$

that reduces to (38) for $\alpha \in \mathbb{N}$. A semigroup built from the two affine branches $z \mapsto 2z \pm 1$ may provide the correct operator calculus.

Problem 13.8 (Formal verification roadmap). A Lean development can be staged as follows:

1. define the Cauchy transform of the canonical up probability measure and prove its density form, natural-domain holomorphy, first derivative, and affine bridge;
2. prove the logarithmic fixed-point identity first on $(1, \infty)$;
3. differentiate under the integral and formalize (37);

4. apply the generic open-domain theorem in `AffineDifferenceOrbit.lean` to obtain the Thue–Morse derivative tree once the preceding resolvent equation is available;
5. define rational moments and the monic Stieltjes procedure;
6. construct finite Jacobi convergents and prove the Gaussian error identity;
7. postpone the Denisov–Rakhmanov limit, which depends on substantial external orthogonal-polynomial theory.

Item 1 is delivered by `CauchyTransform.lean`. Items 2–3 remain open. The generic conditional engine and finite affine-cube induction in item 4 are delivered by `AffineDifferenceOrbit.lean`, but cannot be instantiated for R until item 3 is available.

14 Promising research program

14.1 Analytic program

The Cauchy transform should be treated as a primary Fabius–Rvachev object, on the same level as F , ρ , and Φ . Immediate goals are:

1. derive quantitative bounds for R near the cut from the sharp Fourier decay;
2. study the Hilbert transform (35), including its zeros and endpoint asymptotics;
3. use (37) to continue asymptotic expansions along dyadic affine rays;
4. compare finite-logarithmic and Jacobi rational approximants uniformly near the support.

14.2 Orthogonal-polynomial program

The exact J -fraction opens a separate research direction:

1. prove or disprove Conjecture 13.1;
2. determine strong asymptotics of P_n , h_n , and β_n ;
3. construct Fabius-specific Gauss–Kronrod or nested positive rules;
4. investigate whether the dyadic moment recurrence induces a nonlinear recurrence or integrable system for β_n ;
5. study the arithmetic Galois structure of Gaussian nodes and exact weights.

14.3 Operator and special-function program

The Fredholm determinant and Hurwitz-zeta hierarchy suggest:

1. regularized determinants and spectral shifts for deformed multiplicity sequences;
2. a Barnes-type compression of the determinant rather than only of the scalar product;
3. exact trace identities for logarithmic derivatives at rational and half-integer arguments;
4. meromorphic continuation of dyadically weighted spectral sums with local factors dictated by $\nu_2(n)$.

14.4 Inverse-function program

The pushforward formulas (58)–(61) show that many global integral questions about F^{-1} are easier than pointwise questions. Promising directions include:

1. exact orthogonal expansions of $I(u)$ using its known pushforward moments;
2. entropy and transport functionals of I , reduced to integrals against ρ ;
3. combining the global Cauchy transform with local Lambert– W endpoint inversions;
4. rational and Gaussian quadrature for inverse-function integrals without directly evaluating I .

15 Conclusion

The audited repository already contains most of the classical and modern representation theory one would first seek: convolution laws, sinc and cosine products, integer-zero products, moment and cumulant series, Thue–Morse splines, q-binomial dyadic values, Poisson and Legendre expansions, inverse asymptotics, and Fourier-decay envelopes. The most productive way to extend that corpus is therefore to introduce a new organizing transform rather than another isolated formula.

The Cauchy transform R accomplishes this. It joins the moment sequence to an analytic function, converts the dyadic law into a functional-differential equation, turns finite convolutions into logarithmic Thue–Morse sums, and generates a Jacobi continued fraction with exact rational coefficients. The Fourier image simultaneously acquires an operator determinant and a special-function logarithmic-derivative calculus. Closed one-sided transforms and inverse pushforward identities complete a network in which real-space, frequency-space, moment, orthogonal-polynomial, and operator representations are mutually convertible.

The proved limit $\beta_n \rightarrow 1/4$ anchors the new continued-fraction theory. The exact order-40 data, the monotonicity conjecture, the endpoint-sensitive rate problem, and the regularized Mittag–Leffler problem provide concrete frontier targets suitable for both analysis and formalization.

A Reproducibility instructions

From the directory containing this consolidated report, run

```
python3 assets/Fabius_Rvachev_Representation_Frontiers/
    rvachev_frontier_experiments.py \
    --order 40 --digits 45 --output-dir representation-frontiers-output
```

This regenerates:

- `up_even_moments.csv`;
- `up_jacobi_coefficients.csv`;
- `generated_tables.tex`;
- `numerical_checks.txt`;
- the two PDF/PNG figures.

The exact arithmetic uses Python’s `fractions.Fraction`. The numerical checks use `mpmath`; plotting uses `matplotlib`. The script contains detailed comments, normalization statements, argument validation, and exact orthogonality checks.

B Corpus audit ledger

The pinned 67-path inventory and the additional live draft families used in the audit are listed in the accompanying source fragment `corpus_inventory.tex`.

Pinned 67-source ledger

No.	Path relative to Analysis/FabiusFunction/docs
1	FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2	Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3	Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5	non-formalized-research-frontiers/non-formalized-research-frontiers.tex
6	archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
7	archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex
8	archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
9	archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
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63	papers/arXiv-1502.06738v2/Lacunary_products.tex
64	papers/arXiv-1702.05442v1/09-Function.tex
65	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
66	papers/arXiv-2211.13570v1/document.tex
67	papers/ubiq15/ubiq15-corrected.tex

Live additions checked after the pinned ledger

- non-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/Fabius_Dyadic_Self_Sampling_Frontier_Report.tex
- non-formalized-research-frontiers/drafts/Fabius_Inverse_Frontier_Report_Source_and_PDF/Fabius_Inverse_Frontier_Report.tex
- non-formalized-research-frontiers/drafts/Fabius_Newton_Rvachev_Frontier_Report/Exponent_Sequence_Newton_Rvachev_Frontiers.tex
- non-formalized-research-frontiers/drafts/fabius_frontier_dyadic_inverse_barnes_report/ (report source and auxiliaries)

Additional auxiliary TeX fragments inside these live report packages were checked through their parent report structure. The pinned ledger remains the stable reproducible count; the live tree changes during active research and therefore should not be assigned a permanent count without a fresh commit-pinned traversal.

C Selected exact data

The first eleven even moments and first twelve Jacobi coefficients are typeset in the main text. The CSV files contain the exact data through order 40. Selected late-order decimal values are:

n	β_n	$1/4 - \beta_n$	$n^2(1/4 - \beta_n)/(\log n)^2$
12	0.223692820061698	0.026307179938302	0.6135029154
15	0.229185734718218	0.020814265281782	0.6386018638
20	0.234873012373992	0.015126987626008	0.6742274757
25	0.238343193605764	0.011656806394236	0.7031551379
30	0.240650640426945	0.009349359573055	0.7273789024
35	0.242285352613889	0.007714647386111	0.7476320691
40	0.243489365492140	0.006510634507860	0.7655161747

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Part II

Representation Atlas and New Analytic Bridges for the Fabius and Rvachev Functions

Abstract

This report gives a source-audited atlas of integral, series, product, convolution, probabilistic, spectral, and transform representations of the bounded Fabius function F , Rvachev’s compactly supported even function up , the inverse Fabius function $Q = F^{-1}$, and the entire Fourier image $\Phi(z) = \widehat{\text{up}}(z)$. The baseline material is organized from the current formal exposition and the living non-formalized frontier documents in the ProveIt repository. Previously present results – including the sinc and cosine products, the arithmetic canonical product, Fredholm determinants, half-integer spectral series, q -binomial moment formulae, Thue–Morse splines, the generalized-gamma-convolution dual, and the logistic dual – are explicitly quarantined from novelty claims.

Five new-to-the-audited-corpus theorem packages are developed. First, a meromorphic *dyadic gamma function*

$$\Gamma_{\text{dy}}(z) = \prod_{h \geq 0} \Gamma(1 + 2^{-h}z)$$

factors the Rvachev transform as $\Phi(z) = (\Gamma_{\text{dy}}(z)\Gamma_{\text{dy}}(-z))^{-1}$, with functional equations, canonical products, digamma and heat-trace integrals, and a geometric- q extension. Second, the existing dyadic logistic series is identified in law with an explicit Gaussian variance mixture driven by the existing discrete Thorin random variable; this yields a convolution semigroup, a closed Levy density, complete monotonicity of the density as a function of x^2 , and dimension-free radial positive definiteness. Third, Jensen’s circular mean of $\log |\Phi|$ is evaluated exactly by binary digit sums and a finite factorial product. Fourth, the unbounded zero multiplicities prove that Φ is not D-finite, and neither are the dyadic gamma factors. Fifth, distributional and compact Fourier formulae for F are completed by an odd-harmonic sine series, while Q receives a new nonlinear Fourier transform and sine-series representation in terms of oscillatory integrals of F .

A fully commented Python program independently checks the principal identities, produces four figures, and records truncation diagnostics. “Proved here” means proved in this report and not found by targeted collision searches in the audited repository; it is not a claim of priority over all external literature.

1 Scope, source audit, and status discipline

1.1 Repository boundary

The requested source boundary was the recursive collection of `*.tex` files below

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>.

The primary formal exposition was pinned to repository commit `0770442945d65bac5530c4f214c6c52c4cb9fbdc`, dated 27 August 2026. During the audit the branch changed: in particular, a support-free quantile substitution theorem entered the primary exposition. Consequently, quantile pushforward formulae that looked new against an earlier snapshot are classified here as repository consequences, not new results.

The recursive audit included the current formal exposition, the consolidated frontier notebook, the glossary and Lean walkthrough, the active frontier directories, the embedded literature sources, and the archive provenance map. The active frontier root contained seventeen named draft packages at the time of the audit. One of those packages records a snapshot with sixty-seven distinct TeX paths. Historical byte-identical copies were not counted as independent mathematical sources: the archive provenance file was used to map them to their living successors, after which the successor source was audited at formula level. This avoids manufacturing “novelty” merely by overlooking a renamed or moved draft.

1.2 Collision quarantine

The following structures were already present and are therefore baseline in this report:

- the infinite sinc product, weighted cosine product, and integer-zero canonical product;
- the arithmetic multiplicities $a_n = 1 + v_2(n)$, their Dirichlet series, counting law $\sum_{n \leq N} a_n = 2N - s_2(N)$, lobe signs, heat traces, spectral zeta functions, and Fredholm determinant;
- half-integer cosine expansions, spectral quadrature, sampling and alias formulae;
- moments and cumulants through Bell and Bernoulli numbers, q -binomial and multiple-zeta formulae, Appell sequences, Barnes-type prefixes, and Thue–Morse finite blocks;
- the discrete-Thorin generalized gamma convolution $\mathbb{E}e^{-wV_\tau} = \mathcal{A}(w)^{-\tau}$ and its complete-Bernstein/Stieltjes consequences;
- the dyadic logistic series whose characteristic function is the reciprocal moment generating function of the compact Rvachev law;
- exact negative-Laplace decompositions, periodic lower-Lambert endpoint asymptotics, inverse asymptotics, and the now-formal quantile substitution theorem.

Targeted searches did not find the phrases or exact structures “dyadic gamma”, “Gaussian variance mixture”, “Jensen circular mean”, “D-finite”, the principal-value transform of the global Fabius CDF, or the inverse sine coefficients derived in [Appendix 14](#). These results are marked **[proved here]**. Formulae obtained by combining known repository theorems without a substantively new bridge are marked **[derived synthesis]**.

1.3 Status legend and limits of the novelty statement

Status	Meaning
[repository]	Present in the formal exposition or an audited frontier document.
[derived synthesis]	A direct synthesis or reformulation of repository results; no novelty claim.
[proved here]	Proved in this report and not located by targeted collision searches in the audited corpus. This is a corpus-relative statement, not universal priority.
[conjecture]	A plausible but unproved statement, separated from theorems.

2 Normalization and the master probability diagram

2.1 The compact law and its CDF

Let $(V_j)_{j \geq 1}$ be independent uniform random variables on $[0, 1]$, let $U_j = 2V_j - 1$ be uniform on $[-1, 1]$, and define

$$X = \sum_{j \geq 1} 2^{-j} V_j, \quad Y = 2X - 1 = \sum_{j \geq 1} 2^{-j} U_j. \quad (108)$$

Then $X \in [0, 1]$, $Y \in [-1, 1]$, $X \stackrel{d}{=} 1 - X$, and $Y \stackrel{d}{=} -Y$. The bounded Fabius function and Rvachev density are

$$F(x) = \mathbb{P}(X \leq x), \quad \mathbb{P}(Y \in dy) = \text{up}(y) dy. \quad (109)$$

The normalizations used below are

$$\text{up}(y) = \begin{cases} F(y+1), & -1 \leq y \leq 0, \\ F(1-y) = 1 - F(y), & 0 \leq y \leq 1, \\ 0, & |y| > 1, \end{cases} \quad (110)$$

and

$$F'(x) = 2 \text{up}(2x - 1) \quad (0 < x < 1). \quad (111)$$

Thus up is even, nonnegative, C^∞ , supported on $[-1, 1]$, and has integral one. The density of X is $f_X(x) = 2 \text{up}(2x - 1)$ on $[0, 1]$.

2.2 Fixed-point integral equations

The random affine fixed point

$$Y \stackrel{d}{=} \frac{U + Y'}{2}, \quad (112)$$

where U is uniform on $[-1, 1]$ and Y' is an independent copy of Y , gives the exact refinement equation

$$\boxed{\text{up}(x) = \int_{2x-1}^{2x+1} \text{up}(t) dt.} \quad (113)$$

Differentiation gives

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1). \quad (114)$$

The bounded Fabius function is the fixed point of

$$F(x) = \begin{cases} \int_0^{2x} F(t) dt, & 0 \leq x \leq \frac{1}{2}, \\ 1 - \int_0^{2-2x} F(t) dt, & \frac{1}{2} \leq x \leq 1, \end{cases} \quad (115)$$

with constant continuation outside $[0, 1]$. Hence

$$F'(x) = \begin{cases} 2F(2x), & 0 < x < \frac{1}{2}, \\ 2F(2-2x), & \frac{1}{2} < x < 1. \end{cases} \quad (116)$$

These integral equations are often the most efficient starting point for exact quadrature, monotonicity, and contraction arguments.

2.3 The signed global extension and derivative tower

The repository's signed global extension $\mathcal{F}$ satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x), \quad (117)$$

so iteration gives the all-order identity

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x). \quad (118)$$

After the appropriate unit translation, this yields

$$\text{up}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m(x+1)). \quad (119)$$

Thus every derivative representation of $\mathcal{F}$ immediately becomes one for up, and vice versa. It is important not to apply (117) directly to the bounded CDF outside its valid normalization.

3 Infinite convolutions, finite splines, and head–tail representations

3.1 Box convolution

The density of $2^{-j}U_j$ is $2^{j-1}\mathbf{1}_{[-2^{-j}, 2^{-j}]}$. Therefore

$$\text{up} = \bigstar_{j=1}^{\infty} (2^{j-1}\mathbf{1}_{[-2^{-j}, 2^{-j}]}) . \quad (120)$$

This identity may be read in probability, in L^1 , or through uniform convergence of successive compact splines. It directly explains positivity, compact support, smoothness, and the sinc product of [Appendix 4](#).

Let

$$Y_N = \sum_{j=1}^N 2^{-j}U_j, \quad u_N = \bigstar_{j=1}^N (2^{j-1}\mathbf{1}_{[-2^{-j}, 2^{-j}]}) . \quad (121)$$

Then u_N is a piecewise polynomial of degree $N-1$ and

$$Y = Y_N + 2^{-N}Y', \quad \text{up} = u_N * (2^N \text{up}(2^N \cdot)) . \quad (122)$$

Equivalently,

$$\text{up}(x) = \mathbb{E} [2^N \text{up}(2^N(x - Y_N))] . \quad (123)$$

For the CDF,

$$F(x) = \mathbb{E} \left[F \left(2^N x - \sum_{j=1}^N 2^{N-j} V_j \right) \right], \quad (124)$$

where the inner F is the bounded, clamped CDF. Formula (124) is an exact N -dimensional cube integral and a useful recursive numerical scheme.

3.2 General truncated-power spline

For positive widths $a_1, \dots, a_N$, the density of $\sum_{j=1}^N a_j U_j$ is

$$b_N(x) = \frac{1}{(N-1)! 2^N \prod_{j=1}^N a_j} \sum_{\varepsilon \in \{\pm 1\}^N} \left(\prod_{j=1}^N \varepsilon_j \right) \left(x + \sum_{j=1}^N \varepsilon_j a_j \right)_+^{N-1}. \quad (125)$$

Its CDF is obtained by replacing $(N-1)!$ and the exponent $N-1$ by $N!$ and N . For $a_j = 2^{-j}$, binary indexing turns (125) into the exact Thue–Morse spline

$$u_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{k=0}^{2^N-1} (-1)^{s_2(k)} \left(x + \frac{2^N - 1 - 2k}{2^N} \right)_+^{N-1}. \quad (126)$$

The corresponding finite CDF is

$$U_N(x) = \frac{2^{N(N-1)/2}}{N!} \sum_{k=0}^{2^N-1} (-1)^{s_2(k)} \left(x + \frac{2^N - 1 - 2k}{2^N} \right)_+^N. \quad (127)$$

Prouhet cancellation makes the apparent polynomial tails cancel exactly. These formulae simultaneously represent the finite probability law, a compact spline, and a signed Thue–Morse power sum.

3.3 Hypercube and section representations

For bounded measurable g ,

$$\int_{-1}^1 g(y) \operatorname{up}(y) \, dy = \int_{[-1,1]^{\mathbb{N}}} g \left(\sum_{j \geq 1} 2^{-j} u_j \right) \bigotimes_{j \geq 1} \frac{du_j}{2}. \quad (128)$$

At finite depth this gives a coarea interpretation of u_N as the weighted $(N-1)$ -dimensional volume of the cube section $\sum_{j=1}^N 2^{-j} u_j = x$. The infinite-dimensional expression is often more useful than a closed formula: it converts integrals against up into product quadrature and supports concentration and coupling arguments.

4 Fourier-product atlas and spectral representations

4.1 Conventions and the three canonical products

We use

$$\widehat{g}(z) = \int_{\mathbb{R}} g(x) e^{-2\pi i x z} \, dx, \quad \operatorname{sinc} w = \frac{\sin w}{w}, \quad (129)$$

with the removable value at zero. Put

$$\Phi(z) = \widehat{\operatorname{up}}(z). \quad (130)$$

The repository establishes the entire products

$$\Phi(z) = \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right), \quad (131)$$

$$= \prod_{m=1}^{\infty} \left(\cos \frac{\pi z}{2^m}\right)^m, \quad (132)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{a_n}, \quad a_n = 1 + v_2(n). \quad (133)$$

They satisfy the Mahler-type renormalization

$$\Phi(z) = \operatorname{sinc}(\pi z) \Phi(z/2). \quad (134)$$

The zero at every nonzero integer n has exact order $a_{|n|}$.

4.2 Logarithmic Taylor series and partial fractions

For $|z| < 1$,

$$\log \Phi(z) = - \sum_{r \geq 1} \frac{\zeta(2r)}{r(1-4^{-r})} z^{2r}. \quad (135)$$

Logarithmic differentiation of the canonical product gives, away from the integer zeros,

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \sum_{n \geq 1} \frac{a_n}{n^2 - z^2}. \quad (136)$$

The arithmetic spectrum has Dirichlet generating function

$$\sum_{n \geq 1} \frac{a_n}{n^s} = \frac{\zeta(s)}{1-2^{-s}}, \quad \Re s > 1, \quad (137)$$

and exact counting function

$$A(N) = \sum_{n \leq N} a_n = \sum_{h \geq 0} \left\lfloor N/2^h \right\rfloor = 2N - s_2(N). \quad (138)$$

Thus the zero divisor simultaneously encodes the Riemann zeta function, dyadic valuation, and binary digit sum.

4.3 Linear and quadratic heat traces

Two Lambert/heat kernels occur naturally:

$$L_1(t) = \sum_{n \geq 1} a_n e^{-nt} = \sum_{h \geq 0} \frac{1}{e^{2^h t} - 1}, \quad (139)$$

$$K(t) = \sum_{n \geq 1} a_n e^{-n^2 t} = \sum_{h \geq 0} \sum_{m \geq 1} e^{-4^h m^2 t}. \quad (140)$$

Their Mellin transforms are

$$\int_0^\infty L_1(t) t^{s-1} dt = \Gamma(s) \frac{\zeta(s)}{1-2^{-s}}, \quad \Re s > 1, \quad (141)$$

$$\int_0^\infty K(t) t^{s-1} dt = \Gamma(s) \frac{\zeta(2s)}{1-4^{-s}}, \quad \Re s > \frac{1}{2}. \quad (142)$$

The consolidated frontier work develops theta completion and complex-dimension oscillations for K . The linear kernel L_1 will reappear in the dyadic-gamma integral and in the Levy density of the Gaussian-mixture dual.

4.4 Frullani–Lambert integral for the Fourier product

Summing the elementary Frullani identity over the zero divisor gives the mixed integral/product representation

$$\log \Phi(z) = -2 \int_0^\infty (\cosh(z t) - 1) L_1(t) \frac{dt}{t}, \quad |\Re z| < 1. \quad (143)$$

In particular, for real y ,

$$\log \Phi(iy) = 2 \int_0^\infty (1 - \cos(yt)) L_1(t) \frac{dt}{t}. \quad (144)$$

This writes the Wick-rotated product as the exponential of a positive Lambert-kernel energy. It is also the Levy–Khinchine exponent that underlies [Appendix 10](#).

4.5 Fredholm determinant and spectral zeta

Let D be the positive diagonal operator whose eigenvalue n occurs with multiplicity a_n . Since D^{-2} is trace class,

$$\Phi(z) = \det(I - z^2 D^{-2}). \quad (145)$$

The associated spectral zeta function is

$$\zeta_D(s) = \text{tr}(D^{-s}) = \frac{\zeta(s)}{1 - 2^{-s}}, \quad \Re s > 1. \quad (146)$$

The determinant, zero multiplicities, heat trace, and arithmetic Dirichlet series are thus four views of the same spectrum.

5 Fourier inversion, sampling, and interval series

5.1 Whole-line inversion

Because $\text{up} \in C_c^\infty(\mathbb{R})$, its transform is rapidly decreasing on the real axis and

$$\text{up}(x) = \int_{\mathbb{R}} \Phi(\xi) e^{2\pi i x \xi} d\xi, \quad (147)$$

$$= 2 \int_0^\infty \Phi(\xi) \cos(2\pi x \xi) d\xi. \quad (148)$$

Every derivative may be obtained under the integral:

$$\text{up}^{(m)}(x) = \int_{\mathbb{R}} (2\pi i \xi)^m \Phi(\xi) e^{2\pi i x \xi} d\xi. \quad (149)$$

Combining this with (131), (133), or the dyadic-gamma factorization below gives distinct integral–product representations of the same bump.

5.2 Poisson partition and half-integer spectrum

The integer zeros imply the partition of unity

$$\sum_{k \in \mathbb{Z}} \text{up}(x - k) = 1. \quad (150)$$

On $[-1, 1]$, the half-integer samples give the rapidly convergent cosine expansion

$$\text{up}(t) = \frac{1}{2} + \sum_{k \geq 0} \Phi\left(k + \frac{1}{2}\right) \cos((2k + 1)\pi t). \quad (151)$$

The active half-integer spectral report develops weighted Prouhet identities, exact alias transforms, and spectral quadrature from this series.

5.3 Paley–Wiener sampling of the Fourier image

Since up is supported on $[-1, 1]$, Φ belongs to the Paley–Wiener space of bandwidth one. Shannon sampling at spacing $1/2$ gives

$$\Phi(z) = \sum_{k \in \mathbb{Z}} \Phi(k/2) \text{sinc}(\pi(2z - k)). \quad (152)$$

The rapid decay of the samples makes the series locally uniformly and absolutely convergent. Since $\Phi(m) = 0$ for every nonzero integer m ,

$$\Phi(z) = \text{sinc}(2\pi z) + \sum_{\substack{k \in \mathbb{Z} \\ k \text{ odd}}} \Phi(k/2) \text{sinc}(\pi(2z - k)). \quad (153)$$

Thus the entire transform is reconstructed from its half-integer arithmetic ray.

5.4 Fourier–Legendre representation

The compact interval also supports the uniformly convergent Legendre expansion

$$\text{up}(x) = \sum_{m \geq 0} \frac{2m + 1}{2} \left(\int_{-1}^1 \text{up}(t) P_m(t) dt \right) P_m(x). \quad (154)$$

Evenness removes the odd indices. The coefficient can be expressed either as a moment combination or by inserting (148) and using the Fourier–Bessel transform of P_m . This is a useful bridge between exact moments and orthogonal approximation.

6 Laplace, Mellin, moment, and q -series representations

6.1 Moment-generating and Laplace products

The random series gives the entire moment-generating functions

$$M_Y(t) = \mathbb{E}e^{tY} = \Phi\left(\frac{it}{2\pi}\right) = \prod_{j \geq 1} \frac{\sinh(t/2^j)}{t/2^j}, \quad (155)$$

$$M_X(t) = \mathbb{E}e^{tX} = \prod_{j \geq 1} \frac{e^{t/2^j} - 1}{t/2^j}, \quad (156)$$

$$L_X(s) = \mathbb{E}e^{-sX} = \prod_{j \geq 1} \frac{1 - e^{-s/2^j}}{s/2^j}. \quad (157)$$

For $c > 0$, the CDF has the Bromwich representation

$$F(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{L_X(s)}{s} e^{sx} ds. \quad (158)$$

This contour formula is the natural entrance to the lower-Lambert saddle theory.

6.2 Mellin moments, including negative orders

For $\Re \alpha > 0$,

$$\mathbb{E}X^{-\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty s^{\alpha-1} L_X(s) ds. \quad (159)$$

The endpoint flatness makes every negative moment finite. For $0 < \Re \alpha < 1$,

$$\mathbb{E}X^\alpha = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty (1 - L_X(s)) s^{-\alpha-1} ds. \quad (160)$$

For integer $m \geq 1$, integration by parts gives the survival representation

$$\mathbb{E}X^m = m \int_0^1 x^{m-1} (1 - F(x)) dx = m \int_0^1 x^{m-1} \text{up}(x) dx. \quad (161)$$

6.3 Cumulants, Bernoulli numbers, and Bell polynomials

From (135), the nonzero cumulants of Y are

$$\kappa_{2r}(Y) = \frac{B_{2r}}{2r(1-4^{-r})} = (-1)^{r+1} \frac{(2r)! \zeta(2r)}{r(1-4^{-r})(2\pi)^{2r}}, \quad \kappa_{2r+1}(Y) = 0. \quad (162)$$

Hence

$$\mathbb{E}Y^{2n} = B_{2n}(0, \kappa_2, 0, \kappa_4, \dots, 0, \kappa_{2n}), \quad (163)$$

where the B_{2n} on the right is the complete exponential Bell polynomial, not a Bernoulli number. The first values are

$$\mathbb{E}Y^2 = \frac{1}{9}, \quad \mathbb{E}Y^4 = \frac{19}{675}, \quad \mathbb{E}Y^6 = \frac{583}{59535}. \quad (164)$$

Moments of $X = (1 + Y)/2$ follow by binomial expansion.

6.4 Wick rotation, Newton recurrence, and q -MZV formula

Define

$$\mathcal{A}(w) = \Phi(i\sqrt{w}) = \prod_{n \geq 1} \left(1 + \frac{w}{n^2}\right)^{a_n} = \sum_{r \geq 0} C_r w^r. \quad (165)$$

Then

$$C_r = \frac{(2\pi)^{2r}}{(2r)!} \mathbb{E}Y^{2r}. \quad (166)$$

Writing

$$p_r = \sum_{n \geq 1} \frac{a_n}{n^{2r}} = \frac{\zeta(2r)}{1 - 4^{-r}}, \quad (167)$$

Newton's recurrence is

$$rC_r = \sum_{j=1}^r (-1)^{j+1} p_j C_{r-j}, \quad C_0 = 1. \quad (168)$$

At $q = 1/4$, the active frontier report further expands C_r by partitions. If $\lambda = (\lambda_1, \dots, \lambda_\ell) \vdash r$, $m_j(\lambda)$ is the multiplicity of part j , and

$$Z_\lambda = \sum_{\substack{n_1, \dots, n_\ell \geq 1 \\ n_i \neq n_j}} \frac{1}{n_1^{2\lambda_1} \dots n_\ell^{2\lambda_\ell}}, \quad (169)$$

then

$$C_r = \sum_{\lambda \vdash r} \frac{Z_\lambda}{\prod_{j \geq 1} m_j(\lambda)!} \prod_{i=1}^{\ell(\lambda)} \frac{q^{\lambda_i(\lambda_i-1)/2}}{(q; q)_{\lambda_i}}, \quad q = \frac{1}{4}. \quad (170)$$

Moreover Z_λ is a symmetrized multiple-zeta value. This formula keeps both the integer spectrum and its dyadic layer occupancy visible.

6.5 Appell and logistic representations already in the corpus

The Appell sequence associated with the compact Rvachev law is defined by

$$\frac{e^{xt}}{M_Y(t)} = \sum_{n \geq 0} A_n(x) \frac{t^n}{n!}. \quad (171)$$

The audited Barnes/inverse draft introduces independent standard logistic variables Λ_j and

$$Z = \sum_{j \geq 1} \frac{\Lambda_j}{\pi 2^j}, \quad (172)$$

with

$$\mathbb{E} e^{itZ} = \frac{1}{M_Y(t)}, \quad A_n(x) = \mathbb{E}(x + iZ)^n. \quad (173)$$

This logistic dual is baseline. The new result in [Appendix 10](#) is its exact identification as a Gaussian variance mixture driven by the independently developed Thorin variable.

7 Thue–Morse, dyadic arithmetic, and finite product-to-sum bridges

Let $\varepsilon_k = (-1)^{s_2(k)}$. The finite Prouhet product is

$$\prod_{j=0}^{N-1} (1 - z^{2^j}) = \sum_{k=0}^{2^N-1} \varepsilon_k z^k. \quad (174)$$

With $z = e^t$ this becomes

$$\sum_{k < 2^N} \varepsilon_k e^{kt} = \prod_{j=0}^{N-1} (1 - e^{2^j t}). \quad (175)$$

Coefficient comparison yields

$$\sum_{k < 2^N} \varepsilon_k k^m = 0 \quad (0 \leq m < N), \quad \sum_{k < 2^N} \varepsilon_k k^N = (-1)^N N! 2^{N(N-1)/2}. \quad (176)$$

These cancellations explain the compactness of the truncated-power spline (126); they also power the exact dyadic-value formulae, Gaussian-binomial interpolation, and Richardson extrapolants in the repository.

The finite Fourier product

$$\Phi_N(z) = \prod_{h=0}^{N-1} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right) \quad (177)$$

can be converted into a signed exponential sum by expanding the sine factors. Conversely, Prouhet cancellations force high-order zeros in those finite sums. This is the basic product-to-sum bridge connecting Fourier analysis to exact binary arithmetic.

8 Endpoint Laplace phases, Lambert W , and inverse asymptotics

The active frontier synthesis writes

$$g(s) = \log L_X(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{1}{2} \log s + R(\log_2 s) + E(s), \quad (178)$$

where R is real-analytic and one-periodic and

$$0 \leq E(s) \leq \frac{e^{-s}}{(1 - e^{-s})^2}. \quad (179)$$

The CDF integral (158) has saddle equation

$$x = -g'(s), \quad (180)$$

whose exact phase is solved by a lower branch of Lambert W , perturbed by the periodic function and its derivatives. The periodic term is essential: every nonzero Fourier mode is explicitly nonzero in the formal exposition, so replacing it by a constant loses the claimed sharp error scale.

The inverse $Q = F^{-1}$ consequently lives on the stretched-logarithmic scale

$$Q(y) \asymp \sqrt{\log(1/y)} \exp\left[-\sqrt{2 \log 2 \log(1/y)}\right] \quad (y \downarrow 0), \quad (181)$$

with a sharper periodic Lambert expansion in the audited inverse reports. Reflection gives $Q(1-y) = 1 - Q(y)$. These asymptotic representations complement, rather than replace, the exact Fourier and quantile identities developed below.

New theorem packages

9 The dyadic gamma factorization

9.1 Definition, convergence, and poles

Definition 9.1 (Dyadic gamma function [\[proved here\]](#)). For $z \in \mathbb{C}$, away from the negative integers, define

$$\Gamma_{\text{dy}}(z) = \prod_{h=0}^{\infty} \Gamma(1 + 2^{-h}z). \quad (182)$$

Theorem 9.2 (Meromorphy and Mahler equation [\[proved here\]](#)). The product (182) converges locally uniformly away from its poles and defines a meromorphic function satisfying

$$\Gamma_{\text{dy}}(z) = \Gamma(1 + z)\Gamma_{\text{dy}}(z/2), \quad \Gamma_{\text{dy}}(0) = 1. \quad (183)$$

Its pole at $-n$, $n \geq 1$, has exact order $a_n = 1 + v_2(n)$ and it has no other poles.

Proof. Uniformly on compact sets, $\log \Gamma(1 + w) = -\gamma w + \mathcal{O}(w^2)$ as $w \rightarrow 0$. The sums $\sum_h 2^{-h}$ and $\sum_h 4^{-h}$ converge, so the logarithms form a normally convergent series after finitely many factors are removed. Separating the factor $h = 0$ gives (183). At $z = -n$, the factor indexed by h has a simple gamma pole precisely when $n/2^h$ is a positive integer, namely for $0 \leq h \leq v_2(n)$. Their number is $1 + v_2(n)$. $\square$

9.2 Reflection factorization of the Rvachev transform

Theorem 9.3 (Dyadic gamma factorization [\[proved here\]](#)). For every $z \in \mathbb{C}$,

$$\Phi(z) = \frac{1}{\Gamma_{\text{dy}}(z)\Gamma_{\text{dy}}(-z)}. \quad (184)$$

The equality is entire: at integer z , the poles of the gamma factors produce exactly the zeros of Φ with their arithmetic multiplicities.

Proof. Euler's reflection formula in the shifted form

$$\Gamma(1 + w)\Gamma(1 - w) = \frac{\pi w}{\sin(\pi w)} = \frac{1}{\text{sinc}(\pi w)} \quad (185)$$

gives

$$\frac{1}{\Gamma(1 + 2^{-h}z)\Gamma(1 - 2^{-h}z)} = \text{sinc}(\pi z/2^h).$$

Multiplication over $h \geq 0$ yields the sinc product (131). Normal convergence away from the poles and removable continuation at the integers complete the proof. $\square$

Corollary 9.4 (Gamma integral representations of up and F [\[proved here\]](#)). For real x ,

$$\text{up}(x) = \int_{\mathbb{R}} \frac{e^{2\pi i x \xi}}{\Gamma_{\text{dy}}(\xi)\Gamma_{\text{dy}}(-\xi)} d\xi, \quad (186)$$

and for real x ,

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\sin(2\pi \xi(2x - 1))}{\xi \Gamma_{\text{dy}}(\xi)\Gamma_{\text{dy}}(-\xi)} d\xi. \quad (187)$$

Both integrals are absolutely convergent after the usual removable interpretation at $\xi = 0$.

9.3 Canonical reciprocal product and power series

Let $G_{\text{dy}}(z) = 1/\Gamma_{\text{dy}}(z)$. Euler's canonical product for $1/\Gamma$ and regrouping equal zeros give

$$G_{\text{dy}}(z) = e^{2\gamma z} \prod_{n \geq 1} \left[\left(1 + \frac{z}{n} \right) e^{-z/n} \right]^{a_n}. \quad (188)$$

Consequently

$$\Phi(z) = G_{\text{dy}}(z)G_{\text{dy}}(-z), \quad (189)$$

which splits the even zero divisor into a negative-integer factor and its reflection. For $|z| < 1$,

$$\log \Gamma_{\text{dy}}(z) = -2\gamma z + \sum_{m \geq 2} \frac{(-1)^m \zeta(m)}{m(1 - 2^{-m})} z^m. \quad (190)$$

The even part of (190) is exactly $-\frac{1}{2} \log \Phi(z)$, while the odd part is new information lost in the even Fourier product.

9.4 Dyadic digamma and Stieltjes resolvent

Define

$$\Psi_{\text{dy}}(z) = \frac{\Gamma'_{\text{dy}}(z)}{\Gamma_{\text{dy}}(z)}. \quad (191)$$

Termwise differentiation gives

$$\Psi_{\text{dy}}(z) = \sum_{h \geq 0} 2^{-h} \psi(1 + 2^{-h} z), \quad (192)$$

and the Fourier logarithmic derivative becomes

$$\frac{\Phi'(z)}{\Phi(z)} = \Psi_{\text{dy}}(-z) - \Psi_{\text{dy}}(z) = -2z \sum_{n \geq 1} \frac{a_n}{n^2 - z^2}. \quad (193)$$

Thus the arithmetic Stieltjes resolvent is the odd part of a dyadic digamma function. Differentiating repeatedly produces polygamma representations for every logarithmic jet of Φ .

9.5 Malmsten and heat-trace integrals

For $\Re z > -1$, summing Malmsten's integral for $\log \Gamma(1 + z/2^h)$ gives

$$\log \Gamma_{\text{dy}}(z) = \int_0^\infty \left[2ze^{-t} - \frac{\sum_{h \geq 0} (1 - e^{-2^{-h}zt})}{e^t - 1} \right] \frac{dt}{t}. \quad (194)$$

The inner dyadic exponential sum has the entire expansion

$$\sum_{h \geq 0} (1 - e^{-2^{-h}w}) = \sum_{m \geq 1} \frac{(-1)^{m+1} w^m}{m!(1 - 2^{-m})}. \quad (195)$$

A second integral makes the integer spectrum explicit. From (188) and (139),

$$\log \Gamma_{\text{dy}}(z) = -2\gamma z + \int_0^\infty (zt - (1 - e^{-zt})) L_1(t) \frac{dt}{t}, \quad \Re z > -1. \quad (196)$$

This identity is a direct analytic bridge between the gamma factor, Lambert series, and the zero-divisor heat trace.

9.6 Geometric- q extension

Theorem 9.5 (Geometric gamma factorization [\[proved here\]](#)). *Let $0 < q < 1$ and define*

$$\Gamma_{\langle q \rangle}(z) = \prod_{h \geq 0} \Gamma(1 + q^h z), \quad \Phi_q(z) = \prod_{h \geq 0} \operatorname{sinc}(\pi q^h z). \quad (197)$$

Then

$$\Gamma_{\langle q \rangle}(z) = \Gamma(1 + z) \Gamma_{\langle q \rangle}(qz), \quad (198)$$

$$\Phi_q(z) = \frac{1}{\Gamma_{\langle q \rangle}(z) \Gamma_{\langle q \rangle}(-z)}, \quad (199)$$

$$\log \Gamma_{\langle q \rangle}(z) = -\frac{\gamma z}{1 - q} + \sum_{m \geq 2} \frac{(-1)^m \zeta(m)}{m(1 - q^m)} z^m, \quad |z| < 1. \quad (200)$$

For $q = 1/b$ with integer $b \geq 2$, the pole at $-n$ has order $1 + \max\{h : b^h \mid n\}$.

Proof. The convergence argument is identical to the dyadic case, using $\sum q^h$ and $\sum q^{2h}$. Reflection is applied factor by factor, and the power series follows by summing $\sum_h q^{hm} = (1 - q^m)^{-1}$. $\square$

Remark 9.6. $\Gamma_{\langle q \rangle}$ is not the classical Jackson q -gamma function. It is a geometric Mahler product of ordinary gamma functions. Its denominators $1 - q^m$ nevertheless align exactly with the q -series denominators already pervasive in the Fabius–Rvachev theory.

10 Gaussian variance mixtures and the logistic dual

10.1 The existing Thorin variable

The audited frontier corpus already constructs, for $\tau > 0$, a nonnegative random variable V_τ with

$$\mathbb{E} e^{-w V_\tau} = \mathcal{A}(w)^{-\tau}, \quad (201)$$

and grouped gamma representation

$$V_\tau \stackrel{d}{=} \sum_{n \geq 1} H_{n,\tau}, \quad H_{n,\tau} \sim \text{Gamma}(\tau a_n, \text{rate } n^2), \quad (202)$$

with independent summands. Its discrete Thorin measure is

$$U_\tau = \tau \sum_{n \geq 1} a_n \delta_{n^2}. \quad (203)$$

The construction and its complete-Bernstein consequences are [\[repository\]](#). We now connect it to the independently developed logistic dual.

10.2 Equality in law

Theorem 10.1 (Gaussian variance-mixture bridge [\[proved here\]](#)). *Let*

$$T_\tau = \frac{V_\tau}{4\pi^2}, \quad (204)$$

and let $N \sim \mathcal{N}(0, 1)$ be independent of T_τ . Define

$$Z_\tau = \sqrt{2T_\tau} N = \frac{\sqrt{V_\tau}}{\sqrt{2}\pi} N. \quad (205)$$

Then

$$\mathbb{E}e^{itZ_\tau} = \mathcal{A}\left(\frac{t^2}{4\pi^2}\right)^{-\tau}, \quad (206)$$

$$= \prod_{n \geq 1} \left(1 + \frac{t^2}{4\pi^2 n^2}\right)^{-\tau a_n}, \quad (207)$$

$$= \left[\prod_{j \geq 1} \frac{t/2^j}{\sinh(t/2^j)} \right]^\tau, \quad (208)$$

$$= \left[\Gamma_{\text{dy}}\left(\frac{it}{2\pi}\right) \Gamma_{\text{dy}}\left(-\frac{it}{2\pi}\right) \right]^\tau. \quad (209)$$

For $\tau = 1$,

$$\boxed{\sum_{j \geq 1} \frac{\Lambda_j}{\pi 2^j} \stackrel{\text{d}}{=} \frac{\sqrt{V_1}}{\sqrt{2}\pi} N,} \quad (210)$$

where the Λ_j are independent standard logistic variables.

Proof. Conditioning on T_τ gives

$$\mathbb{E}(e^{it\sqrt{2T_\tau}N} \mid T_\tau) = e^{-t^2 T_\tau}.$$

Taking expectation and applying (201) at $w = t^2/(4\pi^2)$ proves (206); (207) follows from (165). Since $\mathcal{A}(t^2/(4\pi^2)) = M_Y(t)$, the reciprocal hyperbolic-sine product (208) follows from (155). Finally, (184) at $z = it/(2\pi)$ gives (209). At $\tau = 1$, the right-hand side is exactly the characteristic function of the repository's logistic series (172); uniqueness of characteristic functions proves (210). $\square$

The bridge is not a restatement of generalized gamma convolution or of the logistic series. It identifies two previously separate noncompact duals of the compact Rvachev law: one built additively from logistic coordinates and the other obtained by subordinating a Gaussian by the arithmetic Thorin variable.

10.3 Convolution semigroup, density, and moments

The family $(Z_\tau)_{\tau > 0}$ is a symmetric convolution semigroup:

$$Z_{\tau_1} + Z'_{\tau_2} \stackrel{\text{d}}{=} Z_{\tau_1 + \tau_2}. \quad (211)$$

Its density is the exact normal scale mixture

$$\boxed{p_\tau(x) = \mathbb{E} \left[\frac{1}{\sqrt{4\pi T_\tau}} \exp\left(-\frac{x^2}{4T_\tau}\right) \right].} \quad (212)$$

The cumulants are

$$\kappa_{2r}(Z_\tau) = \tau \frac{(2r)!}{r(4\pi^2)^r} \frac{\zeta(2r)}{1 - 4^{-r}}, \quad \kappa_{2r+1}(Z_\tau) = 0. \quad (213)$$

In particular, $\text{Var } Z_\tau = \tau/9$. Conditional Gaussian moments yield

$$\mathbb{E} Z_\tau^{2m} = 2^m (2m-1)!! \mathbb{E} T_\tau^m, \quad (214)$$

where

$$\kappa_r(T_\tau) = \tau(r-1)! \frac{\zeta(2r)}{(4\pi^2)^r (1-4^{-r})}. \quad (215)$$

Thus (214) and complete Bell polynomials give a second moment calculus for the logistic/Appell dual.

10.4 Closed Levy density and self-decomposability

Theorem 10.2 (Levy density of the dual semigroup [proved here]). *The characteristic exponent of Z_τ is*

$$-\log \mathbb{E} e^{itZ_\tau} = 2\tau \int_0^\infty (1 - \cos(tx)) L_1(2\pi x) \frac{dx}{x}. \quad (216)$$

Consequently its symmetric Levy measure has density

$$\nu_\tau(x) = \frac{\tau}{|x|} L_1(2\pi|x|) = \frac{\tau}{|x|} \sum_{h \geq 0} \frac{1}{e^{2^{h+1}\pi|x|} - 1}, \quad x \neq 0. \quad (217)$$

Every Z_τ is self-decomposable.

Proof. Use

$$\log \left(1 + \frac{t^2}{\lambda^2} \right) = 2 \int_0^\infty (1 - \cos(tx)) e^{-\lambda x} \frac{dx}{x}$$

in (207), then sum over n . This proves (216) and (217). A symmetric infinitely divisible law with Levy density $k(|x|)/|x|$ is self-decomposable when k is decreasing. Here $k(x) = \tau L_1(2\pi x)$ is a positive sum of decreasing exponentials. $\square$

10.5 Complete monotonicity in the squared coordinate

Corollary 10.3 (Radial positivity in every dimension [proved here]). *For every $\tau > 0$, the function*

$$g_\tau(s) = p_\tau(\sqrt{s}), \quad s > 0, \quad (218)$$

is completely monotone:

$$(-1)^m g_\tau^{(m)}(s) \geq 0 \quad (m \geq 0, s > 0). \quad (219)$$

Consequently $x \mapsto p_\tau(\|x\|)$ is positive definite on $\mathbb{R}^d$ for every $d \geq 1$.

Proof. Formula (212) writes

$$g_\tau(s) = \mathbb{E} \left[(4\pi T_\tau)^{-1/2} e^{-s/(4T_\tau)} \right],$$

a positive mixture of exponentials in s . Bernstein's theorem gives complete monotonicity, and Schoenberg's theorem converts complete monotonicity of g_τ into radial positive definiteness of $g_\tau(\|x\|^2) = p_\tau(\|x\|)$ in every dimension. $\square$

11 Jensen means, factorial products, and binary digit sums

11.1 Exact circular mean

Define the Jensen mean

$$J(r) = \frac{1}{2\pi} \int_0^{2\pi} \log \left| \Phi(re^{i\theta}) \right| d\theta, \quad r > 0, \quad (220)$$

with the improper interpretation when the circle passes through an integer zero.

Theorem 11.1 (Arithmetic Jensen formula [\[proved here\]](#)). *Let $N = \lfloor r \rfloor$ and $N_j = \lfloor N/2^j \rfloor$. Then*

$$J(r) = 2 \sum_{n \leq N} a_n \log \frac{r}{n}, \quad (221)$$

$$= 2 \sum_{j \geq 0} \left[N_j \log \frac{r}{2^j} - \log \Gamma(N_j + 1) \right]. \quad (222)$$

The second sum is finite. Equivalently,

$$\exp \left(\frac{J(r)}{2} \right) = \prod_{n \leq N} \left(\frac{r}{n} \right)^{a_n} = \prod_{j \geq 0} \frac{(r/2^j)^{N_j}}{N_j!}. \quad (223)$$

Proof. Jensen's formula for the entire function Φ , normalized by $\Phi(0) = 1$, says that the mean is the sum of $\log(r/|\rho|)$ over zeros ρ in the disk, counted with multiplicity. The zeros occur at $\pm n$ with multiplicity a_n , giving (221). Expand $a_n = \sum_{j \geq 0} \mathbf{1}_{2^j | n}$, write $n = 2^j m$, and sum $\log(r/(2^j m))$ for $1 \leq m \leq N_j$. This proves (222) and (223). $\square$

11.2 Digit-sum derivative

Corollary 11.2 (Binary slope law [\[proved here\]](#)). *For $N < r < N + 1$,*

$$J'(r) = \frac{2A(N)}{r} = \frac{2(2N - s_2(N))}{r}. \quad (224)$$

Thus

$$\frac{rJ'(r)}{2} = 2 \lfloor r \rfloor - s_2(\lfloor r \rfloor) \quad (225)$$

recovers the binary digit sum directly from a circular mean of the Fourier image.

Remark 11.3. The zero-counting law $A(N) = 2N - s_2(N)$ was already in the repository. The new point is that it is the logarithmic radial derivative of a two-dimensional complex-analytic average, and that the full mean has the factorial product (223).

11.3 Asymptotic growth

Theorem 11.4 (Leading Jensen asymptotic [\[proved here\]](#)). *As $r \rightarrow \infty$,*

$$J(r) = 4r - \frac{(\log r)^2}{2 \log 2} + \mathcal{O}(\log r). \quad (226)$$

Proof. Let $N = \lfloor r \rfloor$. Partial summation applied to (221) gives

$$\frac{J(r)}{2} = A(N) \log(r/N) + \int_1^N \frac{A(\lfloor t \rfloor)}{t} dt.$$

The first term is $\mathcal{O}(1)$. Since $A(n) = 2n - s_2(n)$, the linear part contributes $2N + \mathcal{O}(\log N)$. The elementary summatory digit estimate

$$\sum_{n \leq x} s_2(n) = \frac{x \log x}{2 \log 2} + \mathcal{O}(x)$$

and another partial summation give

$$\int_1^N \frac{s_2(\lfloor t \rfloor)}{t} dt = \frac{(\log N)^2}{4 \log 2} + \mathcal{O}(\log N).$$

Multiply by two and replace N with r . □

11.4 Subunit circular energy

For $0 \leq r < 1$, Jensen's mean vanishes because the disk contains no zeros. The angular fluctuation nevertheless has an exact zeta-series energy. Put

$$c_m = \frac{\zeta(2m)}{m(1 - 4^{-m})}. \quad (227)$$

Then (135) and trigonometric orthogonality give

$$\boxed{\frac{1}{2\pi} \int_0^{2\pi} \left(\log |\Phi(re^{i\theta})| \right)^2 d\theta = \frac{1}{2} \sum_{m \geq 1} c_m^2 r^{4m}.} \quad (228)$$

More generally, for $0 \leq r, s < 1$,

$$\frac{1}{2\pi} \int_0^{2\pi} \log |\Phi(re^{i\theta})| \log |\Phi(se^{i\theta})| d\theta = \frac{1}{2} \sum_{m \geq 1} c_m^2 (rs)^{2m}. \quad (229)$$

This is a local angular covariance kernel for the Fourier determinant.

12 Non-holonomicity from the arithmetic zero divisor

Theorem 12.1 (The Rvachev transform is not D-finite [proved here]). *The entire function Φ does not satisfy any nontrivial linear differential equation*

$$p_R(z)y^{(R)} + p_{R-1}(z)y^{(R-1)} + \cdots + p_0(z)y = 0 \quad (230)$$

with polynomial coefficients $p_j \in \mathbb{C}[z]$. Equivalently, Φ is not holonomic over $\mathbb{C}(z)$.

Proof. Suppose (230) holds with $p_R \neq 0$. At any ordinary point z_0 with $p_R(z_0) \neq 0$, uniqueness for the associated analytic initial-value problem implies that a nonzero solution cannot have a zero of multiplicity R or larger: if the first R derivatives vanish, the solution is identically zero near z_0 .

The polynomial p_R has only finitely many zeros. But Φ has a zero at 2^k of multiplicity $k + 1$. Choose $k \geq R$ such that $p_R(2^k) \neq 0$. This gives an ordinary zero of multiplicity at least R , a contradiction. □

Corollary 12.2 (Coefficient and algebraic consequences [\[proved here\]](#)). *The Maclaurin coefficient sequence of Φ is not P-recursive. In particular, Φ is not algebraic over $\mathbb{C}(z)$.*

Proof. A power series is D-finite if and only if its coefficient sequence is P-recursive. Every algebraic power series is D-finite. $\square$

Proposition 12.3 (The dyadic gamma factors are not D-finite [\[proved here\]](#)). *Neither Γ_{dy} nor $G_{\text{dy}} = 1/\Gamma_{\text{dy}}$ is D-finite.*

Proof. A D-finite germ has only finitely many possible finite singularities, whereas Γ_{dy} has poles at every negative integer. The reciprocal G_{dy} is entire but has a zero at -2^k of multiplicity $k + 1$, so the preceding multiplicity argument applies. $\square$

Remark 12.4. Non-D-finiteness is much weaker than differential transcendence. The latter is stated only as a conjecture in [Appendix 17](#).

13 Fourier representations of the bounded Fabius function

13.1 The density transform

Let $f_X = F' = 2 \text{up}(2x - 1)$ on $[0, 1]$ and zero outside. A change of variables gives

$$\widehat{f_X}(\xi) = e^{-\pi i \xi} \Phi(\xi/2). \quad (231)$$

Hence the compactly supported pieces

$$F_0(x) = F(x) \mathbf{1}_{[0,1]}(x), \quad S_0(x) = (1 - F(x)) \mathbf{1}_{[0,1]}(x) \quad (232)$$

have, for $\xi \neq 0$,

$$\widehat{F_0}(\xi) = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}, \quad (233)$$

$$\widehat{S_0}(\xi) = \frac{1 - e^{-\pi i \xi} \Phi(\xi/2)}{2\pi i \xi}. \quad (234)$$

Both values at zero are $1/2$. Inserting (131), (133), or (184) turns these into explicit product representations of the Fourier image of the bounded CDF.

13.2 The global CDF as a tempered distribution

Let H be the Heaviside distribution. Since $F = H * f_X$,

$$\widehat{H} = \frac{1}{2} \delta_0 + \frac{1}{2\pi i} \text{pv} \frac{1}{\xi} \quad (235)$$

gives the distributional formula

$$\widehat{F} = \frac{1}{2} \delta_0 + \frac{1}{2\pi i} \text{pv} \left(\frac{e^{-\pi i \xi} \Phi(\xi/2)}{\xi} \right). \quad (236)$$

This is the correct Fourier image of the globally bounded, nonintegrable CDF. Omitting the Dirac mass or principal value changes the distribution.

13.3 Sine integral and odd-harmonic series

Since $F(x) = \frac{1}{2} + \int_0^{2x-1} \text{up}(y) \, dy$, Fourier inversion yields

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Phi(\xi) \frac{\sin(2\pi\xi(2x-1))}{\xi} \, d\xi. \quad (237)$$

This holds for all real x under the bounded-CDF convention. Integrating the interval cosine series (151) gives a second representation.

Theorem 13.1 (Odd-harmonic Fabius series [\[proved here\]](#)). For $0 \leq x \leq 1$,

$$F'(x) = 1 - 2 \sum_{k \geq 0} \Phi\left(k + \frac{1}{2}\right) \cos(2\pi(2k+1)x), \quad (238)$$

$$F(x) = x - \frac{1}{\pi} \sum_{k \geq 0} \frac{\Phi(k + 1/2)}{2k+1} \sin(2\pi(2k+1)x). \quad (239)$$

Both series converge absolutely and uniformly together with every differentiated series allowed by the displayed factors.

Proof. Substitute $t = 2x - 1$ into (151) and multiply by two. Since

$$\cos((2k+1)\pi(2x-1)) = -\cos(2\pi(2k+1)x),$$

we obtain (238). Integrating from 0 to x and using $F(0) = 0$ gives (239). Rapid decay of the half-integer samples follows from $\text{up} \in C_c^\infty$. $\square$

13.4 Gamma versions

The new factorization makes the preceding formulae explicitly gamma-theoretic:

$$\widehat{f_X}(\xi) = \frac{e^{-\pi i \xi}}{\Gamma_{\text{dy}}(\xi/2) \Gamma_{\text{dy}}(-\xi/2)}, \quad (240)$$

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\sin(2\pi\xi(2x-1))}{\xi \Gamma_{\text{dy}}(\xi) \Gamma_{\text{dy}}(-\xi)} \, d\xi. \quad (241)$$

Thus ordinary gamma functions, assembled on a dyadic scale, provide direct integral and Fourier images for the Fabius CDF.

14 Quantile and inverse-Fabius transform representations

14.1 Quantile pushforward and product transforms

Let $Q = F^{-1}$ be the clamped inverse. The current formal exposition proves the support-free quantile substitution theorem

$$\int_0^1 \varphi(Q(y)) \, dy = \mathbb{E} \varphi(X) \quad (242)$$

for a broad Banach-valued class of integrands. Therefore

$$\int_0^1 e^{-sQ(y)} dy = \prod_{j \geq 1} \frac{1 - e^{-s/2^j}}{s/2^j}, \quad (243)$$

$$\int_0^1 e^{s(Q(y)-1/2)} dy = \prod_{j \geq 1} \frac{\sinh(s/2^{j+1})}{s/2^{j+1}}, \quad (244)$$

$$\int_0^1 e^{-2\pi i \xi Q(y)} dy = e^{-\pi i \xi} \Phi(\xi/2). \quad (245)$$

For $0 < \Re \alpha < 1$,

$$\int_0^1 Q(y)^\alpha dy = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty \left[1 - \prod_{j \geq 1} \frac{1 - e^{-t/2^j}}{t/2^j} \right] t^{-\alpha-1} dt. \quad (246)$$

These are **[derived synthesis]** consequences of the now-formal quantile theorem.

14.2 A nonlinear Fourier transform of the inverse

Theorem 14.1 (Inverse-transform identity **[proved here]**). For $\eta \in \mathbb{C} \setminus \{0\}$,

$$\int_0^1 Q(y) e^{-2\pi i \eta y} dy = \int_0^1 x F'(x) e^{-2\pi i \eta F(x)} dx, \quad (247)$$

$$= \frac{\int_0^1 e^{-2\pi i \eta F(x)} dx - e^{-2\pi i \eta}}{2\pi i \eta}. \quad (248)$$

The value at $\eta = 0$ is $1/2$ by continuity.

Proof. Apply $y = F(x)$, $dy = F'(x) dx$, and $Q(F(x)) = x$ to obtain (247). Since $d e^{-2\pi i \eta F(x)} = -2\pi i \eta F'(x) e^{-2\pi i \eta F(x)} dx$, integration by parts gives (248). $\square$

14.3 Sine series of the inverse in terms of F

Theorem 14.2 (Nonlinear inverse-Fabius sine series **[proved here]**). Put

$$c_n = \frac{1}{\pi n} \int_0^1 \cos(2\pi n F(x)) dx, \quad n \geq 1. \quad (249)$$

Then

$$\boxed{Q(y) = y + \sum_{n \geq 1} c_n \sin(2\pi n y), \quad 0 \leq y \leq 1.} \quad (250)$$

The series converges in $L^2(0, 1)$ and pointwise at every $y \in [0, 1]$. Moreover $c_n = o(1/n)$.

Proof. The function $g(y) = Q(y) - y$ is continuous, vanishes at both endpoints, and is of bounded variation. The symmetry $Q(1-y) = 1 - Q(y)$ gives $g(1-y) = -g(y)$, so only the modes $\sin(2\pi n y)$ occur. Their coefficients are

$$2 \int_0^1 g(y) \sin(2\pi n y) dy.$$

For the Q term, use $y = F(x)$ and integrate by parts:

$$\begin{aligned}\int_0^1 Q(y) \sin(2\pi n y) \, dy &= \int_0^1 x F'(x) \sin(2\pi n F(x)) \, dx \\ &= -\frac{1}{2\pi n} + \frac{1}{2\pi n} \int_0^1 \cos(2\pi n F(x)) \, dx.\end{aligned}$$

Since $\int_0^1 y \sin(2\pi n y) \, dy = -1/(2\pi n)$, the difference is exactly (249). Dirichlet–Jordan convergence applies to the bounded-variation function g . Finally,

$$\int_0^1 \cos(2\pi n F(x)) \, dx = \int_0^1 \cos(2\pi n y) Q'(y) \, dy$$

with $Q' \in L^1(0, 1)$; the Riemann–Lebesgue lemma gives $c_n = o(1/n)$. $\square$

Remark 14.3. Formula (250) is not an asymptotic reversion. It is a global interval representation of the inverse in which every coefficient is a nonlinear oscillatory integral of the original CDF. It therefore complements the repository’s local Lambert–Bell inverse expansions.

15 Compact representation atlas

The following table is a lookup index rather than a proof substitute. It collects the main representations developed above and the principal established forms in the audited corpus. The domain column records the natural interpretation: pointwise, entire, distributional, probabilistic, or asymptotic.

Object	Representation	Domain	Status
F	$F(x) = \mathbb{P}(\sum_{j \geq 1} 2^{-j} V_j \leq x)$	all $x \in \mathbb{R}$	[repo]
up	$\mathbb{P}(Y \in dy) = \text{up}(y) \, dy, Y = \sum 2^{-j} U_j$	probability	[repo]
F, up	$F'(x) = 2 \text{up}(2x - 1)$	$0 < x < 1$	[repo]
up	$\text{up}(x) = \int_{2x-1}^{2x+1} \text{up}(t) \, dt$	pointwise	[repo]
up'	$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1)$	pointwise	[repo]
F	fixed integral equation (115)	$[0, 1]$	[repo]
$\mathcal{F}^{(m)}$	$2^{\binom{m+1}{2}} \mathcal{F}(2^m x)$	all orders	[repo]
up	$*_{j \geq 1} (2^{j-1} \mathbf{1}_{[-2^{-j}, 2^{-j}]})$	L^1 /probability	[repo]
u_N	finite box convolution (121)	compact spline	[repo]
u_N	Thue–Morse truncated powers (126)	pointwise	[repo]
up	$u_N * (2^N \text{up}(2^N \cdot))$	exact head–tail	[repo]
F	expectation/cube recursion (124)	exact	[repo]
Φ	$\prod_{h \geq 0} \text{sinc}(\pi z / 2^h)$	entire	[repo]
Φ	$\prod_{m \geq 1} (\cos(\pi z / 2^m))^m$	entire	[repo]
Φ	$\prod_{n \geq 1} (1 - z^2 / n^2)^{1+v_2(n)}$	entire	[repo]
Φ	$[\Gamma_{\text{dy}}(z) \Gamma_{\text{dy}}(-z)]^{-1}$	entire	[new]
Φ	$\det(I - z^2 D^{-2})$	Fredholm	[repo]
$\log \Phi$	zeta series (135)	$ z < 1$	[repo]
Φ' / Φ	arithmetic resolvent (136)	off zeros	[repo]
$\log \Phi$	Frullani–Lambert integral (143)	$ \Re z < 1$	[derived]
up	Fourier inversion (147)	all real x	[repo]
up	cosine integral (148)	all real x	[repo]
up	half-integer cosine series (151)	$[-1, 1]$	[repo]

Object	Representation	Domain	Status
Φ	Shannon sampling (153)	all $z \in \mathbb{C}$	[repo]
up	$\sum_{k \in \mathbb{Z}} \text{up}(x - k) = 1$	all x	[repo]
up	Legendre series (154)	$[-1, 1]$	[repo]
M_Y	$\prod_{j \geq 1} \sinh(t/2^j)/(t/2^j)$	entire	[repo]
M_X	$\prod_{j \geq 1} (e^{t/2^j} - 1)/(t/2^j)$	entire	[repo]
L_X	$\prod_{j \geq 1} (1 - e^{-s/2^j})/(s/2^j)$	entire in s	[repo]
F	Bromwich integral (158)	$c > 0$	[repo]
$\mathbb{E}X^{-\alpha}$	Mellin–Laplace integral (159)	$\Re \alpha > 0$	[derived]
$\mathbb{E}X^\alpha$	subtracted Mellin integral (160)	$0 < \Re \alpha < 1$	[derived]
$\mathbb{E}Y^{2n}$	Bernoulli cumulants and Bell polynomial (163)	exact	[repo]
C_r	Newton recurrence (168)	exact	[repo]
C_r	q -binomial/MZV partition sum (170)	$q = 1/4$	[repo]
F	density transform (231)	entire in ξ	[new]
F_0	compact transform (233)	$\xi \in \mathbb{R}$	[new]
F	tempered transform (236)	$\mathcal{S}'$	[new]
F	sine integral (237)	all real x	[new]
F	odd-harmonic sine series (239)	$[0, 1]$	[new]
Γ_{dy}	gamma product (182)	meromorphic	[new]
$1/\Gamma_{\text{dy}}$	arithmetic canonical product (188)	entire	[new]
Γ_{dy}	heat-trace integral (196)	$\Re z > -1$	[new]
V_τ	gamma convolution (202)	probability	[repo]
Z	dyadic logistic sum (172)	probability	[repo]
Z_τ	Gaussian variance mixture (205)	probability	[new]
Z_τ	gamma/sinc characteristic products (208)–(209)	real t	[new]
Z_τ	normal-mixture density (212)	real x	[new]
Z_τ	Levy density (217)	$x \neq 0$	[new]
$J(r)$	zero sum and factorial product (221)–(223)	$r > 0$	[new]
$J'(r)$	$(4 \lfloor r \rfloor - 2s_2(\lfloor r \rfloor))/r$	noninteger r	[new]
$J(r)$	$4r - (\log r)^2/(2 \log 2) + \mathcal{O}(\log r)$	$r \rightarrow \infty$	[new]
Q	quantile pushforward (242)	measurable tests	[repo]
Q	Laplace/product transform (243)	complex s	[derived]
Q	nonlinear Fourier transform (248)	$\eta \neq 0$	[new]
Q	sine series (250)	$[0, 1]$	[new]
F, Q	periodic lower-Lambert endpoint expansions	endpoint	[repo]

16 Numerical experiments and reproducibility

The package contains `numerical_experiments.py`, written to be run without network access. It uses double precision, a fixed seed 20260827, vectorized dyadic products, and explicit tail diagnostics. The program creates the figures in this section and writes a plain-text audit trail to `figures/numerical_results.txt`.

16.1 Dyadic gamma factorization

Three complex test points were evaluated with sixty product layers. The absolute errors in (184) were between 1.8×10^{-15} and 3.6×10^{-15} ; the Mahler functional equation (183) was verified at comparable accuracy. The finite-layer identity is exact mathematically, so the plateau in Fig. 3 is floating-point accumulation rather than truncation error.

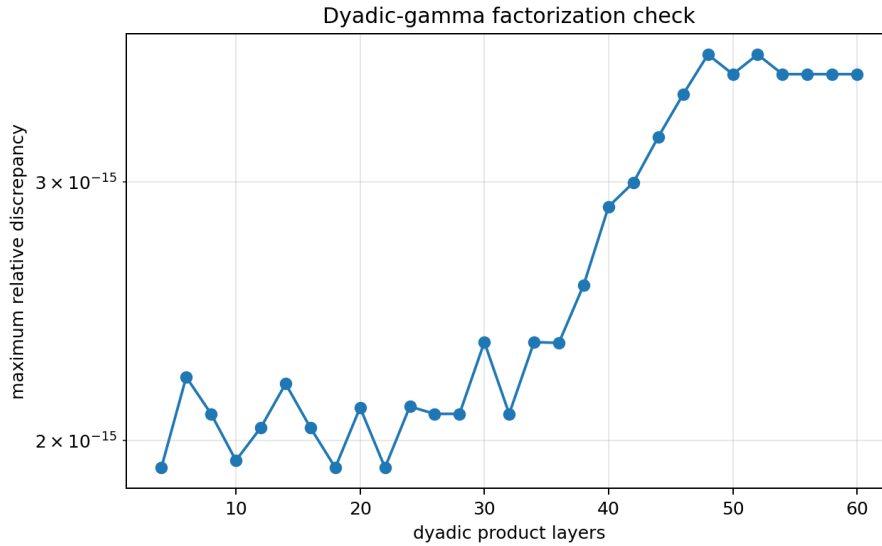


Figure 3: Finite-layer check of the dyadic-gamma reflection factorization.

16.2 Jensen mean

Angular trapezoidal quadrature with 8192 nodes was compared with the exact zero-divisor formula on 76 radii. The maximum absolute discrepancy was 7.11×10^{-15} . Representative values are

r	angular quadrature	exact formula	error
0.6	-3.6082×10^{-16}	0	-3.61×10^{-16}
1.4	0.6729444732424255	0.6729444732424258	-3.33×10^{-16}
3.7	5.496849257625430	5.496849257625430	0
10.3	26.15154490220755	26.15154490220754	3.55×10^{-15}

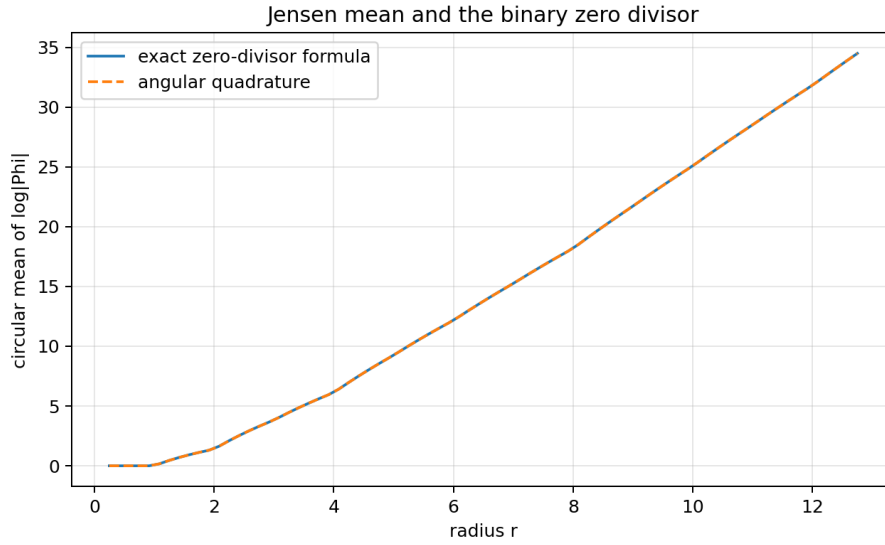


Figure 4: Jensen’s angular mean versus the exact factorial/zero-divisor expression.

16.3 Logistic series versus Gaussian mixture

The simulation used 100,000 samples. The logistic series retained sixteen dyadic layers; its omitted standard deviation was 5.09×10^{-6} . The gamma mixture retained the first 250 spectral sites and added the exact mean of the positive tail. The tail standard deviation was 2.07×10^{-4} , negligible compared with the exact standard deviation $1/3$ of Z_1 . The two-sample Kolmogorov–Smirnov statistic was 0.00347 with p-value 0.5825. Sample variances were 0.1114725 and 0.1110627, compared with the exact $1/9 = 0.111111 \dots$

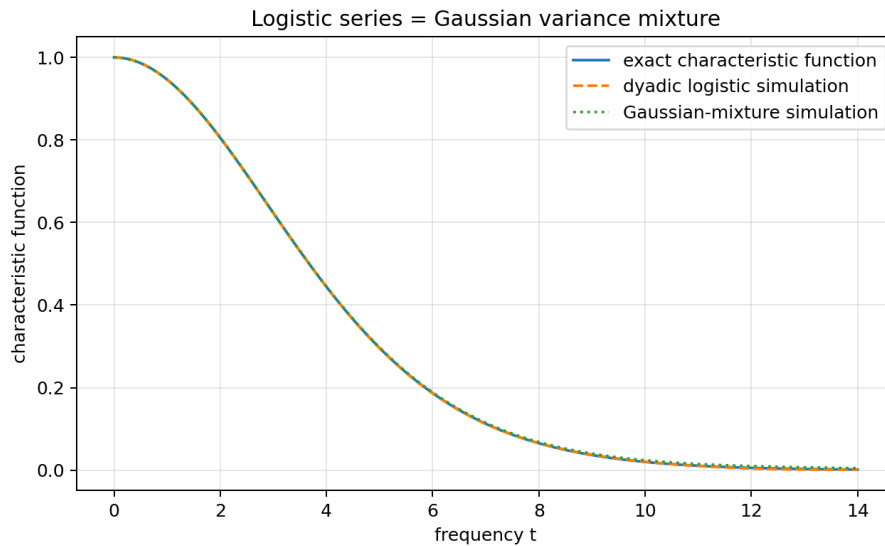


Figure 5: The exact characteristic function and two independent simulations of the law in (210).

16.4 Fabius sine series

The series (239) was compared with the independent Fourier integral (237). At $x = 0.25$, for example,

$$F(0.25) = 0.06944444445136150 \quad (\text{integral}), \quad 0.06944444444444434 \quad (128 \text{ harmonics}).$$

The discrepancy is governed by the numerical quadrature floor after roughly thirty-two harmonics. The rapid early convergence is visible in Fig. 6.

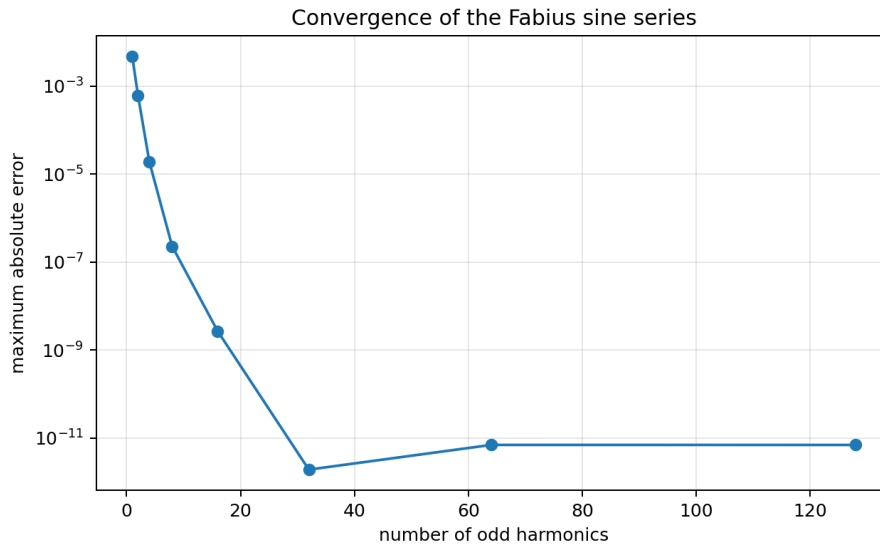


Figure 6: Maximum error of the odd-harmonic Fabius series against Fourier inversion on ten sample points.

16.5 Reproduction command

From the directory containing this consolidated report, run

```
python3 assets/fabius_rvachev_representation_frontier/numerical_experiments.py
```

The script creates no hidden cache and writes all outputs beneath `assets/fabius_rvachev_representation_frontier/figures/`. The source comments explain the product truncations, gamma-tail correction, and quadrature splitting at integer zeros.

17 Conjectures and testable frontier statements

Conjecture 17.1 (Differential transcendence of the Fourier determinant **[conjecture]**). The functions Φ , G_{dy} , and Γ_{dy} are differentially transcendental over $\mathbb{C}(z)$: none satisfies a nonzero algebraic differential equation with coefficients in $\mathbb{C}(z)$.

The non-D-finite theorem is evidence but does not approach the full differential-algebraic claim. A promising route is to combine the Mahler equation (183) with differential Galois theory for mixed differential–difference systems and the nonperiodic pole multiplicities.

Conjecture 17.2 (Full Delange expansion of the Jensen mean **[conjecture]**). There exist explicit continuous one-periodic functions P_1, P_0 such that

$$J(r) = 4r - \frac{(\log r)^2}{2 \log 2} + (\log r)P_1(\log_2 r) + P_0(\log_2 r) + o(1), \quad (251)$$

away from a controlled boundary layer around the integers. The functions should be computable from the Delange Fourier series for the binary digit sum and from the fractional part $r - \lfloor r \rfloor$.

Conjecture 17.3 (Asymptotics of inverse Fourier coefficients **[conjecture]**). Let c_n be defined by (249). Then there is a bounded, nonconstant, phase-dependent function P such that

$$\pi n c_n = Q(1/n) (P(\log_2 \lambda_n) + o(1)), \quad (252)$$

where λ_n is the exact lower-Lambert saddle phase for $F(1/n)$. In particular, $\log |n c_n|$ should have leading scale $-\sqrt{2 \log 2 \log n}$ with the same periodic modulation that governs the inverse endpoint.

This statement is testable by computing the oscillatory integral in (249) with the repository’s certified endpoint saddle machinery. It would turn the local inverse asymptotic into a global spectral law for Q .

Conjecture 17.4 (Hyperbolic complete monotonicity of the mixture density **[conjecture]**). For every $\tau > 0$, the function $s \mapsto p_\tau(\sqrt{s})$ is hyperbolically completely monotone, not merely completely monotone. Equivalently, the normal variance mixture may lie in the Bondesson class with a stronger multiplicative total-positivity structure inherited from the discrete Thorin measure.

Conjecture 17.5 (Arithmetic special values of the dyadic gamma function **[conjecture]**). At rational arguments with odd denominator, suitable products of $\Gamma_{\text{dy}}(r)$, powers of 2π , and classical Barnes multiple-gamma constants reduce to a finite family of dyadic L -values and logarithmic q -products. The functional equation (183) should produce explicit distribution relations indexed by the binary orbit of r .

Conjecture 17.6 (Optimal rational approximation from the Stieltjes resolvent **[conjecture]**). Diagonal Padé approximants obtained from $\Psi_{\text{dy}}(-z) - \Psi_{\text{dy}}(z)$ provide alternating, globally sharp rational bounds for $\Phi(iy)$ and for the characteristic function of Z_τ on $y > 0$, with error constants controlled explicitly by the next unused spectral atom.

18 Promising research directions

18.1 Formalization roadmap

The shortest Lean route to the new results is modular.

1. Define Γ_{dy} as a normally convergent product of meromorphic gamma factors and prove (183) and (184) by the existing infinite product API.
2. Reuse the formal zero-multiplicity theorem to prove non-D-finiteness in a small analytic-ODE module.
3. Build Z_τ by conditioning a standard Gaussian on the already developed V_τ , then identify characteristic functions.
4. Formalize Jensen’s formula for the canonical product, followed by the finite factorial regrouping. The digit-sum derivative then becomes arithmetic simplification.
5. Derive the compact and distributional Fourier transforms of F from the existing transform of up and a formal Heaviside distribution library.
6. Use the support-free quantile theorem and bounded-variation Fourier convergence to formalize (248) and (250).

18.2 Spectral geometry and complex analysis

The exact Jensen mean suggests treating Φ as an arithmetic spectral determinant in the complex plane, not only on the real axis. Natural next objects include logarithmic capacity, angular large deviations of $\log |\Phi|$, Nevanlinna characteristics refined by binary digit sums, and de Branges spaces generated by the reflected factors $G_{\text{dy}}(z)$ and $G_{\text{dy}}(-z)$.

18.3 Probability and semigroups

The family Z_τ gives a canonical subordinate-Brownian semigroup attached to the Rvachev zero divisor. Its transition density, fractional generator, entropy production, and small-time scaling should be expressible through the two heat kernels L_1 and K . The explicit Levy density (217) makes simulation, self-decomposability, and potential theory accessible without reconstructing the gamma series term by term.

18.4 Base- b and geometric- q families

The geometric gamma factorization (199) should be developed in parallel with the repository's base-generic Gaussian-binomial interpolation. For integer b , the collision multiplicity of gamma poles is a whole-base valuation; for nonintegral q^{-1} , the pole orbits are typically noncoincident. Comparing these regimes may separate which phenomena are genuinely dyadic from those caused only by geometric scaling.

18.5 Inverse spectral analysis

The coefficients (249) provide a concrete meeting point for global Fourier analysis and endpoint Lambert asymptotics. Certified steepest-descent evaluation of these coefficients could reveal periodic modulation, sign patterns, and optimal approximation rates for the inverse. A parallel study of $\int_0^1 e^{-2\pi i \eta F(x)} dx$ would produce a nonlinear Fourier theory of the CDF itself.

18.6 Computer-assisted certification

The existing Python checks are exploratory. A rigorous package should combine interval arithmetic for gamma products, exact rational moment recurrences, certified quadrature split at the integer zero set, and explicit tail bounds from the rapid Fourier decay. Such a tool could test the conjectural Jensen fluctuation and inverse-coefficient phases at hundreds of dyadic scales without confusing truncation error with genuine periodic oscillation.

19 Conclusion

The Fabius–Rvachev system is unusually rich because the same object can be read as a smooth compact bump, a CDF, an infinite uniform convolution, a Thue–Morse spline limit, an entire sinc determinant, an arithmetic spectrum, a q -series, and a Lambert-saddle problem. The source audit shows that many attractive formulae already coexist in the repository; treating them as separate discoveries would obscure rather than advance the subject.

The new results in this report add bridges rather than isolated identities. The dyadic gamma function splits the even Fourier determinant into reflected arithmetic gamma factors. The Gaussian-mixture theorem joins the logistic/Appell dual to the Thorin/gamma dual and exposes a new Levy semigroup. Jensen's formula turns binary digit sums into a radial complex-analytic observable. Unbounded

multiplicities settle non-holonomicity in one stroke. Finally, the bounded CDF and its inverse acquire transform formulae that are global, explicit, and compatible with the existing endpoint theory. Each bridge supplies both a theorem and a research program, with a comparatively short path to formal verification.

A Audited living document families

The active draft root at the audited snapshot contained the following document families. Their exact names are retained to make future collision searches reproducible.

Draft family	Principal audited themes
Fabius_Arithmetic_Rays_Frontier_Report	arithmetic rays, exact values, spectral sampling
Fabius_Dyadic_Self_Sampling_Frontier_Package	dyadic self-sampling and reconstruction
Fabius_Half_Integer_Spectral_Frontier_Report	half-integer cosine spectrum, Prouhet identities, alias transforms
Fabius_Inverse_Frontier_Report_Source_and_PDF	inverse germs, Catalan and Lambert–Bell asymptotics
Fabius_Newton_Rvachev_Frontier_Report	Newton recurrences, exponent sequences, exact FFT checks
Fabius_Rvachev_Frontier_Report-2	arithmetic spectrum, zero multiplicities, heat trace, lobe signs
Fabius_Rvachev_Frontier_Report	tail quadrature, exact dyadic error laws, q -Richardson acceleration
Fabius_Rvachev_New_Frontiers	q -MZV moments, Thorin dual, complete Bernstein theory, theta completion
Fabius_Rvachev_Thue_Morse_Frontier_Results	finite blocks, deconvolution, zeta–Lambert tails
Spectral_Arithmetic_Pascal_Rvachev_Hierarchy	spectral arithmetic and Pascal-type hierarchies
Thue_Morse_Formula_Atlas	broad Thue–Morse product, series, automata, and arithmetic atlas
fabius_frontier_dyadic_inverse_barnes_report	dyadic inverse, Barnes prefixes, Appell and logistic dual
fabius_frontier_new_results	logarithmic q -Richardson and Lambert phase locking
fabius_frontier_results	q -Richardson sinc products and acceleration
fabius_frontier_results_bundle	Fredholm determinant, spectral zeta, zero jets, q -Euler determinant
fabius_frontier_spectral_endpoint_report_bundle	endpoint spectral products, Mellin and saddle analysis
rvachev_up_fourier_decay	successive Fourier-decay audits and certified bounds

The source audit also included the primary formal article, the consolidated non-formalized-research-frontiers.tex, glossary, Lean walkthrough, non-elementarity document, embedded papers, and the archive provenance map. The archive map was essential for identifying superseded or byte-identical historical variants without treating them as independent sources.

B Numerical package contents

The reproducibility assets retained for this part live under assets/fabius_rvachev_representation_frontier/:

- `numerical_experiments.py`, with detailed comments and deterministic seed;
- `figures/numerical_results.txt`;
- the four PNG figures embedded in the report;
- `SOURCE_AUDIT.md`, a concise audit and novelty-collision record;
- `README.md`, reproduction instructions and file descriptions.

C Selected derivations in one line

For convenient reuse, here are the central bridges in compressed form:

$$\begin{aligned}
 \Gamma(1+w)\Gamma(1-w) &= 1/\operatorname{sinc}(\pi w) & \implies & \Phi(z) = 1/[\Gamma_{\mathrm{dy}}(z)\Gamma_{\mathrm{dy}}(-z)], \\
 \mathbb{E}e^{-wV_\tau} &= \mathcal{A}(w)^{-\tau}, \quad \mathbb{E}(e^{it\sqrt{2T}N} \mid T) = e^{-t^2T} & \implies & Z_1 \stackrel{\mathrm{d}}{=} \sqrt{2T_1}N, \\
 \operatorname{zeros}(\Phi) &= \{\pm n \text{ with multiplicity } a_n\} & \implies & J(r) = 2 \sum_{n \leq r} a_n \log(r/n), \\
 a_{2^k} &= k+1 \rightarrow \infty & \implies & \Phi \text{ is not D-finite}, \\
 F'(x) &= 2 \operatorname{up}(2x-1), \quad \widehat{\operatorname{up}} = \Phi & \implies & \widehat{F'}(\xi) = e^{-\pi i \xi} \Phi(\xi/2), \\
 y &= F(x), \quad Q(F(x)) = x & \implies & \widehat{Q}(\eta) = \frac{\int_0^1 e^{-2\pi i \eta F(x)} dx - e^{-2\pi i \eta}}{2\pi i \eta}.
 \end{aligned}$$

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Part III

Dyadic Multiresolution and Product-Series Representations in the Fabius–Rvachev System

Abstract

The existing ProveIt corpus develops the Fabius function, its inverse, Rvachev’s compactly supported up-function, the Thue–Morse sequence, exact dyadic values, q -binomial formulas, infinite sinc products, Bell and Bernoulli polynomial machinery, and phase-resolved Lambert– W endpoint asymptotics in considerable depth. This report first organizes those results into a representation atlas and then develops a complementary multiresolution layer that was not located in the audited LaTeX corpus.

Let $p = F'$ and let $\Delta_{N,k} = F((k+1)2^{-N}) - F(k2^{-N})$ be the exact dyadic cell masses. The same rational projective array produces four different finite models. It produces a fixed-cell uniform mixture for p and up, an exact rational unnormalized Haar expansion of p , a rational Faber–Schauder expansion of F , and the finite Walsh–Hadamard transform of p . Their Fourier images yield locally uniform limits of rational exponential polynomials multiplied by sinc windows. The Walsh transform of every basis function is itself a finite product of sines and cosines. Consequently the infinite sinc product admits a new series whose summands are finite trigonometric products and whose coefficients are exact Walsh transforms of dyadic Fabius masses.

A Walsh mode indexed by m has exactly $\text{wt}_2(m)$ vanishing algebraic moments. Writing $\sigma(m) = \sum_j j b_j$ for the positions of the nonzero binary digits, its first nonzero moment is

$$\int_0^1 x^{\text{wt}_2(m)} w_m(x) dx = (-1)^{\text{wt}_2(m)} \text{wt}_2(m)! 2^{-2 \text{wt}_2(m) - \sigma(m)}.$$

Summing over a fixed binary weight produces the Gaussian binomial coefficient $\begin{bmatrix} N \\ r \end{bmatrix}_{1/2}$. Higher Walsh moments are encoded by a finite Bell-polynomial expression whose logarithmic coefficients are explicit Bernoulli-number sums. This gives a direct bridge among Walsh cancellation, Thue–Morse signs, q -binomial coefficients, and the Bell–Bernoulli structure of the Fabius moments.

A second positive approximation is obtained by differentiating the Bernstein polynomial of F : p is the uniform limit of finite mixtures of beta densities with rational weights $\Delta_{N,k}$. Its moment-generating and Fourier transforms are therefore limits of positive rational mixtures of confluent hypergeometric functions. A dual construction uses dyadic values of the inverse $G = F^{-1}$: equal probability cells with variable spatial widths give a finite sinc-mixture formula for $\widehat{\text{up}}$, as well as polynomial limits for all moments.

Finally, the exact Haar detail energy E_N is rational and has an all-orders continuum expansion

$$E_N \sim \frac{h^2}{16} \|p'\|_2^2 - \frac{h^4}{384} \|p''\|_2^2 + \frac{h^6}{20480} \|p'''\|_2^2 - \cdots, \quad h = 2^{-N}.$$

The complete coefficient sequence is generated by sinc^4 , hence by complete Bell polynomials in Bernoulli numbers. The beyond-all-orders aliasing remainder provides a new place where the known log-periodic Fourier decay should reappear. The report distinguishes repository input, results proved here, numerical checks, and conjectures; “new” always means new relative to the audited repository corpus, not an assertion of worldwide priority.

1 Audit protocol, scope, and novelty boundary

1.1 Pinned corpus and cumulative audit

The documentation tree

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>

was audited recursively. The Git tree contained 89 files with extension `.tex` at the source-level inventory used in the audit. During this investigation the branch advanced; the report is pinned to commit `924652f4a9a656fd957d2936b2f053747c6f6c47`. The five-commit delta from the preceding pin modified the primary exposition, its coverage ledger, and an integral-transform report, but added no new TeX path. The delta was checked before the novelty boundary below was frozen.

The audit is cumulative rather than a naive count of duplicated revisions. Earlier source-level passes covered the complete path inventory; the present pass reread the canonical primary exposition, current audit ledgers, active frontier syntheses, and every recent report capable of overlapping the proposed results. Historical copies under `archive/` were normalized against their live successors. Imported papers were used as literature baselines, not counted as independent repository discoveries.

What the current formal source already contains

The current primary exposition and Lean modules include the probability law of weighted uniform series, finite atomic and spline approximants, exact dyadic values, quantile transport, sinc and cosine products, Fourier inversion and Poisson summation, moment recurrences, a Fourier–Legendre expansion, normalized integer Volterra calculus, inverse-Fabius infrastructure, and the proved Lambert-phase endpoint machinery. These are inputs below. Repackaging one of those statements with different notation is not counted as a new result.

1.2 Explicit exclusion map

The following recent manuscripts were treated as an exclusion map, not as a source of claims to relabel.

Already developed in the corpus	Excluded from novelty claims here
Primary Fabius–Rvachev synthesis	Function triad, probability construction, refinement laws, exact dyadic arithmetic, sinc product, Fourier inversion, moments, Fourier–Legendre series, and formal Lambert asymptotics.
Integral and transform reports	Weighted antiderivatives, inverse-pair area identities, Mellin/Newton continuation, negative moments, fractional primitives, order statistics, Stieltjes/Cauchy transforms, autocorrelation, and Gini-type integrals.
Spectral determinant and Barnes reports	Zero multiplicities, spectral zeta, Fredholm determinants, heat traces, Barnes deconvolution, reciprocal-sinc poles, and coefficient-threshold asymptotics.
Confluent digital extrapolation report	Reciprocal-moment Appell systems, Strang–Fix/cardinal reproduction, q -binomial scale closure, confluent Richardson filters, and Lambert-phase tomography.

Already developed in the corpus	Excluded from novelty claims here
Gamma-duality/total-positivity report	Laguerre–Pólya and PF_∞ structure, generalized gamma-convolution duality, Thorin measure, theta completion, mod-Gamma endpoint tilting, and large-order transfer.
Fourier-decay campaign	Exact zero orders, Thue–Morse lobe signs, sharp global envelopes, log-periodic fluctuations, and phase-sensitive real-axis decay.

Targeted searches were then made for the exact structures used below: Faber–Schauder expansions of F or G , Haar expansions from dyadic Fabius masses, Walsh–Hadamard expansions of p , finite-product Fourier transforms of Walsh modes, Bernstein beta-mixtures, and the sinc^4 Haar-energy expansion. No corresponding package was located in the audited TeX corpus. General Haar, Walsh, Schauder, and Bernstein theory is classical; the frontier content is the exact specialization and synthesis for the Fabius–Rvachev system.

1.3 Status ledger

ID	Statement	Status
A	Projective rational dyadic mass array and fixed-cell uniform-mixture representations, including locally uniform entire transform limits.	proved here
B	Rational unnormalized Haar and Faber–Schauder representations, with an orthonormal Haar/Fourier product expansion.	proved here
C	Walsh–Hadamard representation, reflection selection rule, and a series for $\widehat{\text{up}}$ whose terms are finite trigonometric products.	proved here
D	Exact Hamming-weight cancellation order, first nonzero Walsh moment, and $q = 1/2$ Gaussian-binomial aggregation.	proved here
E	Bell–Bernoulli formula for every higher Walsh moment.	proved here
F	Positive beta-density mixture and hypergeometric transform limits.	proved here
G	Equal-weight inverse-quantile chord mixtures and their finite sinc-product transform formula.	proved here
H	All-orders Haar-energy expansion generated by sinc^4 , with a superalgebraically small alias remainder.	proved here
I	Lambert-phase asymptotics for the boundary Schauder coefficients of $G = F^{-1}$ and boundary domination of the global interpolation error.	conjectural
J	A log-periodic transseries for the beyond-all-orders Haar alias remainder.	conjectural

2 Normalization and the function triad

2.1 Bounded Fabius function, density, and Rvachev function

Throughout, $F : \mathbb{R} \rightarrow [0, 1]$ is the bounded Fabius distribution function,

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1),$$

and up is Rvachev's even C^∞ function supported on $[-1, 1]$. On their common fundamental intervals,

$$F(x) = \text{up}(x - 1), \quad 0 \leq x \leq 1, \quad (253)$$

$$F'(x) = 2 \text{up}(2x - 1) =: p(x), \quad 0 < x < 1, \quad (254)$$

$$F(x) + F(1 - x) = 1, \quad 0 \leq x \leq 1, \quad (255)$$

$$\text{up}(x) = 1 - F(x) = F(1 - x), \quad 0 \leq x \leq 1. \quad (256)$$

The density p , extended by zero outside $[0, 1]$, is a compactly supported C^∞ function satisfying

$$p(x) = p(1 - x), \quad \int_0^1 p(x) \, dx = 1. \quad (257)$$

The ordinary inverse of $F|_{[0,1]}$ is denoted

$$G = F^{-1} : [0, 1] \longrightarrow [0, 1], \quad G(1 - u) = 1 - G(u). \quad (258)$$

The signed global extension $\mathcal{F}$, important in the repository's repeated-integration theory, is not needed for the multiresolution proofs; it appears only in the baseline atlas.

2.2 Fourier convention

For an integrable function f we use angular frequency,

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-itx} \, dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(t) e^{itx} \, dt. \quad (259)$$

Set

$$\Phi(t) = \widehat{\text{up}}(t), \quad M(z) = \int_{-1}^1 \text{up}(x) e^{zx} \, dx = \Phi(iz). \quad (260)$$

The compact-interval Fourier image of the bounded Fabius function is

$$\mathcal{F}_F(t) = \int_0^1 F(x) e^{-itx} \, dx. \quad (261)$$

This notation avoids the meaningless full-line Fourier transform of a function that equals one on $[1, \infty)$.

2.3 Probabilistic model

Let $U_1, U_2, \dots$ be independent uniform random variables on $[-1, 1]$ and let $V_j = (U_j + 1)/2$, uniform on $[0, 1]$. Then

$$Y = \sum_{j \geq 1} 2^{-j} U_j, \quad X = \frac{Y + 1}{2} = \sum_{j \geq 1} 2^{-j} V_j. \quad (262)$$

The law of Y has density up, and F is the distribution function of X . Consequently

$$X \stackrel{d}{=} G(U), \quad Y \stackrel{d}{=} 2G(U) - 1, \quad U \sim \text{Unif}[0, 1]. \quad (263)$$

3 A representation atlas from the existing theory

This section collects a broad common baseline. Most formulas are already present, sometimes in stronger form, in the audited repository. They are included so that the genuinely new representations later can be compared against one normalization without forcing the reader to move among several manuscripts.

3.1 Infinite convolution, cube integrals, and finite heads

Let ν_a denote the uniform probability density on $[-a, a]$. Independence in (262) gives the infinite convolution

$$\text{up} = \nu_{1/2} * \nu_{1/4} * \nu_{1/8} * \cdots. \quad (264)$$

In distributional notation this is the infinite-cube integral

$$\text{up}(x) = \int_{[-1,1]^{\mathbb{N}}} \delta\left(x - \sum_{j \geq 1} 2^{-j} u_j\right) \prod_{j \geq 1} \frac{du_j}{2}, \quad (265)$$

and

$$F(x) = \int_{[0,1]^{\mathbb{N}}} \mathbf{1}_{\{\sum_{j \geq 1} 2^{-j} v_j \leq x\}} \prod_{j \geq 1} dv_j. \quad (266)$$

For the finite head

$$Y_N = \sum_{j=1}^N 2^{-j} U_j, \quad A_N = \sum_{j=1}^N 2^{-j} = 1 - 2^{-N},$$

write B_N for its density. Inclusion–exclusion for a box spline gives

$$B_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{\varepsilon \in \{0,1\}^N} (-1)^{|\varepsilon|} \left(x + A_N - 2 \sum_{j=1}^N \varepsilon_j 2^{-j} \right)_+^{N-1}. \quad (267)$$

The signs $(-1)^{|\varepsilon|}$ are the finite Thue–Morse signs. Likewise, for $X_N = \sum_{j=1}^N 2^{-j} V_j$,

$$\mathbb{P}(X_N \leq x) = \frac{2^{N(N+1)/2}}{N!} \sum_{\varepsilon \in \{0,1\}^N} (-1)^{|\varepsilon|} \left(x - \sum_{j=1}^N \varepsilon_j 2^{-j} \right)_+^N. \quad (268)$$

Taking $N \rightarrow \infty$ gives $B_N \rightarrow \text{up}$ and the finite CDFs tend to F .

The tail is an independent scaled copy:

$$X \stackrel{d}{=} X_N + 2^{-N} X', \quad Y \stackrel{d}{=} Y_N + 2^{-N} Y'. \quad (269)$$

Hence the exact finite-scale identities

$$F(x) = \mathbb{E} F(2^N(x - X_N)), \quad (270)$$

$$\text{up}(x) = 2^N \int_{\mathbb{R}} B_N(y) \text{up}(2^N(x - y)) \, dy. \quad (271)$$

These formulas combine a finite Thue–Morse spline with a shrunken copy of the same limiting function.

3.2 Sinc, hyperbolic-sinc, and renormalization products

The Fourier and moment-generating products are

$$\Phi(t) = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{t}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z}, \quad (272)$$

$$M(z) = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j}. \quad (273)$$

They converge locally uniformly on $\mathbb{C}$, and

$$\Phi(2t) = \operatorname{sinc}(t)\Phi(t), \quad M(2z) = \frac{\sinh z}{z}M(z). \quad (274)$$

Iteration at any finite depth gives

$$\Phi(t) = \Phi(t/2^N) \prod_{j=1}^N \operatorname{sinc}(t/2^j), \quad (275)$$

which is the transform counterpart of (271).

Euler's product for sine yields two canonical products:

$$\Phi(t) = \prod_{m=1}^{\infty} \left(\frac{t^2}{4\pi^2 m^2}; \frac{1}{4} \right)_{\infty}, \quad (276)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{t^2}{4\pi^2 n^2} \right)^{1+v_2(n)}. \quad (277)$$

Thus $2\pi n$ is a zero of multiplicity $1 + v_2(n)$. For $|t| < 2\pi$,

$$\log \Phi(t) = - \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r\pi^{2r}(4^r - 1)} t^{2r}. \quad (278)$$

The Wick-rotated determinant

$$\mathcal{A}(s) = \Phi(i\sqrt{s}) = \prod_{n=1}^{\infty} \left(1 + \frac{s}{4\pi^2 n^2} \right)^{1+v_2(n)} \quad (279)$$

underlies the spectral-zeta, total-positivity, and generalized-gamma-convolution results already developed in the corpus.

3.3 Moments, cumulants, Bell polynomials, and Bernoulli numbers

Write

$$\mu_n = \int_{-1}^1 x^n \operatorname{up}(x) \, dx = \mathbb{E}Y^n. \quad (280)$$

Odd moments vanish. The even cumulants are

$$\kappa_{2r}(Y) = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Y) = 0. \quad (281)$$

Therefore

$$\mu_n = Y_n(0, \kappa_2, 0, \kappa_4, \dots, \kappa_n), \quad (282)$$

where Y_n is the complete exponential Bell polynomial, and

$$\Phi(t) = \sum_{r=0}^{\infty} \frac{(-1)^r \mu_{2r}}{(2r)!} t^{2r}. \quad (283)$$

For $m_n = \mathbb{E}X^n$ the smoothing equation $X \stackrel{d}{=} (V + X')/2$ gives

$$(2^n - 1)m_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{m_k}{n - k + 1}, \quad m_0 = 1. \quad (284)$$

All m_n are rational, and

$$\mu_{2r} = \frac{2m_{2r+1}}{2r + 1}. \quad (285)$$

3.4 Fourier inversion, twisted Poisson summation, and cosine series

Since Φ is rapidly decreasing on the real axis,

$$\text{up}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(t) e^{itx} dt = \frac{1}{\pi} \int_0^\infty \Phi(t) \cos(tx) dt. \quad (286)$$

The distribution function of the centered law gives, for $0 \leq x \leq 1$,

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Phi(t) \frac{\sin((2x-1)t)}{t} dt. \quad (287)$$

Combining this with $F(x) = \text{up}(x-1)$ produces the integral identity

$$\int_0^\infty \Phi(t) \cos((1-x)t) dt = \frac{\pi}{2} + \int_0^\infty \Phi(t) \frac{\sin((2x-1)t)}{t} dt. \quad (288)$$

The partition of unity $\sum_{k \in \mathbb{Z}} \text{up}(x-k) = 1$ kills all nonzero integer Fourier modes. Twisting by $(-1)^k$ leaves the half-integer modes:

$$\sum_{k \in \mathbb{Z}} (-1)^k \text{up}(x-k) = \sum_{m \in \mathbb{Z}} \Phi((2m+1)\pi) e^{i(2m+1)\pi x}. \quad (289)$$

For $x \in [0, 1]$ the left side is $1 - 2F(x)$. With

$$A_m = \Phi((2m+1)\pi), \quad (290)$$

we obtain the rapidly convergent series

$$\text{up}(x) = \frac{1}{2} + \sum_{m=0}^{\infty} A_m \cos((2m+1)\pi x), \quad |x| \leq 1, \quad (291)$$

$$F(x) = \frac{1}{2} - \sum_{m=0}^{\infty} A_m \cos((2m+1)\pi x), \quad 0 \leq x \leq 1, \quad (292)$$

$$p(x) = \sum_{m=0}^{\infty} A_m (2m+1)\pi \sin((2m+1)\pi x). \quad (293)$$

Every derivative may be taken termwise.

3.5 Fourier images of p , F , and $1 - F$

The affine relation between p and up gives

$$\widehat{p}(t) = e^{-it/2} \Phi(t/2). \quad (294)$$

Integration by parts then yields the exact compact-interval transforms

$$\mathcal{F}_F(t) = \frac{e^{-it/2} \Phi(t/2) - e^{-it}}{it}, \quad (295)$$

$$\int_0^1 (1 - F(x)) e^{-itx} dx = \frac{1 - e^{-it/2} \Phi(t/2)}{it}. \quad (296)$$

The singularities at $t = 0$ are removable. After Wick rotation,

$$\int_0^1 e^{zx} F(x) dx = \frac{e^z - e^{z/2} M(z/2)}{z}, \quad (297)$$

$$\int_0^1 e^{zx} (1 - F(x)) dx = \frac{e^{z/2} M(z/2) - 1}{z}. \quad (298)$$

3.6 Quantile, Stieltjes, and correlation representations

Quantile transport (263) gives

$$\Phi(t) = \int_0^1 \exp(-it(2G(u) - 1)) \, du, \quad (299)$$

$$M(z) = \int_0^1 \exp(z(2G(u) - 1)) \, du, \quad (300)$$

$$\mu_n = \int_0^1 (2G(u) - 1)^n \, du. \quad (301)$$

For $z \notin [-1, 1]$, the Cauchy transform is

$$\mathcal{S}(z) = \int_{-1}^1 \frac{\text{up}(x)}{z - x} \, dx = \int_0^1 \frac{du}{z - (2G(u) - 1)}. \quad (302)$$

If $\Im z > 0$,

$$\mathcal{S}(z) = -i \int_0^\infty e^{itz} \Phi(t) \, dt. \quad (303)$$

The autocorrelation and convolution powers are

$$\int_{\mathbb{R}} \text{up}(x) \text{up}(x + y) \, dx = \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(t)^2 e^{ity} \, dt, \quad (304)$$

$$\text{up}^{*r}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(t)^r e^{itx} \, dt, \quad r \in \mathbb{N}. \quad (305)$$

3.7 Legendre and Chebyshev coefficient integrals

The repository contains a Fourier–Legendre expansion. Fourier inversion gives a useful transform-side form of its coefficients. If

$$\text{up}(x) = \sum_{n \geq 0} \lambda_n P_n(x), \quad \lambda_n = \frac{2n+1}{2} \int_{-1}^1 \text{up}(x) P_n(x) \, dx, \quad (306)$$

then odd coefficients vanish and, with j_n the spherical Bessel function,

$$\lambda_{2r} = \frac{4r+1}{\pi} (-1)^r \int_0^\infty \Phi(t) j_{2r}(t) \, dt. \quad (307)$$

Indeed, $\int_{-1}^1 e^{itx} P_n(x) \, dx = 2i^n j_n(t)$.

A parallel Chebyshev representation, apparently not stated in the audited corpus, is

$$\text{up}(x) = \frac{a_0}{2} + \sum_{n \geq 1} a_n T_n(x), \quad a_n = \frac{2}{\pi} \int_0^\pi \text{up}(\cos \theta) \cos(n\theta) \, d\theta. \quad (308)$$

The odd coefficients vanish and

$$a_{2r} = \frac{2(-1)^r}{\pi} \int_0^\infty \Phi(t) J_{2r}(t) \, dt. \quad (309)$$

The function $\theta \mapsto \text{up}(\cos \theta)$ is smooth and periodic, so the Chebyshev coefficients decrease faster than every power.

4 The projective dyadic mass array

4.1 Definition and exact algebra

Put

$$n_N = 2^N, \quad h_N = 2^{-N}, \quad I_{N,k} = [kh_N, (k+1)h_N), \quad (310)$$

with the last interval closed at 1, and define

$$\Delta_{N,k} = F((k+1)h_N) - F(kh_N), \quad 0 \leq k < n_N. \quad (311)$$

These are the probabilities that X lies in the dyadic cells.

Theorem 4.1 (Projective rational mass array). *For every $N \geq 0$, put*

$$\Delta_N = (\Delta_{N,0}, \dots, \Delta_{N,n_N-1}).$$

This vector has the following properties:

$$\Delta_{N,k} \in \mathbb{Q}_{>0}, \quad (312)$$

$$\sum_{k=0}^{n_N-1} \Delta_{N,k} = 1, \quad (313)$$

$$\Delta_{N,n_N-1-k} = \Delta_{N,k}, \quad (314)$$

$$\Delta_{N,k} = \Delta_{N+1,2k} + \Delta_{N+1,2k+1}. \quad (315)$$

Consequently the mass polynomial

$$D_N(z) = \sum_{k=0}^{n_N-1} \Delta_{N,k} z^k \quad (316)$$

has rational coefficients, satisfies $D_N(1) = 1$, and is palindromic:

$$D_N(z) = z^{n_N-1} D_N(z^{-1}). \quad (317)$$

Proof. Rationality is the exact dyadic-value theorem from the repository. Positivity comes from strict increase of F on $(0, 1)$. The sum telescopes. Reflection (314) follows from $F(1-x) = 1 - F(x)$, and (315) is additivity after bisecting a cell. The polynomial statements are immediate. $\square$

The array is therefore a rational projective system of probability vectors. It is useful to regard it as the common discrete skeleton behind all subsequent representations. No approximation has yet been made.

4.2 Fixed-cell uniform mixtures

Define the dyadic conditional-expectation density

$$p_N(x) = n_N \Delta_{N,k}, \quad x \in I_{N,k}. \quad (318)$$

Equivalently,

$$p_N(x) dx = \sum_{k=0}^{n_N-1} \Delta_{N,k} \text{Unif}(I_{N,k}). \quad (319)$$

This is a finite mixture of equal-width uniform laws with variable rational weights. On the centered up scale, put

$$u_{N,k} = \frac{2k+1}{n_N} - 1. \quad (320)$$

The corresponding approximation is

$$u_{p_N}(u) = \frac{1}{2} p_N \left(\frac{u+1}{2} \right) = \frac{n_N}{2} \Delta_{N,k}, \quad |u - u_{N,k}| < \frac{1}{n_N}. \quad (321)$$

Theorem 4.2 (Martingale and uniform convergence). *Let $\mathcal{D}_N$ be the dyadic sigma-algebra on $[0, 1]$. Then*

$$p_N = \mathbb{E}[p \mid \mathcal{D}_N]. \quad (322)$$

Moreover,

$$\|p_N - p\|_\infty \longrightarrow 0, \quad \|p_N - p\|_{L^q(0,1)} \longrightarrow 0 \quad (0 < q < \infty), \quad (323)$$

and $u_{p_N} \rightarrow u_p$ in the corresponding centered norms. If $p \in C^1$, then

$$\|p_N - p\|_\infty \leq h_N \|p'\|_\infty. \quad (324)$$

Proof. The average of p on $I_{N,k}$ is $h_N^{-1} \int_{I_{N,k}} p = n_N \Delta_{N,k}$, proving (322). Uniform continuity gives uniform convergence of cell averages; the displayed bound follows from the mean-value theorem. Finite-measure L^q convergence follows from uniform convergence. $\square$

Remark 4.3. Although p_N is only piecewise constant, all its coefficients are exact. It is not the finite-head spline B_N of (267): B_N is obtained by truncating the random series, whereas p_N is obtained by conditioning the limiting density on a spatial dyadic partition. The two approximations therefore probe different structures.

4.3 Rational exponential-polynomial limits for the transforms

The transform of (319) can be written directly in terms of the mass polynomial.

Theorem 4.4 (Primal dyadic transform representations). *For every $z \in \mathbb{C}$,*

$$\widehat{p}_N(z) = \frac{n_N(1 - e^{-iz/n_N})}{iz} D_N(e^{-iz/n_N}). \quad (325)$$

Consequently

$$\Phi_N^{\text{cell}}(t) := e^{it} \widehat{p}_N(2t) \quad (326)$$

$$= \text{sinc}(t/n_N) \sum_{k=0}^{n_N-1} \Delta_{N,k} \exp \left[-it \left(\frac{2k+1}{n_N} - 1 \right) \right]. \quad (327)$$

The limits

$$\widehat{p}(z) = \lim_{N \rightarrow \infty} \frac{n_N(1 - e^{-iz/n_N})}{iz} D_N(e^{-iz/n_N}), \quad (328)$$

$$\Phi(t) = \lim_{N \rightarrow \infty} \text{sinc}(t/n_N) \sum_{k=0}^{n_N-1} \Delta_{N,k} e^{-itu_{N,k}} \quad (329)$$

are locally uniform on $\mathbb{C}$. Thus the infinite sinc product is a locally uniform limit of rational exponential polynomials times one elementary sinc window.

Proof. Integrating p_N cell by cell gives

$$\int_{kh_N}^{(k+1)h_N} n_N \Delta_{N,k} e^{-izx} dx = \Delta_{N,k} \frac{n_N(1 - e^{-iz/n_N})}{iz} e^{-izk/n_N},$$

which sums to (325). The affine change from p to up gives (327). If $K \subset \mathbb{C}$ is compact, then

$$\sup_{z \in K} |\widehat{p}_N(z) - \widehat{p}(z)| \leq e^{\sup_{z \in K} |\Im z|} \|p_N - p\|_1,$$

so convergence is locally uniform. The same argument applies after the affine change of variables. $\square$

Deleting the sinc window gives the midpoint-atomic representation

$$\Phi(t) = \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} e^{-itu_{N,k}}. \quad (330)$$

The cell version is usually more accurate because it integrates the test function across each cell rather than concentrating the mass at its midpoint.

After Wick rotation one obtains the moment-generating limit

$$M(z) = \lim_{N \rightarrow \infty} \frac{\sinh(z/n_N)}{z/n_N} \sum_{k=0}^{n_N-1} \Delta_{N,k} \exp \left[z \left(\frac{2k+1}{n_N} - 1 \right) \right]. \quad (331)$$

Equations (329) and (331) are the same finite mixture on the imaginary and real axes.

4.4 A second-order quadrature principle

The conditional-expectation property gives better weak accuracy than the pointwise bound (324) suggests.

Proposition 4.5 (Cell-average quadrature). *If $p, g \in C^1[0, 1]$, then*

$$\left| \int_0^1 g(x)(p(x) - p_N(x)) dx \right| \leq h_N^2 \|g'\|_\infty \|p'\|_\infty. \quad (332)$$

In particular, on each compact set $|z| \leq R$,

$$\widehat{p}_N(z) - \widehat{p}(z) = \mathcal{O}_R(4^{-N}), \quad \Phi_N^{\text{cell}}(z) - \Phi(z) = \mathcal{O}_R(4^{-N}). \quad (333)$$

Proof. On a cell I choose any $x_I \in I$. Since $\int_I (p - p_N) = 0$,

$$\int_I g(p - p_N) = \int_I (g - g(x_I))(p - p_N).$$

Now $\text{osc}_I g \leq h_N \|g'\|_\infty$ and $\int_I |p - p_N| \leq h_N \text{osc}_I p \leq h_N^2 \|p'\|_\infty$. Summing $n_N = h_N^{-1}$ cells proves (332). For exponential test functions, their derivatives are uniformly bounded on compact subsets of $\mathbb{C}$. $\square$

The 4^{-N} behavior is visible in the numerical experiment of Appendix 13. The atomic midpoint formula is also second order for a smooth density and smooth test function, but the sinc-window formula has a clearer probabilistic interpretation: it is exactly the transform of the conditional-expectation density.

4.5 Moments, L^q norms, and discrete entropies

For $q > 0$, uniform convergence gives

$$\int_0^1 p(x)^q dx = \lim_{N \rightarrow \infty} n_N^{q-1} \sum_{k=0}^{n_N-1} \Delta_{N,k}^q. \quad (334)$$

For $q = 2$ this is an exact rational approximation to the collision integral,

$$\int_{-1}^1 \text{up}(x)^2 dx = \frac{1}{2} \lim_{N \rightarrow \infty} n_N \sum_{k=0}^{n_N-1} \Delta_{N,k}^2. \quad (335)$$

Differentiating (334) at $q = 1$ gives the Shannon asymptotic

$$-\sum_{k=0}^{n_N-1} \Delta_{N,k} \log \Delta_{N,k} = \log n_N - \int_0^1 p(x) \log p(x) dx + o(1). \quad (336)$$

Here $p \log p$ extends continuously by zero at the endpoints. Thus the continuous entropy of the Fabius density is the constant term in the entropy of its exact rational dyadic mass vectors.

For ordinary moments, integrating a monomial over each uniform cell gives

$$m_r = \int_0^1 x^r p(x) dx = \lim_{N \rightarrow \infty} \frac{1}{(r+1)n_N^r} \sum_{k=0}^{n_N-1} \Delta_{N,k} ((k+1)^{r+1} - k^{r+1}). \quad (337)$$

Every finite approximant is rational.

5 Haar and Faber–Schauder representations

5.1 Rational unnormalized Haar increments

Let

$$\eta_{n,k} = \mathbf{1}_{I_{n+1,2k}} - \mathbf{1}_{I_{n+1,2k+1}}, \quad 0 \leq k < 2^n, \quad (338)$$

and let $h_{n,k} = 2^{n/2} \eta_{n,k}$ be the orthonormal Haar functions. Define

$$\gamma_{n,k} = 2^n (\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) \in \mathbb{Q}, \quad (339)$$

$$c_{n,k} = \langle p, h_{n,k} \rangle = 2^{n/2} (\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) = 2^{-n/2} \gamma_{n,k}. \quad (340)$$

Thus the coefficients of the unnormalized Haar representation are rational, while the orthonormal coefficients lie in $2^{n/2} \mathbb{Q}$.

Theorem 5.1 (Exact Haar expansion). *For every $N \geq 1$,*

$$p_N(x) = 1 + \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} \gamma_{n,k} \eta_{n,k}(x) = 1 + \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} c_{n,k} h_{n,k}(x). \quad (341)$$

The power-of-two partial sums converge uniformly and in every finite L^q :

$$p(x) = 1 + \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} \gamma_{n,k} \eta_{n,k}(x). \quad (342)$$

Reflection gives

$$\gamma_{n,2^n-1-k} = -\gamma_{n,k}, \quad c_{n,2^n-1-k} = -c_{n,k}. \quad (343)$$

Proof. The difference $p_{n+1} - p_n$ is constant with opposite signs on the two children of each parent cell. On the left child it equals $2^n(\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) = \gamma_{n,k}$, so $p_{n+1} - p_n = \sum_k \gamma_{n,k} \eta_{n,k}$. Telescoping from $p_0 \equiv 1$ proves (341). Convergence is Theorem 4.2; antisymmetry follows from (314). $\square$

The coefficients can also be read directly from three dyadic values:

$$c_{n,k} = 2^{n/2} \left[2F\left(\frac{2k+1}{2^{n+1}}\right) - F\left(\frac{k}{2^n}\right) - F\left(\frac{k+1}{2^n}\right) \right]. \quad (344)$$

This is a discrete second derivative of F at a dyadic midpoint.

5.2 Fourier image of the Haar expansion

A direct integration gives

$$\hat{h}_{n,k}(z) = 2^{n/2} e^{-izk/2^n} \frac{(1 - e^{-iz/2^{n+1}})^2}{iz}. \quad (345)$$

Therefore:

Theorem 5.2 (Haar product-series transform). *Locally uniformly on $\mathbb{C}$,*

$$\hat{p}(z) = \frac{1 - e^{-iz}}{iz} + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} c_{n,k} 2^{n/2} e^{-izk/2^n} \frac{(1 - e^{-iz/2^{n+1}})^2}{iz}. \quad (346)$$

Equivalently,

$$\Phi(t) = \text{sinc}(t) + e^{it} \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} c_{n,k} 2^{n/2} e^{-2itk/2^n} \frac{(1 - e^{-2it/2^{n+1}})^2}{2it}. \quad (347)$$

Proof. Transform the finite identity (341) and let $N \rightarrow \infty$. Local uniform convergence follows from uniform L^1 convergence of p_N to p . Absolute convergence by scale also follows directly: smoothness gives $c_{n,k} = \mathcal{O}(2^{-3n/2})$ uniformly in k , while $\hat{h}_{n,k}(z) = \mathcal{O}_K(2^{-3n/2})$ on a compact K ; summing 2^n locations leaves a convergent $\mathcal{O}_K(4^{-n})$ scale contribution. $\square$

The factor $(1 - e^{-iz/2^{n+1}})^2$ is the frequency signature of one vanishing moment. In Appendix 6 it will be replaced by a product with as many sine factors as there are nonzero binary digits in the Walsh index.

5.3 Faber–Schauder series of the Fabius function

Let $\psi_{n,k}$ be the peak-one tent supported on $I_{n,k}$, with apex at

$$m_{n,k} = \frac{2k+1}{2^{n+1}}. \quad (348)$$

Define

$$d_{n,k} = F(m_{n,k}) - \frac{1}{2}F(k2^{-n}) - \frac{1}{2}F((k+1)2^{-n}) \quad (349)$$

$$= \frac{1}{2}(\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) = 2^{-n/2-1} c_{n,k} \in \mathbb{Q}. \quad (350)$$

Theorem 5.3 (Rational Faber–Schauder representation). *The piecewise-linear interpolant $L_N F$ through the dyadic values $F(k2^{-N})$ is*

$$L_N F(x) = x + \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} d_{n,k} \psi_{n,k}(x). \quad (351)$$

It converges uniformly to F , hence

$$F(x) = x + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} d_{n,k} \psi_{n,k}(x), \quad (352)$$

where the series is understood by complete dyadic levels. The coefficients are rational and obey

$$d_{n,2^n-1-k} = -d_{n,k}. \quad (353)$$

Moreover,

$$\|F - L_N F\|_{\infty} \leq \frac{\|F''\|_{\infty}}{8} 4^{-N}. \quad (354)$$

Proof. The classical Faber–Schauder construction says that the coefficient added at a new midpoint is its value minus the chord average, giving (349). Equation (350) follows by expanding the two child masses. Uniform convergence follows from continuity; the displayed error is the standard linear-interpolation bound on intervals of length 2^{-N} . Symmetry follows from $F(1-x) = 1-F(x)$. $\square$

Because F is convex on $[0, 1/2]$ and concave on $[1/2, 1]$, one also has

$$d_{n,k} < 0 \quad (0 \leq k < 2^{n-1}), \quad d_{n,k} > 0 \quad (2^{n-1} \leq k < 2^n), \quad (355)$$

with the $n = 0$ coefficient zero by symmetry.

5.4 A sinc-squared series for the Fourier image of F

The transform of a peak-one tent of width 2^{-n} is

$$\int_0^1 \psi_{n,k}(x) e^{-izx} dx = \frac{2^{-n}}{2} e^{-izm_{n,k}} \operatorname{sinc}^2\left(\frac{z}{2^{n+2}}\right). \quad (356)$$

Since

$$\int_0^1 x e^{-izx} dx = \frac{e^{-iz}(1+iz) - 1}{z^2}, \quad (357)$$

we obtain a representation of the Fourier image of the Fabius function itself.

Corollary 5.4 (Schauder–sinc representation of $\mathcal{F}_F$). *Locally uniformly on $\mathbb{C}$,*

$$\mathcal{F}_F(z) = \frac{e^{-iz}(1+iz) - 1}{z^2} + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} d_{n,k} 2^{-n-1} e^{-izm_{n,k}} \operatorname{sinc}^2\left(\frac{z}{2^{n+2}}\right). \quad (358)$$

All coefficients $d_{n,k}$ are rational. Equation (358) is identical, after summation, to the compact product formula (295).

The identity equates a single infinite sinc product to a doubly indexed sum of translated sinc^2 atoms. It is a time–frequency multiresolution counterpart of the canonical zero product (277).

6 Walsh–Hadamard and Thue–Morse product representations

6.1 Paley Walsh functions and the finite Hadamard transform

Let

$$r_j(x) = (-1)^{\lfloor 2^{j+1}x \rfloor}, \quad j \geq 0, \quad (359)$$

away from dyadic endpoints. If

$$m = \sum_{j \geq 0} b_j 2^j, \quad b_j \in \{0, 1\}, \quad (360)$$

then the Paley Walsh function is

$$w_m(x) = \prod_{j \geq 0} r_j(x)^{b_j}. \quad (361)$$

Write

$$r(m) = \text{wt}_2(m) = \sum_j b_j, \quad \sigma(m) = \sum_j j b_j. \quad (362)$$

The top mode w_{2^N-1} on a depth- N partition is the signed Thue–Morse step function.

Define the Walsh coefficient of the Fabius density by

$$a_m = \int_0^1 p(x) w_m(x) dx. \quad (363)$$

If $m < n_N$, then w_m is constant on every depth- N cell. With

$$\varepsilon_{m,N}(k) = w_m\left(\frac{k + 1/2}{n_N}\right) \in \{\pm 1\}, \quad (364)$$

we have the exact finite formula

$$a_m = \sum_{k=0}^{n_N-1} \varepsilon_{m,N}(k) \Delta_{N,k}. \quad (365)$$

Thus every a_m is rational.

Let $H_N = (\varepsilon_{m,N}(k))_{0 \leq m, k < n_N}$. In vector notation,

$$a^{(N)} = H_N \Delta_N, \quad \Delta_N = n_N^{-1} H_N a^{(N)}, \quad H_N H_N^\top = n_N I. \quad (366)$$

This is an exact rational Walsh–Hadamard transform of the dyadic mass vector.

Theorem 6.1 (Walsh expansion and finite Parseval). *For every N ,*

$$p_N(x) = \sum_{m=0}^{n_N-1} a_m w_m(x). \quad (367)$$

Consequently the power-of-two Walsh partial sums converge uniformly:

$$p(x) = \lim_{N \rightarrow \infty} \sum_{m=0}^{n_N-1} a_m w_m(x). \quad (368)$$

At each finite level,

$$\sum_{m=0}^{n_N-1} a_m^2 = n_N \sum_{k=0}^{n_N-1} \Delta_{N,k}^2 = \|p_N\|_2^2. \quad (369)$$

Hence

$$\|p\|_2^2 = \sum_{m=0}^{\infty} a_m^2, \quad \|\text{up}\|_2^2 = \frac{1}{2} \sum_{m=0}^{\infty} a_m^2. \quad (370)$$

Proof. On a depth- N cell, (367) is exactly the inverse Hadamard formula in (366). Uniform convergence is Theorem 4.2. Parseval follows from orthonormality of the Walsh functions or directly from $H_N H_N^\top = n_N I$. $\square$

6.2 Reflection selection rules

Binary reflection complements every Rademacher sign, modulo a null set:

$$w_m(1-x) = (-1)^{r(m)} w_m(x). \quad (371)$$

Since $p(1-x) = p(x)$, this gives a sharp parity filter.

Proposition 6.2 (Hamming-parity selection). *The density coefficients satisfy*

$$a_m = 0 \quad \text{whenever } r(m) \text{ is odd.} \quad (372)$$

If

$$b_m = \int_0^1 \left(F(x) - \frac{1}{2} \right) w_m(x) dx, \quad (373)$$

then

$$b_m = 0 \quad \text{whenever } r(m) \text{ is even.} \quad (374)$$

Proof. Substitute $x \mapsto 1-x$. For a_m the integral is multiplied by $(-1)^{r(m)}$; for b_m , the antisymmetry $F(1-x) - 1/2 = -(F(x) - 1/2)$ contributes one additional minus sign. $\square$

Thus the Walsh support of the density occupies only even Hamming layers, while the centered distribution function occupies only odd layers. This is a digital analogue of the disappearance of odd ordinary Fourier and polynomial moments under reflection symmetry.

6.3 Finite-product Fourier transform of a Walsh mode

Set

$$W_m(z) = \int_0^1 w_m(x) e^{-izx} dx. \quad (375)$$

For $m > 0$, let

$$N(m) = 1 + \lfloor \log_2 m \rfloor. \quad (376)$$

Theorem 6.3 (Walsh finite-product transform). *For $m = 0$,*

$$W_0(z) = e^{-iz/2} \frac{2 \sin(z/2)}{z}. \quad (377)$$

For $m > 0$, with $N = N(m)$ and $r = r(m)$,

$$W_m(z) = \frac{1 - e^{-iz/2^N}}{iz} \prod_{j=0}^{N-1} \left(1 + (-1)^{b_j} e^{-iz/2^{j+1}} \right) \quad (378)$$

$$= e^{-iz/2} \frac{2^{N+1} i^r}{z} \sin\left(\frac{z}{2^{N+1}}\right) \prod_{b_j=0} \cos\left(\frac{z}{2^{j+2}}\right) \prod_{b_j=1} \sin\left(\frac{z}{2^{j+2}}\right). \quad (379)$$

Both right sides have removable singularities and define the same entire function.

Proof. At depth N , integrate the constant value of w_m on each cell. The common cell integral is $(1 - e^{-iz/2^N})/(iz)$, and the signed geometric sum factorizes according to the binary digits of the cell index:

$$\sum_{k=0}^{2^N-1} \varepsilon_{m,N}(k) q^k = \prod_{j=0}^{N-1} \left(1 + (-1)^{b_j} q^{2^{N-1-j}} \right), \quad q = e^{-iz/2^N}.$$

This proves (378). Pairing exponentials into sines and cosines gives (379). $\square$

6.4 A series of finite products for the infinite sinc product

Transforming the Walsh partial sums gives a representation of Φ that is qualitatively different from the canonical product: the coefficients encode Fabius arithmetic, while each basis transform is a finite product.

Theorem 6.4 (Walsh product-series for Φ). *Locally uniformly on $\mathbb{C}$,*

$$\Phi(t) = e^{it} \sum_{m=0}^{\infty} a_m W_m(2t), \quad (380)$$

where the sum is taken through complete blocks $m < 2^N$. More explicitly,

$$\begin{aligned} \Phi(t) = \operatorname{sinc}(t) + \sum_{\substack{m \geq 1 \\ r(m) \text{ even}}} a_m \frac{2^{N(m)} (-1)^{r(m)/2}}{t} \sin\left(\frac{t}{2^{N(m)}}\right) \\ \times \prod_{b_j=0} \cos\left(\frac{t}{2^{j+1}}\right) \prod_{b_j=1} \sin\left(\frac{t}{2^{j+1}}\right). \end{aligned} \quad (381)$$

Each coefficient a_m is rational and is given exactly by (365).

Proof. Since $p_N \rightarrow p$ in L^1 and all functions are supported in $[0, 1]$, their Fourier transforms converge locally uniformly. Substitute the finite Walsh expansion (367) and use $\Phi(t) = e^{it} \widehat{p}(2t)$. Equation (381) follows from Theorem 6.3 and the parity filter (372). $\square$

The corresponding hyperbolic product-series is obtained by $t \mapsto iz$:

$$M(z) = e^{-z} \sum_{m=0}^{\infty} a_m W_m(2iz), \quad (382)$$

so each term becomes a finite product of sinh and cosh factors. Equations (272) and (381) therefore give a product-versus-product-series duality for the same entire function.

6.5 Exact Hamming-weight cancellation

The number of sine factors in (379) is not cosmetic: it is the exact number of vanishing algebraic moments.

Theorem 6.5 (Walsh cancellation order and first moment). *For $m > 0$, put $r = r(m)$ and $\sigma = \sigma(m)$. Then*

$$\int_0^1 x^q w_m(x) dx = 0, \quad 0 \leq q < r, \quad (383)$$

and

$$\int_0^1 x^r w_m(x) dx = (-1)^r r! 2^{-2r-\sigma}. \quad (384)$$

Thus W_m has a zero of exact order $r(m)$ at the origin.

Proof. Extract the small factors from (379). One obtains

$$W_m(z) = i^r 2^{-2r-\sigma} z^r R_m(z), \quad R_m(0) = 1, \quad (385)$$

where R_m is entire. On the other hand,

$$W_m(z) = \sum_{q=0}^{\infty} \frac{(-iz)^q}{q!} \int_0^1 x^q w_m(x) dx. \quad (386)$$

Comparing the first nonzero Taylor coefficient proves both statements. $\square$

For the top Thue–Morse mode, $m = 2^N - 1$, one has $r = N$ and $\sigma = N(N-1)/2$. Hence

$$\int_0^1 x^q w_{2^N-1}(x) dx = 0 \quad (q < N), \quad \int_0^1 x^N w_{2^N-1}(x) dx = (-1)^N N! 2^{-N(N+3)/2}. \quad (387)$$

This is the continuous, normalized Prouhet cancellation attached to the Thue–Morse sign pattern.

6.6 The Gaussian-binomial cancellation spectrum

The first nonzero moments aggregate exactly by Hamming weight.

Theorem 6.6 ($q = 1/2$ Gaussian-binomial aggregation). *For $0 \leq r \leq N$,*

$$\sum_{\substack{0 \leq m < 2^N \\ r(m)=r}} 2^{-\sigma(m)} = 2^{-r(r-1)/2} \left[\begin{matrix} N \\ r \end{matrix} \right]_{1/2}. \quad (388)$$

Consequently

$$\sum_{\substack{0 \leq m < 2^N \\ r(m)=r}} \int_0^1 x^r w_m(x) dx = (-1)^r r! 2^{-2r-r(r-1)/2} \left[\begin{matrix} N \\ r \end{matrix} \right]_{1/2}. \quad (389)$$

Proof. The finite q -binomial theorem gives

$$\prod_{j=0}^{N-1} (1 + y 2^{-j}) = \sum_{r=0}^N 2^{-r(r-1)/2} \left[\begin{matrix} N \\ r \end{matrix} \right]_{1/2} y^r.$$

The coefficient of y^r on the left is the sum of $2^{-\sigma(m)}$ over all subsets of r bit positions. Combine with (384). $\square$

The appearance of $q = 1/2$, rather than the $q = 1/4$ common in the even-moment and scale-energy theory, has a simple origin: it records bit *positions*, not squared dyadic lengths. Squaring amplitudes or passing to even cumulants naturally changes $1/2$ to $1/4$.

6.7 All higher Walsh moments: a Bell–Bernoulli formula

The product formula encodes every moment, not only the first nonzero one. Define the even entire factor

$$E_m(z) = \operatorname{sinc}\left(\frac{z}{2^{N+1}}\right) \prod_{b_j=0} \cos\left(\frac{z}{2^{j+2}}\right) \prod_{b_j=1} \operatorname{sinc}\left(\frac{z}{2^{j+2}}\right), \quad (390)$$

where $N = N(m)$. Then

$$R_m(z) = e^{-iz/2} E_m(z) \quad (391)$$

in (385). Write

$$E_m(z) = \sum_{j=0}^{\infty} e_{m,j} z^{2j}. \quad (392)$$

For $q \geq 1$, set

$$\lambda_{2q}(m) = -\frac{2^{2q-1}|B_{2q}|}{q} \left[2^{-2q(N+1)} + \sum_{b_j=1} 2^{-2q(j+2)} + (2^{2q} - 1) \sum_{b_j=0} 2^{-2q(j+2)} \right]. \quad (393)$$

The classical expansions of log sinc and log cos show that

$$\log E_m(z) = \sum_{q=1}^{\infty} \lambda_{2q}(m) \frac{z^{2q}}{(2q)!}. \quad (394)$$

Therefore

$$e_{m,j} = \frac{1}{(2j)!} Y_{2j}(0, \lambda_2(m), 0, \lambda_4(m), \dots, \lambda_{2j}(m)). \quad (395)$$

Theorem 6.7 (Bell–Bernoulli formula for Walsh moments). *Let $m > 0$, $r = r(m)$, and $\ell \geq 0$. Then*

$$\int_0^1 x^{r+\ell} w_m(x) dx = (-1)^r \frac{(r+\ell)!}{2^{2r+\sigma(m)}} \sum_{j=0}^{\lfloor \ell/2 \rfloor} \frac{(-1)^j e_{m,j}}{2^{\ell-2j}(\ell-2j)!}, \quad (396)$$

with $e_{m,j}$ given explicitly by (395) and (393).

Proof. Use $W_m(z) = i^r 2^{-2r-\sigma} z^r e^{-iz/2} E_m(z)$, expand the exponential and the even series (392), and compare with (386). Formula (395) is the standard exponential Bell-polynomial coefficient formula applied to (394). $\square$

This theorem places the Bernoulli numbers in the *basis moments*, while the rational coefficients a_m carry the Fabius-specific arithmetic. The ordinary Fabius moments can therefore be reconstructed as

$$m_q = \lim_{N \rightarrow \infty} \sum_{m=0}^{2^N-1} a_m \int_0^1 x^q w_m(x) dx. \quad (397)$$

Only modes with even $r(m) \leq q$ contribute. For example, since $m_2 = 5/18$,

$$\sum_{\substack{m \geq 1 \\ r(m)=2}} 2^{-\sigma(m)} a_m = -\frac{4}{9}. \quad (398)$$

Indeed, the constant Walsh mode contributes $\int_0^1 x^2 dx = 1/3$, and (384) contributes $2^{-3-\sigma(m)} a_m$ from every weight-two mode.

Remark 6.8. Equation (397) is naturally ordered by dyadic blocks, not by arbitrary rearrangement. Blockwise convergence follows immediately from uniform convergence of the Walsh conditional expectations. Establishing absolute convergence after regrouping by Hamming weight is a separate and interesting coefficient-decay problem.

7 Bernstein polynomials and positive beta-mixture limits

7.1 Rational Bernstein approximants

For $n = n_N = 2^N$, define the Bernstein polynomial

$$\mathcal{B}_N F(x) = \sum_{k=0}^n F(k/n) \binom{n}{k} x^k (1-x)^{n-k}. \quad (399)$$

Every coefficient in the Bernstein basis is a rational dyadic Fabius value. The power-basis form is also rational:

$$\mathcal{B}_N F(x) = \sum_{j=0}^n \binom{n}{j} \left(\sum_{k=0}^j (-1)^{j-k} \binom{j}{k} F(k/n) \right) x^j. \quad (400)$$

Thus

$$F(x) = \lim_{N \rightarrow \infty} \mathcal{B}_N F(x) \quad (401)$$

uniformly, as a limit of explicit rational polynomials.

The dyadic Faber interpolants of [Appendix 5](#) converge at order 4^{-N} because they exploit the C^2 smoothness on a mesh of width 2^{-N} . The Bernstein approximants use a binomial cloud of width $n^{-1/2}$ and have the classical expansion

$$\mathcal{B}_N F(x) - F(x) = \frac{x(1-x)}{2n} F''(x) + \mathcal{O}(n^{-2}) \quad (402)$$

under the available smoothness. Their advantage is positivity and the probabilistic mixture structure of the derivative.

7.2 Derivative as a beta-density mixture

Differentiating (399) gives

$$p_N^B(x) := (\mathcal{B}_N F)'(x) \quad (403)$$

$$= n \sum_{k=0}^{n-1} \Delta_{N,k} \binom{n-1}{k} x^k (1-x)^{n-1-k} \quad (404)$$

$$= \sum_{k=0}^{n-1} \Delta_{N,k} \frac{x^k (1-x)^{n-1-k}}{B(k+1, n-k)}. \quad (405)$$

Theorem 7.1 (Positive rational beta-mixture representation). *The functions p_N^B are probability densities and are finite mixtures of $\text{Beta}(k+1, n-k)$ densities with the exact rational weights $\Delta_{N,k}$. Moreover,*

$$p_N^B \longrightarrow p \quad \text{uniformly on } [0, 1]. \quad (406)$$

Consequently p is a uniform limit of positive polynomial densities whose mixture weights are exactly the dyadic Fabius masses.

Proof. Formula (405) proves positivity and unit mass. The classical derivative theorem for Bernstein polynomials states that $(\mathcal{B}_n f)' \rightarrow f'$ uniformly when $f \in C^1[0, 1]$. $\square$

This representation is structurally different from the uniform-cell mixture:

model	component family	weights
p_N	disjoint equal-width uniforms	$\Delta_{N,k}$
p_N^B	overlapping beta densities	$\Delta_{N,k}$

The same rational vector is therefore a mixing law for both a local and a global positive basis.

7.3 Hypergeometric transform representations

The moment-generating function of a $\text{Beta}(a, b)$ variable is ${}_1F_1(a; a + b; z)$. Hence:

Theorem 7.2 (Confluent-hypergeometric mixture limits). *Locally uniformly on $\mathbb{C}$,*

$$\int_0^1 p(x)e^{zx} dx = \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} {}_1F_1(k+1; n_N+1; z), \quad (407)$$

$$\Phi(t) = e^{it} \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} {}_1F_1(k+1; n_N+1; -2it), \quad (408)$$

$$M(z) = e^{-z} \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} {}_1F_1(k+1; n_N+1; 2z). \quad (409)$$

Thus the infinite sinc and hyperbolic-sinc products are limits of positive rational mixtures of confluent hypergeometric functions.

Proof. Integrate (405) against e^{zx} and use uniform convergence. The affine transform relations $\Phi(t) = e^{it} \hat{p}(2t)$ and $M(z) = e^{-z} \int_0^1 p(x)e^{2zx} dx$ give the remaining formulas. $\square$

The beta moments give another rational moment limit:

$$m_r = \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} \frac{(k+1)_r}{(n_N+1)_r}, \quad r \in \mathbb{N}, \quad (410)$$

where $(a)_r$ is the rising factorial. Comparison of (410) with the cell formula (337) gives two positive rational quadrature schemes for the same exact Fabius moments.

8 Dual quantile cells and equal-weight sinc mixtures

8.1 The inverse dyadic width array

The primal mass array fixes spatial widths and varies probabilities. The inverse construction fixes probabilities and varies spatial widths. Put

$$g_{N,k} = G(k/n_N), \quad q_{N,k} = g_{N,k+1} - g_{N,k} > 0, \quad 0 \leq k < n_N. \quad (411)$$

Then

$$\sum_{k=0}^{n_N-1} q_{N,k} = 1, \quad (412)$$

$$q_{N,n_N-1-k} = q_{N,k}, \quad (413)$$

$$q_{N,k} = q_{N+1,2k} + q_{N+1,2k+1}. \quad (414)$$

Unlike $\Delta_{N,k}$, these widths are not expected to be rational in general. They are exact inverse-Fabius data.

Let $G_N = L_N G$ be the piecewise-linear interpolant through the points $(k/n_N, g_{N,k})$. Pushing uniform measure forward by G_N gives the density

$$p_N^Q(x) = \frac{1}{n_N q_{N,k}}, \quad g_{N,k} < x < g_{N,k+1}. \quad (415)$$

Each interval has probability $1/n_N$. Thus p_N^Q is a finite mixture of variable-width uniforms with *equal* weights.

Primal–dual cell dictionary

	spatial interval	probability weight
primal Fabius cells	fixed width $1/n_N$	variable $\Delta_{N,k}$
dual quantile cells	variable width $q_{N,k}$	fixed $1/n_N$

The two projective arrays are interchanged by F and G : the first samples increments of F on a uniform spatial grid, the second samples increments of G on a uniform probability grid.

8.2 Exact chord integration and the Fourier image

On the k th probability cell, G_N moves linearly from $g_{N,k}$ to $g_{N,k+1}$. Integrating an exponential along that chord gives a sinc factor.

Theorem 8.1 (Inverse-quantile chord representation). *Locally uniformly on $\mathbb{C}$,*

$$\Phi(t) = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} e^{-it(g_{N,k}+g_{N,k+1}-1)} \operatorname{sinc}(tq_{N,k}), \quad (416)$$

$$M(z) = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} e^{z(g_{N,k}+g_{N,k+1}-1)} \frac{\sinh(zq_{N,k})}{zq_{N,k}}. \quad (417)$$

Equivalently, Φ is the locally uniform limit of the Fourier transforms of finite equal-weight uniform mixtures on the centered intervals

$$[2g_{N,k} - 1, 2g_{N,k+1} - 1]. \quad (418)$$

Proof. Uniform continuity of G gives $G_N \rightarrow G$ uniformly. Therefore

$$\int_0^1 e^{-it(2G_N(u)-1)} du \longrightarrow \int_0^1 e^{-it(2G(u)-1)} du = \Phi(t)$$

locally uniformly in t . On a probability cell of length $1/n_N$, direct integration of the affine phase gives

$$\frac{1}{n_N} e^{-it(g_{N,k}+g_{N,k+1}-1)} \operatorname{sinc}(tq_{N,k}).$$

Summing proves (416); Wick rotation gives (417). $\square$

Removing the chord average gives the simpler midpoint Riemann representation

$$\Phi(t) = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} \exp \left[-it \left(2G \left(\frac{k+1/2}{n_N} \right) - 1 \right) \right]. \quad (419)$$

The chord formula is preferable when inverse values at endpoints are already available, because it integrates the linear interpolant exactly.

8.3 Moment and Laplace limits from inverse dyadic values

For the uncentered law, exact integration of powers along each chord yields:

Corollary 8.2 (Inverse-dyadic moment polygons). *For every integer $r \geq 0$,*

$$m_r = \lim_{N \rightarrow \infty} \frac{1}{n_N(r+1)} \sum_{k=0}^{n_N-1} \frac{g_{N,k+1}^{r+1} - g_{N,k}^{r+1}}{g_{N,k+1} - g_{N,k}} \quad (420)$$

$$= \lim_{N \rightarrow \infty} \frac{1}{n_N(r+1)} \sum_{k=0}^{n_N-1} \sum_{j=0}^r g_{N,k}^j g_{N,k+1}^{r-j}. \quad (421)$$

For arbitrary $z \in \mathbb{C}$,

$$\int_0^1 p(x) e^{zx} dx = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} e^{z(g_{N,k} + g_{N,k+1})/2} \operatorname{sinhc}\left(\frac{z q_{N,k}}{2}\right), \quad (422)$$

where $\operatorname{sinhc} w = \sinh(w)/w$.

The first line is a limit of complete homogeneous symmetric polynomials in adjacent inverse-dyadic values. It complements the repository's local inverse series and exact inverse values at special dyadic levels: local inverse arithmetic enters a global moment formula through a simple polygonal limit.

8.4 Dual L^q spectra and entropy constants

Because $q_{N,k} = \int_{k/n_N}^{(k+1)/n_N} G'(u) du$, the slope $n_N q_{N,k}$ is the dyadic conditional expectation of G' . This gives a discrete counterpart of the inverse-derivative norm threshold.

Theorem 8.3 (Primal–dual Rényi spectra). *For $s > 0$,*

$$n_N^{s-1} \sum_k \Delta_{N,k}^s \longrightarrow \int_0^1 p(x)^s dx. \quad (423)$$

For $0 < s < 1$,

$$n_N^{s-1} \sum_k q_{N,k}^s \longrightarrow \int_0^1 G'(u)^s du = \int_0^1 p(x)^{1-s} dx. \quad (424)$$

At $s = 1$ both normalized sums equal one. For $s > 1$, the dual normalized sum tends to $+\infty$, consistently with $G' \notin L^s(0, 1)$.

Proof. The primal statement is (334). On each probability cell, $n_N q_{N,k} = \mathbb{E}[G' \mid \mathcal{D}_N]$. The dyadic martingale converges to G' in L^1 . For $0 < s < 1$, the inequality $|a^s - b^s| \leq |a - b|^s$ and Hölder's inequality give convergence of the s th powers in L^1 . The change of variables $u = F(x)$ gives

$$\int_0^1 G'(u)^s du = \int_0^1 p(x)^{1-s} dx.$$

For $s > 1$, conditional-expectation L^s norms increase to the infinite L^s norm of G' . $\square$

The one-sided derivatives at $s = 1$ give entropy constants. In addition to (336), the endpoint asymptotics imply $\log p \in L^1(0, 1)$, and

$$- \sum_{k=0}^{n_N-1} q_{N,k} \log q_{N,k} = \log n_N + \int_0^1 \log p(x) dx + o(1). \quad (425)$$

Thus the two exact partitions retain different continuous entropies:

$$H(\Delta_N) - \log n_N \longrightarrow - \int p \log p, \quad (426)$$

$$H(q_N) - \log n_N \longrightarrow \int \log p. \quad (427)$$

The second constant is sensitive to endpoint flatness, whereas the first is weighted by the probability density itself.

9 Haar scale energies and the sinc^4 continuum expansion

9.1 Three exact forms of the energy

Define the detail energy

$$E_n = \|p_{n+1} - p_n\|_2^2 = \sum_{k=0}^{2^n-1} c_{n,k}^2. \quad (428)$$

Using (340),

$$E_n = 2^n \sum_{k=0}^{2^n-1} (\Delta_{n+1,2k} - \Delta_{n+1,2k+1})^2 \in \mathbb{Q}. \quad (429)$$

Thus a sum of squares of generally quadratic Haar coefficients is rational. Orthogonality gives

$$\|p_N\|_2^2 = 1 + \sum_{n=0}^{N-1} E_n, \quad \|p\|_2^2 = 1 + \sum_{n=0}^{\infty} E_n. \quad (430)$$

The Haar and Walsh filtrations are the same nested dyadic subspaces. Hence:

Proposition 9.1 (Haar–Walsh shell identity). *For every $n \geq 0$,*

$$E_n = \sum_{m=2^n}^{2^{n+1}-1} a_m^2. \quad (431)$$

Only even-Hamming-weight indices contribute to the right side.

Proof. Both sides are the squared norm of the orthogonal projection of p onto $V_{n+1} \ominus V_n$, where V_n is the space of functions constant on depth- n dyadic cells. The Haar basis organizes this complement by location; the Walsh basis organizes it by sequency. $\square$

Thus the same rational E_n is simultaneously a local curvature statistic of F , a sibling-mass contrast, and a Walsh spectral-shell energy.

9.2 Local Taylor expansion of a Haar coefficient

Let $h = 2^{-n}$ and let $m_{n,k} = (k + 1/2)h$ be the center of a parent cell. Flatness of p at the endpoints permits uniform Taylor expansion to arbitrary order. Direct integration of odd Taylor terms gives

$$c_{n,k} \sim - \sum_{r=0}^{\infty} \frac{h^{2r+3/2}}{2^{2r+1}(2r+2)!} p^{(2r+1)}(m_{n,k}). \quad (432)$$

The first terms are

$$c_{n,k} = -h^{3/2} \left[\frac{p'(m_{n,k})}{4} + \frac{h^2 p'''(m_{n,k})}{192} + \frac{h^4 p^{(5)}(m_{n,k})}{23040} + \mathcal{O}(h^6) \right]. \quad (433)$$

This explains the scale $E_n \asymp 4^{-n}$ despite the infinite smoothness of p : the Haar wavelet has only one vanishing moment. Higher-order multiwavelets would expose the superalgebraic smoothness more efficiently.

9.3 A frequency multiplier and a superalgebraic alias remainder

Let

$$K_h(x) = h^{-1/2} (\mathbf{1}_{[-h/2,0]}(x) - \mathbf{1}_{[0,h/2]}(x)). \quad (434)$$

Its squared Fourier modulus is

$$|\widehat{K}_h(\xi)|^2 = \frac{16}{h\xi^2} \sin^4\left(\frac{h\xi}{4}\right). \quad (435)$$

The coefficients $c_{n,k}$ are samples of a translate of $p * K_h$ at the cell centers. Poisson summation for the squared samples separates a zero alias from all nonzero aliases.

Theorem 9.2 (sinc^4 multiplier formula). *For $h = 2^{-n}$,*

$$E_n = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{p}(\xi)|^2 \frac{h^2 \xi^2}{16} \text{sinc}^4\left(\frac{h\xi}{4}\right) d\xi + R(h), \quad (436)$$

where for every $M > 0$,

$$R(h) = \mathcal{O}_M(h^M) \quad (h \downarrow 0). \quad (437)$$

Proof. Apply Poisson summation to the samples of $|p * K_h|^2$ on the shifted lattice $h(\mathbb{Z} + 1/2)$. The zero Fourier mode is

$$\frac{1}{h} \|p * K_h\|_2^2 = \frac{1}{2\pi h} \int |\widehat{p}(\xi)|^2 |\widehat{K}_h(\xi)|^2 d\xi,$$

which simplifies to the integral in (436). Every nonzero alias contains a factor $\widehat{p}(\xi + 2\pi\ell/h)$ or its conjugate. Because the zero extension of p is $C_c^\infty(\mathbb{R})$, $\widehat{p}$ is Schwartz; the nonzero aliases are therefore smaller than every power of h after summation over $\ell \neq 0$. $\square$

9.4 All orders and Bell–Bernoulli coefficients

Write

$$\text{sinc}^4 z = \sum_{j=0}^{\infty} \alpha_j z^{2j}, \quad \alpha_0 = 1, \quad \alpha_1 = -\frac{2}{3}, \quad \alpha_2 = \frac{1}{5}, \quad \alpha_3 = -\frac{34}{945}, \dots \quad (438)$$

Expanding the multiplier in (436) and using Parseval gives:

Theorem 9.3 (All-orders Haar-energy expansion). *As $n \rightarrow \infty$, with $h = 2^{-n}$,*

$$E_n \sim \sum_{j=0}^{\infty} \frac{\alpha_j}{16^{j+1}} h^{2j+2} \|p^{(j+1)}\|_2^2. \quad (439)$$

In particular,

$$E_n \sim \frac{h^2}{16} \|p'\|_2^2 - \frac{h^4}{384} \|p''\|_2^2 + \frac{h^6}{20480} \|p'''\|_2^2 - \frac{17h^8}{30965760} \|p^{(4)}\|_2^2 + \dots \quad (440)$$

Proof. Taylor-expand $\text{sinc}^4(h\xi/4)$ to an arbitrary fixed order in (436). The Schwartz decay of $\widehat{p}$ justifies termwise integration and bounds the remainder. Parseval identifies $(2\pi)^{-1} \int \xi^{2j+2} |\widehat{p}(\xi)|^2 d\xi$ with $\|p^{(j+1)}\|_2^2$. The alias term is smaller than every algebraic order. $\square$

The coefficients themselves are Bell–Bernoulli quantities. Since

$$\log \text{sinc}^4 z = - \sum_{r=1}^{\infty} \frac{2^{2r+1} |B_{2r}|}{r(2r)!} z^{2r}, \quad (441)$$

put

$$\beta_{2r} = - \frac{2^{2r+1} |B_{2r}|}{r}, \quad \beta_{2r+1} = 0. \quad (442)$$

Then

$$\alpha_j = \frac{1}{(2j)!} Y_{2j}(0, \beta_2, 0, \beta_4, \dots, \beta_{2j}). \quad (443)$$

Thus the same Bell–Bernoulli mechanism that governs the moment expansion of Φ also governs the continuum expansion of its dyadic Haar energies, but through the fourth power of a single-scale sinc filter.

9.5 Derivative norms from half-integer samples of the sinc product

The cosine series (291) gives exact rapidly convergent sums for every derivative norm:

$$\|p^{(r)}\|_2^2 = 2^{2r+1} \sum_{m=0}^{\infty} A_m^2 ((2m+1)\pi)^{2r}. \quad (444)$$

Substitution into (439) expresses the rational scale energies asymptotically through discrete samples of the infinite sinc product at the odd half-lattice. The first numerical values are reported in [Appendix 13](#).

9.6 A meromorphic generating function for the rational energies

The complete scale expansion has a strong analytic consequence. Define

$$\mathcal{H}(z) = \sum_{n=0}^{\infty} E_n z^n. \quad (445)$$

Put

$$C_j = \frac{\alpha_j}{16^{j+1}} \|p^{(j+1)}\|_2^2. \quad (446)$$

Then (439) says, at every fixed depth J ,

$$E_n = \sum_{j=0}^J C_j 4^{-(j+1)n} + \mathcal{O}_J(4^{-(J+2)n}). \quad (447)$$

Theorem 9.4 (Meromorphic Haar-energy generating function). *The power series $\mathcal{H}$ has radius of convergence 4 and admits a meromorphic continuation to the whole complex plane. Its only possible finite poles are simple poles at*

$$z = 4, 16, 64, \dots, 4^{j+1}, \dots \quad (448)$$

Near $z = 4^{j+1}$ its principal part is

$$\frac{C_j}{1 - z/4^{j+1}}, \quad (449)$$

and hence the residue is

$$\operatorname{Res}_{z=4^{j+1}} \mathcal{H}(z) = -\frac{\alpha_j}{4^{j+1}} \|p^{(j+1)}\|_2^2. \quad (450)$$

In particular,

$$\operatorname{Res}_{z=4} \mathcal{H}(z) = -\frac{1}{4} \|p'\|_2^2. \quad (451)$$

Proof. The positive leading asymptotic $E_n \sim C_0/4^n$, with $C_0 = \|p'\|_2^2/16 > 0$, gives radius 4. For a fixed J , subtract from $\mathcal{H}$ the finite rational function

$$\sum_{j=0}^J \frac{C_j}{1 - z/4^{j+1}}.$$

By (447), the remaining coefficient sequence is $\mathcal{O}_J(4^{-(J+2)n})$, so its power series is holomorphic in $|z| < 4^{J+2}$. Letting J grow gives compatible continuations to every finite disc. The principal parts and residues are immediate. $\square$

This is a new $q = 1/4$ spectral object attached to the dyadic multiresolution, but unlike the zero determinant already present in the corpus its coefficients are rational sibling-mass energies. The pole at 4^{j+1} directly records the Sobolev norm of derivative order $j + 1$, multiplied by the Bell–Bernoulli coefficient α_j .

9.7 Beyond all orders: a phase-locked alias frontier

The remainder $R(2^{-n})$ is superalgebraic but not expected to be a featureless error term. Its Poisson representation samples cross-products of $\widehat{p}$ at frequencies separated by $2\pi 2^n \ell$. The corpus proves that Φ has a log-square envelope with a nonconstant periodic phase in $\log_2 |t|$. Along a dyadic scale sequence, the phase of each fixed alias ℓ is locked at the fractional part of $\log_2(2\pi|\ell|)$.

Conjecture 9.5 (Phase-locked Haar alias transseries). After subtraction of any fixed number of terms in (439), the first nonzero Poisson aliases admit a transseries whose exponential scale is determined by the proved real-axis decay of $\Phi(2\pi 2^n \ell)$ and whose amplitudes depend on the corresponding locked Lambert phase. In particular, the optimally truncated remainder should be asymptotic to a finite sum of $\ell = \pm 1$ alias contributions except at isolated cancellation phases.

A proof would require inserting the phase-resolved Fourier asymptotic into the exact Poisson cross-correlation integral, not merely into a pointwise sample. This is a concrete meeting point between the repository’s Fourier-decay campaign and the new multiresolution energy theory.

10 Multiresolution of the inverse Fabius function

10.1 Faber–Schauder series and reflection

The inverse G is continuous and strictly increasing, so it has the universal Faber–Schauder expansion

$$G(u) = u + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} e_{n,k} \psi_{n,k}(u), \quad (452)$$

where

$$e_{n,k} = G\left(\frac{2k+1}{2^{n+1}}\right) - \frac{1}{2}G\left(\frac{k}{2^n}\right) - \frac{1}{2}G\left(\frac{k+1}{2^n}\right). \quad (453)$$

The sum is taken through complete dyadic levels and converges uniformly. From $G(1-u) = 1-G(u)$,

$$e_{n,2^n-1-k} = -e_{n,k}. \quad (454)$$

Since F is convex on the left half and concave on the right half, G is concave on $[0, 1/2]$ and convex on $[1/2, 1]$; hence

$$e_{n,k} > 0 \quad (k < 2^{n-1}), \quad e_{n,k} < 0 \quad (k \geq 2^{n-1}). \quad (455)$$

The coefficients are generally not expected to be rational. They package the inverse-dyadic values studied elsewhere in the corpus into a global basis.

10.2 Fourier and Walsh images of the inverse

Transforming the tents gives, levelwise and locally uniformly,

$$\int_0^1 G(u) e^{-iz u} du = \frac{e^{-iz}(1+iz) - 1}{z^2} + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} e_{n,k} 2^{-n-1} e^{-iz m_{n,k}} \operatorname{sinc}^2\left(\frac{z}{2^{n+2}}\right). \quad (456)$$

This is the inverse analogue of (358). It is particularly well suited to low-frequency moments, because derivatives at $z = 0$ produce finite polynomial combinations of inverse-dyadic coefficients at each level.

The centered inverse $G(u) - 1/2$ is antisymmetric under $u \mapsto 1-u$. Therefore its Walsh coefficients

$$g_m = \int_0^1 \left(G(u) - \frac{1}{2}\right) w_m(u) du \quad (457)$$

vanish for even Hamming weight. The inverse and the density thus have complementary Walsh parity:

$$a_m \neq 0 \Rightarrow r(m) \text{ even}, \quad g_m \neq 0 \Rightarrow r(m) \text{ odd}. \quad (458)$$

The finite-product transform (379) applies unchanged to the Walsh expansion of $G - 1/2$.

10.3 Boundary coefficients and slow variation

The leftmost coefficient is

$$e_{n,0} = G(2^{-n-1}) - \frac{1}{2}G(2^{-n}). \quad (459)$$

The phase-resolved Lambert endpoint theory in the repository implies the leading inverse scale

$$\log \frac{1}{G(y)} \sim \sqrt{2 \log 2 \log \frac{1}{y}} \quad (y \downarrow 0), \quad (460)$$

and, more strongly, the fixed-ratio slow-variation law

$$\frac{G(cy)}{G(y)} \rightarrow 1 \quad (y \downarrow 0), \quad c > 0. \quad (461)$$

Applying $c = 1/2$ to (459) gives a rigorous first-order consequence.

Proposition 10.1 (Boundary Schauder scale). *Assuming the proved phase-resolved inverse endpoint expansion from the primary corpus,*

$$e_{n,0} \sim \frac{1}{2}G(2^{-n}) = 2^{-\sqrt{2n}+o(\sqrt{n})}. \quad (462)$$

Thus the inverse Schauder coefficients at the boundary decay faster than every power of n but slower than every geometric sequence in the level n .

The crude leading model $G(2^{-n}) \approx 2^{-\sqrt{2n}}$ predicts one further term:

$$\frac{e_{n,0}}{G(2^{-n})} \approx \frac{1}{2} - \frac{\log 2}{\sqrt{2n}}. \quad (463)$$

The actual correction must include the lower-Lambert logarithmic terms and the nonconstant phase already known for the endpoint problem.

Conjecture 10.2 (Phase-resolved boundary coefficient). There is an explicit expansion

$$\frac{e_{n,0}}{G(2^{-n})} = \frac{1}{2} - \frac{\log 2}{\sqrt{2n}} + \frac{P_1(\log n, \Theta_n)}{n} + \frac{P_2(\log n, \Theta_n)}{n^{3/2}} + \dots, \quad (464)$$

where Θ_n is the exact lower-Lambert phase inherited from the endpoint expansion and each P_j is explicitly computable from its Bell-polynomial coefficient calculus.

10.4 Boundary domination of inverse interpolation

Let $L_N G$ denote the polygon through the inverse dyadic values $G(k2^{-N})$. Uniform continuity gives $L_N G \rightarrow G$, but the usual C^2 error bound fails because G' diverges at both endpoints. Concavity on the left shows that the interpolation error there is positive, and the first cell already has midpoint error $e_{N,0}$.

Conjecture 10.3 (Boundary-dominated inverse interpolation). The global polygonal error is asymptotically attained in the two boundary cells:

$$\|G - L_N G\|_\infty \sim e_{N,0} \sim \frac{1}{2} G(2^{-N}). \quad (465)$$

In particular,

$$\log_2 \|G - L_N G\|_\infty = -\sqrt{2N} + o(\sqrt{N}). \quad (466)$$

A proof should combine monotonicity of the inverse curvature on successive endpoint cells with the phase-resolved ratio asymptotics $G(2^{-N-j})/G(2^{-N})$. Even a two-sided constant-factor result would provide a sharp complexity theorem for inverse-Fabius interpolation.

11 Comparison of the representation families

The formulas developed above are not redundant. They expose different parts of the same function and have different numerical and formal strengths.

representation	finite data		building blocks	principal strength
Infinite convolution	dyadic scales	uniform	compact uniforms	Probability, support, smoothing equations, finite head–tail factorization.
Sinc product	no sampled values		$\text{sinc}(t/2^j)$	Zeros, entire-function structure, Fourier decay, canonical products.
Half-integer cosine	$\Phi((2m+1)\pi)$		cosines	Fast point evaluation, derivatives, Sobolev norms, twisted Poisson summation.

representation	finite data	building blocks	principal strength
Dyadic cell mixture	rational $\Delta_{N,k}$	equal-width uni-forms	Positive local model, 4^{-N} transform quadrature, exact mass polynomial.
Haar	rational $\gamma_{n,k}$	local sign steps	Curvature by scale and location, exact rational energies, meromorphic energy generating function.
Faber–Schauder	rational $d_{n,k}$	tents and sinc^2 atoms	Uniform piecewise-linear approximation of F and a Fourier series of translated sinc^2 functions.
Walsh–Hadamard	rational a_m	digital characters	Exact Hamming cancellation, Thue–Morse top mode, finite trigonometric-product transforms.
Bernstein/beta	rational $\Delta_{N,k}$	beta densities, ${}_1F_1$	Positive polynomial approximation and hypergeometric transform mixtures.
Quantile chords	inverse values $g_{N,k}$	variable-width uni-forms	Equal probability cells, direct use of inverse Fabius values, finite sinc mixtures.
Legendre–Chebyshev	Bessel integrals of Φ	orthogonal polynomials	Global spectral approximation and transform-to-polynomial bridges.

11.1 A commuting diagram of finite transforms

At level N , one rational vector Δ_N enters three exact linear maps:

$$\Delta_N \xrightarrow{H_N} a^{(N)}, \quad \Delta_N \xrightarrow{\text{sibling differences}} \gamma_n, \quad \Delta_N \xrightarrow{\text{beta kernel}} p_N^B. \quad (467)$$

The same vector also determines the mass polynomial D_N . The resulting functions all integrate to one and converge to p , but their Fourier images factor differently:

$$\widehat{p}_N(z) = \text{exponential polynomial} \times \text{sinc window}, \quad (468)$$

$$\widehat{p}_N(z) = \sum_{m < 2^N} a_m \times \text{finite sine-cosine product}, \quad (469)$$

$$\widehat{p}_N^B(z) = \sum_k \Delta_{N,k} \times {}_1F_1(k+1; 2^N+1; -iz). \quad (470)$$

This makes it possible to transfer a question among a local partition, a digital frequency basis, and a positive global polynomial basis without changing the exact arithmetic input.

11.2 Computational observations

For direct evaluation of $\Phi(t)$ at moderate t , the canonical sinc product remains the simplest representation. The new formulas are useful for other purposes:

- the cell and Haar formulas expose exact rationality and give certified dyadic approximation errors;

- the Walsh formula isolates cancellation order before numerical evaluation, avoiding subtraction of low moments;
- the beta mixture preserves positivity and is convenient for rigorous interval bounds on the real Laplace axis;
- the quantile chord formula is naturally adapted to a table of inverse values and does not require differentiating G ;
- the energy expansion converts fine-scale rational data into derivative norms and vice versa.

12 Further deductions and frontier directions

12.1 Exact arithmetic of multiresolution coefficients

The formulas

$$\begin{aligned}\gamma_{n,k} &= 2^n (\Delta_{n+1,2k} - \Delta_{n+1,2k+1}), \\ d_{n,k} &= \frac{1}{2} (\Delta_{n+1,2k} - \Delta_{n+1,2k+1}), \\ a_m &= \sum_k \varepsilon_{m,N}(k) \Delta_{N,k}\end{aligned}$$

reduce the arithmetic of three basis systems to exact dyadic Fabius values. The existing two-adic theory should therefore induce valuation laws for all three coefficient arrays.

Open problem 12.1 (Two-adic multiresolution spectrum). Determine sharp formulas or lower envelopes for $v_2(\gamma_{n,k})$, $v_2(d_{n,k})$, and $v_2(a_m)$ in terms of binary carries, Hamming weight, and the known valuations of dyadic Fabius values. In particular, classify the indices for which sibling subtraction creates additional two-adic cancellation.

A promising approach is to compute the exact mass vectors by the repository's binary reduction algorithm and apply the fast Walsh–Hadamard transform over $\mathbb{Q}_2$, retaining valuations before rational reconstruction.

12.2 Mass-polynomial renormalization

The palindromic rational polynomials D_N are projectively compatible, but (315) alone does not close them under a scalar renormalization. The functional equation of F supplies additional relations among sibling masses.

Open problem 12.2 (Closed recursion for D_N). Find a matrix or q -difference recursion that computes D_{N+1} from a small number of transforms of D_N and finitely many boundary values, preferably in a form compatible with the existing q -binomial dyadic formulas. Such a recursion would turn (329) into an autonomous sequence of rational entire approximants.

A two-component recursion separating the left and right halves is more plausible than a scalar recursion, because $F'(x) = 2F(2x)$ on the left couples a cell mass to an integral of F , not merely to another mass.

12.3 Hamming layers and Walsh pressure

Let

$$\mathcal{W}_{N,r} = \sum_{\substack{m < 2^N \\ r(m)=r}} a_m^2. \quad (471)$$

The exact parity filter removes odd r , and [Theorem 6.6](#) controls the corresponding basis moments. The coefficient energy itself is a new digital spectrum.

Conjecture 12.3 (Walsh Hamming-layer pressure). There exists a limiting pressure function $P(\theta)$ for the bivariate sums

$$\sum_{m < 2^N} |a_m|^2 e^{\theta r(m)}, \quad (472)$$

possibly after subtraction of a deterministic quadratic scale term. Its Legendre transform should describe the distribution of Walsh energy across Hamming layers and should be related, but not identical, to the multifractal pressure already found for Fourier lobes.

The exact shell identity [\(431\)](#) supplies the marginal in binary length; the new problem is to resolve each shell by Hamming weight.

12.4 Higher-order rational multiwavelets

Haar has only one vanishing moment, explaining the geometric energy scale. Alpert-type multiwavelets on dyadic cells have any prescribed number r of vanishing moments. Their coefficients are finite linear combinations of integrals $\int_I x^j p(x) dx$, which the repository's antiderivative and exact dyadic machinery can evaluate rationally.

Open problem 12.4 (Fabius–Alpert hierarchy). Construct an exact rational multiwavelet hierarchy for p and derive its Fourier product-series. For order r , the detail energy should start at $h^{2r} \|p^{(r)}\|_2^2$ with a multiplier determined by the corresponding finite filter. Determine the Bell–Bernoulli coefficients and the meromorphic pole lattice of its scale-energy generating function.

This would replace the single sinc^4 multiplier by squared higher-order finite-difference symbols and produce a family of rational spectral objects.

12.5 Beta mixtures and confluent dyadic extrapolation

The beta mixture converges at the binomial scale $n^{-1} = 2^{-N}$, slower than the cell transform's 4^{-N} rate. Its complete Voronovskaya expansion is in integer powers of 2^{-N} , so the repository's confluent geometric filters can be applied without abandoning positivity at the base level.

Open problem 12.5 (Accelerated hypergeometric mixtures). Derive the full operator expansion of $\sum_k \Delta_{N,k} F_1(k+1; 2^N+1; z)$ on compact z -sets and construct $q = 1/2$ extrapolation filters that cancel the first r Bernstein corrections. Quantify how acceleration affects positivity, conditioning, and zero locations.

12.6 Quantile-width extremes and Lambert phases

The largest quantile width is the boundary width $q_{N,0} = G(2^{-N})$, up to symmetry. Hence the extremes of the projective array q_N are governed by the same Lambert phase as the endpoint inverse. The dual Rényi spectrum diverges for $s > 1$ because these boundary widths are much larger than 2^{-N} .

Conjecture 12.6 (Extreme-width partition function). For $s > 1$,

$$\sum_{k=0}^{2^N-1} q_{N,k}^s \sim 2G(2^{-N})^s \mathcal{C}_s(\Theta_N), \quad (473)$$

where $\mathcal{C}_s$ is a positive phase-dependent correction obtained by summing a finite boundary layer $G((j+1)2^{-N}) - G(j2^{-N})$ and Θ_N is the lower-Lambert phase.

This would turn inverse endpoint asymptotics into a discrete partition function and provide a phase-resolved refinement of the L^s threshold for G' .

12.7 Base- b and nonuniform scale systems

The multiresolution arguments do not fundamentally require base two. A base- b weighted-uniform series would produce b -adic mass arrays, b -ary Haar systems, and characters of $(\mathbb{Z}/b\mathbb{Z})^N$. The Walsh moment cancellation would be replaced by digit-character cancellation, while q -binomial aggregates would naturally involve $q = 1/b$.

Open problem 12.7 (Base- b cancellation spectrum). For the geometric scale b^{-j} , derive the analogue of [Theorems 6.5](#) and [6.6](#), identify the first nonzero moments of b -ary characters, and determine which Gaussian or multinomial q -coefficients aggregate the digit-position factors.

For nongeometric weights the finite character product still exists, but the clean q -binomial aggregation is replaced by an elementary symmetric function of the scales.

12.8 Formalization route

Most new statements separate into finite algebra plus standard convergence:

1. define dyadic mass vectors and prove projective consistency;
2. formalize the finite Hadamard matrix and the identity $H_N H_N^T = 2^N I$;
3. identify Walsh partial sums with dyadic conditional expectations;
4. prove the finite product [\(378\)](#) by induction on bit length;
5. extract vanishing moments from the order of zero at the origin;
6. use the finite q -binomial theorem for [\(388\)](#);
7. obtain beta mixtures by differentiating the already elementary Bernstein polynomial;
8. formalize quantile chord integration as an affine integral on each cell.

The analytic bottleneck is the all-orders alias estimate, which requires a Schwartz-space Poisson-summation API. The finite identities and all rationality claims should be substantially easier to integrate into Lean.

13 Numerical and exact-symbolic verification

The accompanying script `numerical_experiments.py` is not used as a substitute for proof. It serves four narrower purposes:

1. independently check the Fourier conventions and finite-product signs;
2. verify the exact Walsh moment and q -binomial identities over a large finite range using rational arithmetic;
3. compare four finite transform representations against the canonical sinc product;
4. test the constants in the Haar-energy expansion.

The default run uses 55 decimal digits, 160 sinc factors, and 200 half-integer cosine modes. It completes in a few seconds in the supplied environment and writes both a plain-text ledger and two figures.

13.1 Exact checks

SymPy rational arithmetic checked all vanishing moments and first nonzero moments for every nonzero Walsh index of binary length at most seven:

$$1016 \text{ exact moment identities checked, } \quad 0 \text{ failures.} \quad (474)$$

It also checked all 35 pairs (N, r) with $1 \leq N \leq 7$ in (388), again with zero failures. Selected exact moments are

m	binary m	$r(m)$	$\int_0^1 x^{r(m)} w_m(x) dx$
1	1	1	$-1/4$
2	10	1	$-1/8$
3	11	2	$1/16$
5	101	2	$1/32$
6	110	2	$1/64$
7	111	3	$-3/256$
15	1111	4	$3/2048$
31	11111	5	$-15/131072$

At the complex test point $z = 3.7 + 0.4i$, the largest discrepancy between the cell-by-cell integral of W_m and the finite product (379), for $0 \leq m \leq 15$, was 3.75×10^{-56} . At dyadic level seven, the largest numerically computed coefficient with odd Hamming weight was 1.73×10^{-54} , while finite Hadamard Parseval had zero residual at the working precision.

As a normalization check, the cosine evaluator gives

$$F(1/4) = 0.06944444444444445713\dots, \quad \frac{5}{72} = 0.06944444444444444\dots;$$

the 1.27×10^{-16} difference is the deliberately finite half-integer Fourier tail, not a failure of the exact dyadic value.

13.2 Transform convergence

The test frequencies were $t = 0.7, 3.1, 7.3, 11.2$. The table reports the maximum absolute error relative to $\Phi(t) = \prod_{j \geq 1} \text{sinc}(t/2^j)$.

N	cell/Walsh	midpoint atomic	beta mixture	Walsh product series
2	$9.720 \cdot 10^{-2}$	$8.750 \cdot 10^{-1}$	$3.312 \cdot 10^{-1}$	$9.720 \cdot 10^{-2}$
3	$2.780 \cdot 10^{-2}$	$1.416 \cdot 10^{-2}$	$1.985 \cdot 10^{-1}$	$2.780 \cdot 10^{-2}$
4	$7.012 \cdot 10^{-3}$	$3.517 \cdot 10^{-3}$	$1.098 \cdot 10^{-1}$	$7.012 \cdot 10^{-3}$
5	$1.760 \cdot 10^{-3}$	$8.805 \cdot 10^{-4}$	$5.796 \cdot 10^{-2}$	$1.760 \cdot 10^{-3}$
6	$4.403 \cdot 10^{-4}$	$2.202 \cdot 10^{-4}$	$2.980 \cdot 10^{-2}$	$4.403 \cdot 10^{-4}$
7	$1.101 \cdot 10^{-4}$	$5.505 \cdot 10^{-5}$	$1.512 \cdot 10^{-2}$	$1.101 \cdot 10^{-4}$

The equality of the cell and Walsh errors is exact, not accidental: the power-of-two Walsh partial sum is precisely p_N , whose transform is the cell mixture. The observed factor near four per level agrees with (333). The Bernstein/beta approximation displays its slower factor near two per level.

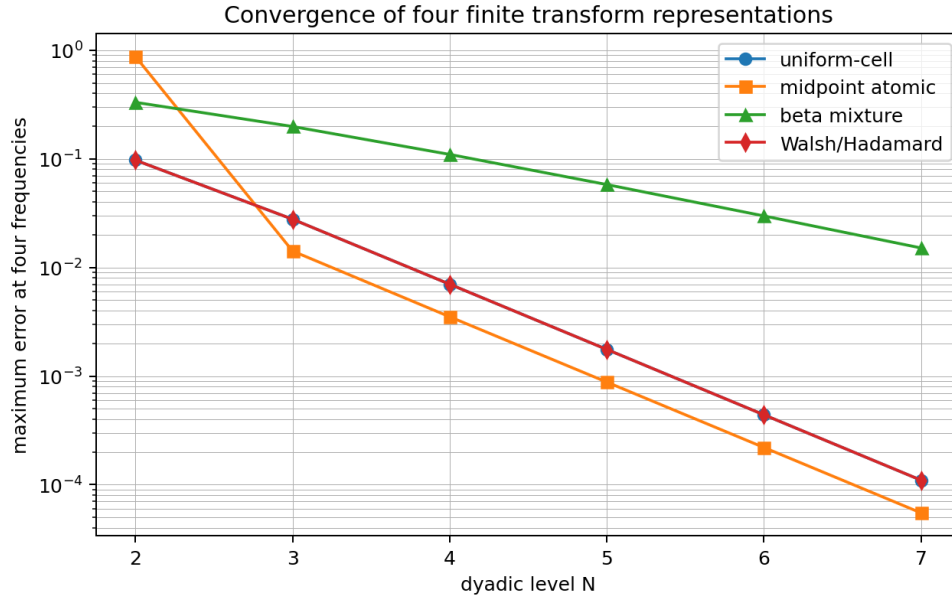


Figure 7: Convergence of the fixed-cell, midpoint-atomic, beta-mixture, and Walsh–Hadamard transform representations at four sample frequencies.

13.3 Haar energy asymptotics

The half-integer Fourier samples give

$$\|p'\|_2^2 = 25.8839531509750580138\dots, \quad (475)$$

$$\|p''\|_2^2 = 1656.57300166240371284\dots, \quad (476)$$

$$\|p'''\|_2^2 = 424082.688425575063737\dots \quad (477)$$

The exact rational energies and the three-term approximation from (440) behave as follows.

n	E_n	$16 \cdot 4^n E_n$	relative error of three terms
2	$6.2500000 \cdot 10^{-2}$	16.0000000000	$4.290 \cdot 10^{-1}$
3	$2.5292927 \cdot 10^{-2}$	25.8999572626	$3.914 \cdot 10^{-2}$
4	$6.2471084 \cdot 10^{-3}$	25.5881558642	$1.220 \cdot 10^{-3}$
5	$1.5757619 \cdot 10^{-3}$	25.8172829143	$1.627 \cdot 10^{-5}$
6	$3.9470093 \cdot 10^{-4}$	25.8671202723	$4.231 \cdot 10^{-8}$
7	$9.8723379 \cdot 10^{-5}$	25.8797415033	$4.139 \cdot 10^{-11}$

The scaled energy converges to $\|p'\|_2^2$, while the three-term expansion quickly becomes much more accurate than the displayed leading convergence.

13.4 Reproducibility files

The archive contains:

file	purpose
numerical_experiments.py	fully commented exact and numerical checks
numerical_results.txt	complete deterministic output ledger
multiresolution_convergence.pdf/.png	transform convergence figure
haar_energy_asymptotics.pdf/.png	scale-energy figure

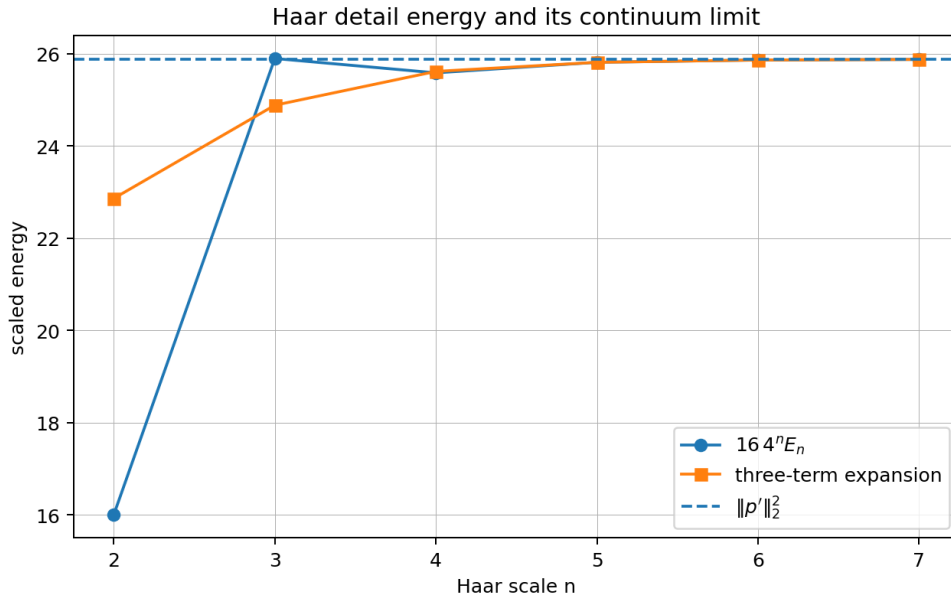


Figure 8: The scaled exact Haar energy, its three-term continuum expansion, and the limiting derivative norm.

The script contains independent direct and product implementations of the Walsh transform, rather than comparing a formula with a rearrangement of itself.

14 Conclusions

The central organizing object of this report is the exact dyadic mass array $\Delta_{N,k}$. It simultaneously yields:

1. a positive uniform-cell approximation and a rational exponential-polynomial transform limit;
2. rational unnormalized Haar coefficients and rational Faber–Schauder coefficients;
3. exact Walsh–Hadamard coefficients, with a finite sine–cosine product for every basis transform;
4. a positive beta-density mixture and a confluent-hypergeometric transform limit;
5. exact rational scale energies whose generating function has poles at the geometric lattice $4, 16, 64, \dots$

The dual inverse array $q_{N,k}$ exchanges fixed widths and variable weights for fixed weights and variable widths. Its chord integrals give finite sinc mixtures for Φ and complete homogeneous polynomial limits for the Fabius moments.

The strongest new structural bridge is

$$\begin{aligned} \text{binary Hamming weight} &\longleftrightarrow \text{Walsh vanishing moments} \\ &\longleftrightarrow \left[\begin{matrix} N \\ r \end{matrix} \right]_{1/2} \longleftrightarrow \text{Bell–Bernoulli higher moments.} \end{aligned}$$

It complements the repository’s existing $q = 1/4$ moment and scale structures. The strongest analytic bridge is

$$\begin{aligned} \text{rational Haar detail energies} &\longleftrightarrow \text{sinc}^4 \text{ multiplier} \\ &\longleftrightarrow \|p^{(r)}\|_2^2 \longleftrightarrow \text{meromorphic poles at } 4^r. \end{aligned}$$

Both bridges are finite and exact before limits are taken, making them good candidates for formalization.

The most promising next steps are the two-adic classification of the coefficient arrays, a higher-order rational multiwavelet hierarchy, phase-resolved asymptotics of the inverse boundary coefficients, and an exact analysis of the first Poisson aliases beyond the sinc^4 energy expansion.

A Compact formula index

object	equation	representation type
up	(264)	infinite convolution of dyadic uniforms
up	(265)	infinite-dimensional delta integral
up	(267)	limit of finite Thue–Morse splines
up	(271)	finite spline convolved with a scaled copy
Φ	(272)	infinite sinc product
Φ	(276)	q -Pochhammer product at $q = 1/4$
Φ	(277)	canonical zero product with $1 + v_2(n)$ multiplicity
$\log \Phi$	(278)	zeta power series
up	(286)	Fourier cosine integral
F	(287)	sine-over-frequency integral
F	(292)	half-integer cosine series
p	(293)	half-integer sine series
$\mathcal{F}_F$	(295)	compact transform through Φ
Φ	(299)	inverse-Fabius quantile integral
up	(307)	Fourier–Legendre/Bessel coefficient integral
up	(309)	Chebyshev/Bessel coefficient integral
p	(342)	rational unnormalized Haar series
F	(352)	rational Faber–Schauder series
$\mathcal{F}_F$	(358)	translated sinc^2 product-series
p	(368)	rational Walsh–Hadamard series
Φ	(381)	series of finite sine–cosine products
Walsh moments	(396)	Bell–Bernoulli finite-product coefficients
F	(401)	limit of rational Bernstein polynomials
p	(405)	positive rational beta mixture
Φ	(408)	confluent-hypergeometric mixture limit
Φ	(329)	fixed-width rational sinc mixture
Φ	(416)	equal-weight variable-width sinc mixture
Fabius moments	(337)	rational fixed-cell polynomial limit
Fabius moments	(410)	rational beta-moment limit
Fabius moments	(421)	inverse-dyadic polygon limit
E_n	(436)	sinc^4 Fourier multiplier plus aliases
E_n	(439)	Bell–Bernoulli Sobolev expansion

object	equation	representation type
$\mathcal{H}$	(445)	meromorphic rational-energy generating series
G	(452)	inverse Faber–Schauder series
$\widehat{G}$	(456)	inverse translated-sinc ² series

B Proof details for two auxiliary identities

B.1 Derivation of the local Haar expansion

Let m be the center of a parent interval of length h . Then

$$\begin{aligned}
 c_{n,k} &= h^{-1/2} \left(\int_{m-h/2}^m p(x) \, dx - \int_m^{m+h/2} p(x) \, dx \right) \\
 &= -h^{-1/2} \sum_{r \geq 0} \frac{2p^{(2r+1)}(m)}{(2r+1)!} \frac{(h/2)^{2r+2}}{2r+2} \\
 &= - \sum_{r \geq 0} \frac{h^{2r+3/2}}{2^{2r+1}(2r+2)!} p^{(2r+1)}(m),
 \end{aligned}$$

which is (432). Squaring and summing midpoint samples recovers the first coefficients in (440); the multiplier proof packages all integrations by parts into one sinc⁴ symbol.

B.2 Absolute convergence of the first-moment Hamming layers

Letting $N \rightarrow \infty$ in (388) gives

$$\sum_{\substack{m \geq 0 \\ r(m)=r}} 2^{-\sigma(m)} = \frac{2^{-r(r-1)/2}}{(1/2; 1/2)_r}. \quad (478)$$

Since $|a_m| \leq \int_0^1 p = 1$, the weight- r contribution formed from the first nonzero Walsh moments is absolutely summable. In particular, the sum rule (398) is independent of the blockwise ordering.

C Reproducibility ledger

Dyadic Multiresolution and Product-Series Representations
Numerical and exact-symbolic verification

=====

```

mpmath decimal digits : 55
sinc-product factors : 160
half-integer Fourier modes : 200

```

1. Normalizations

```

F(1/4) = 0.06944444444444445713227823382417
Expected F(1/4) = 0.069444444444444444444444444444444
absolute residual = 1.2687834e-16
sum of level-7 masses = 1.0

```

2. Exact Walsh moment and q-binomial checks

Walsh moment identities : 1016 checked, 0 failed
 q-binomial aggregates : 35 checked, 0 failed

Selected first nonzero Walsh moments:

m binary r integral $x^r w_m(x) dx$
 1 1 1 -1/4
 2 10 1 -1/8
 3 11 2 1/16
 5 101 2 1/32
 6 110 2 1/64
 7 111 3 -3/256
 15 1111 4 3/2048
 23 10111 4 3/4096
 31 11111 5 -15/131072

3. Walsh finite-product transform

max residual m=0,...,15 at z=3.7+0.4i: 3.745842e-56

4. Symmetry selection rule for Walsh coefficients

max $|a_m|$, odd Hamming weight (should be 0): 1.720445e-54
 max $|a_m|$, nonzero even weight : 0.722222222222221715
 Hadamard Parseval residual, level 7 : 0.0

5. Transform convergence

level max uniform-cell error max atomic error max beta-mixture error max Walsh error
 2 0.09719740288 0.8749804357 0.3311617664 0.09719740288
 3 0.0278015789 0.01416477865 0.1984606384 0.0278015789
 4 0.007011640078 0.003516689896 0.1097810782 0.007011640078
 5 0.001759556884 0.0008804669754 0.05795858183 0.001759556884
 6 0.0004403020799 0.0002201940863 0.02980477899 0.0004403020799
 7 0.0001101013485 5.505336511e-5 0.01511621869 0.0001101013485

6. Haar energy and all-orders continuum expansion

$||p'||_2^2 = 25.8839531509750580137998$
 $||p''||_2^2 = 1656.57300166240371284162$
 $||p'''||_2^2 = 424082.688425575063736863$
 $n E_n 16^{4^n} E_n \text{ rel.error(3-term)}$
 2 0.0625 16.000000000000006 0.42900991
 3 0.025292927014271 25.899957262613 0.039135771
 4 0.0062471083652827 25.58815586419799 0.0012204175
 5 0.0015757618966264 25.81728291432679 1.6265272e-5
 6 0.00039470093189003 25.86712027234482 4.2313174e-8
 7 9.872337914768e-5 25.87974150328945 4.1389921e-11

7. Low-level rational-structure samples (high-precision decimals)

level 1: Haar $c_{\{n,k\}}$ = -0.510688230857, 0.510688230857
 Schauder $d_{\{n,k\}}$ = -0.180555555556, 0.180555555556
 level 2: Haar $c_{\{n,k\}}$ = -0.125, -0.125, 0.125, 0.125
 Schauder $d_{\{n,k\}}$ = -0.03125, -0.03125, 0.03125, 0.03125
 level 3: Haar $c_{\{n,k\}}$ = -0.00943081845124, -0.0789575291971, -0.0789575291971, -0.00943081845124,
 0.00943081845124, 0.0789575291971, 0.0789575291971, 0.00943081845124
 Schauder $d_{\{n,k\}}$ = -0.00166714891975, -0.0139578510802, -0.0139578510802, -0.00166714891975,
 0.00166714891975, 0.0139578510802, 0.0139578510802, 0.00166714891975
 level 4: Haar $c_{\{n,k\}}$ = -0.000271267361112, -0.00808376736111, -0.0231662326389, -0.0309787326389,
 -0.0309787326389, -0.0231662326389, -0.00808376736111, -0.00027126736111, 0.00027126736111,
 0.00808376736111, 0.0231662326389, 0.0309787326389, 0.0309787326389, 0.0231662326389,
 0.00808376736111, 0.000271267361112
 Schauder $d_{\{n,k\}}$ = -3.3908420139e-5, -0.00101047092014, -0.00289577907986, -0.00387234157986,
 -0.00387234157986, -0.00289577907986, -0.00101047092014, -3.39084201388e-5, 3.39084201388e-5,
 0.00101047092014, 0.00289577907986, 0.00387234157986, 0.00387234157986,
 0.00289577907986, 0.00101047092014, 3.3908420139e-5

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